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Geometric Estimation of Autoregressive Processes and Spectral Scale Selection for Kernel Density Estimation

Abschlussarbeit an der Universität Ulm

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Abstract

1 Introduction

2 Geometric Estimation of Autoregressive Processes with bounded innovations

The original goal of this chapter was to apply the spectral scale selection developed in [4] to recover the order p of autoregressive time series. Even though the scale selection principle is primarily developed for applications in TDA, the hope was to use the well known sliding window embedding as a bridge between TDA and time series analysis. In the process of studying this problem, an alternative convex-geometric direction emerged, while the TDA approach turned out to be unfruitful. This chapter presents a new AR order selection method using ideas from convex geometry.

In the first section, we give an overview of the theory of time series needed for our purposes. In particular, we briefly review classical order selection and AR coefficient estimation. Next, we introduce the embedding of time series as point clouds in $\mathbb{R}^k$. We show consistency for a certain kind of convex-geometric functional, the minimum width, that can recover the correct AR order p. The projected sub-gradient descent is used to compute the empirical minimum width. A discussion of different kinds of innovation distributions is given, where the focus lies primarily on bounded distributions, of which the Beta distribution is the main example. Finally, we present our numerical results and interpret them.

The main goal of this chapter is to study the connection between time series analysis and convex geometry through sliding window analysis.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be the underlying probability space.

2.1 Preliminaries on time series

For information on autoregressive moving-average (ARMA) processes, see Brockwell et al [7, Chapter 3].

Definition 1. *Let $\{Z_t\}_{t\in\mathbb{Z}}$ be a family of independent and identically distributed random variables with finite second moment. A process $\{X_t\}$ is an ARMA(p, q) process if*

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + Z_t + \sum_{j=1}^{q} \theta_j Z_{t-j}, \qquad t \in \mathbb{Z}, \tag{2.1}$$

where the polynomials $\phi(z) := 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) := 1 + \theta_1 z + \cdots + \theta_q z^q$ are required to have no common factors. Additionally, $\phi_p \neq 0$ and $\theta_q \neq 0$.

In case of $\theta \equiv 1$ we recover the autoregressive (AR) process of order $p \in \mathbb{N}$ and X_t has the form

$$X_t = \sum_{i=1}^{p} \phi_i X_{t-i} + Z_t, \tag{2.2}$$

where $\phi_i \in \mathbb{R}$, $i = 1, \ldots, p$ are the coefficients with $\phi_p \neq 0$.

It is well known that there exists a stationary process $\{X_t\}_{t\in\mathbb{Z}}$ that can be written in the form (2.1) if $\phi(z) \neq 0$ for all $z \in \mathbb{C}, |z| = 1$. Another important property that we will need for the geometric interpretation of the embedded AR process is causality: An ARMA process $\{X_t\}_{t\in\mathbb{Z}}$ is called causal if there exist constants ψ_j, $j \geq 0$ with $\sum_{j\geq 0} |\psi_j| < +\infty$, such that $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ for all $t \in \mathbb{Z}$. It turns out that this property is stronger than stationarity, as an ARMA process is causal if and only if $\phi(z) \neq 0$ for all $z \in \mathbb{C}, |z| \leq 1$. In what follows, we will exclusively work with causal ARMA processes.

Define ψ_j as the coefficients of $1/\phi$

An important second order object for stationary sequences X is its auto covariance function $\gamma_X(h) = \mathrm{Cov}[X_{t+h}, X_t]$. The auto covariance matrix Γ_k of $(X_t, \ldots, X_{t+k-1})$

is

$$\Gamma_k = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(k-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(k-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(k-1) & \gamma(k-2) & \dots & \gamma(0) \end{pmatrix}$$

The empirical analog of Γ_k is obtained by plugging in the sample auto covariance function $\hat{\gamma}$. The sample covariance matrix $\hat{\Gamma}_k$ is positive semi-definite.

2.2 Estimation methods

We present the two classical coefficient estimation methods Yule-Walker and Maximum Likelihood. As a third estimation method we present Horváth's L^∞ estimator which is specifically designed for AR processes with innovations having uniform distribution.

Yule-Walker

The Yule-Walker Equations are given by

$$\Gamma_p \phi = \gamma_p, \quad \sigma^2 = \gamma(0) - \phi^T \gamma_p, \quad \gamma_p = \begin{pmatrix} \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix}$$

We obtain estimates $\hat{\phi}$ and $\hat{\sigma}^2$ by plugging in the empirical analogues $\hat{\Gamma}_p$ and $\hat{\gamma}_p$ into the Yule-Walker equations and solving for ϕ and σ^2.

In particular, the Yule-Walker estimate $\hat{\phi}$ is obtained by solving the linear system of equations $\hat{\Gamma}_p \hat{\phi} = \hat{\gamma}_p$. This can be solved efficiently by using the Durbin-Levinson algorithm which exploits the Toeplitz structure of $\hat{\Gamma}_k$.

Maximum-Likelihood

Write the joint density of the realization of an AR(p) process

$$f(x_1,\dots,x_T) = f_{(X_1,\dots,X_p)}(x_1,\dots,x_p)\prod_{t=p+1}^{T}\underbrace{f_{X_t|X_1,\dots,X_{t-1}}(x_t|x_1,\dots,x_{t-1})}_{=f_{X_t|X_{t-p},\dots,X_{t-1}}(x_t|x_{t-p},\dots,x_{t-1})}$$

Conditioning on the first p observations gives

$$\begin{aligned} f_{X_{p+1},\dots,X_T|X_1,\dots,X_p}(x_{p+1},\dots,x_T|x_1,\dots,x_p) &= \frac{f(x_1,\dots,x_T)}{f_{(X_1,\dots,X_p)}(x_1,\dots,x_p)} \\ &= \prod_{t=p+1}^{T} f_{X_t|X_{t-p},\dots,X_{t-1}}(x_t|x_{t-p},\dots,x_{t-1}) \end{aligned} \tag{2.3}$$

We use conditional ML because otherwise we would have to specify the joint distribution of $(X_1,\dots,X_p)$.

Now we know that $X_t|X_{t-p},\dots,X_{t-1}$ is distributed as $Z_t+\mu_t$, where $\mu_t = \sum_{j=1}^{p}\phi_j X_{t-j}$. At this point, we can assume different distribution classes for Z_t and continue to evaluate with the assumed distribution.

Assuming that the innovations are normally distributed, we obtain the quasi-conditional maximum likelihood function

$$f(x_{p+1},\dots,x_T|x_1,\dots,x_p) = \prod_{t=p+1}^{T} f_{\mu_t,\sigma^2}(x_t), \quad f_{\mu_t,\sigma^2} \sim N(\mu_t,\sigma^2).$$

The conditional log-likelihood function is then

$$l(\phi,\sigma^2) = -\frac{T-p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}(x_t-\mu_t)^2$$

Note that conditional quasi maximum likelihood is equivalent to least squares. Under fixed σ^2 maximizing the expression above is equivalent to minimizing

$$\sum_{t=p+1}^{T}\left(x_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right)^2$$

Horváth

Of course we are in the scenario where we assume that the innovations are bounded. Horváth [21] assumed further that the innovations are $\mathcal{U}[-h,h]$ and found that the conditional likelihood function can then be written as a Linear program.

The Estimation of the coefficients of an AR process with uniform innovations was already considered by Horváth [21] in 1989. His starting point is Equation (2.3), but instead of using the Gaussian density he plugs in the uniform density. The resulting conditional likelihood function becomes

$$l_{x_{p+1},\dots,x_T|x_1,\dots,x_p}(\phi,h) = \frac{1}{(2h)^{T-p}}\mathbf{1}\left\{\left|y_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right| \leq h \text{ for all } t = p+1,\dots,T\right\}. \tag{2.4}$$

This can be written as a linear programming problem, as demonstrated below.

We find a vector $b = (b_0, b_1, \dots, b_n)$ that minimizes $e_1^T b = b_0$ subject to

$$\begin{pmatrix} A \\ -A \end{pmatrix}\begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{pmatrix} \leq \mathbf{1}, \quad A = \begin{pmatrix} x_{p+1} & \dots & x_1 \\ \vdots & & \vdots \\ x_T & \dots & x_{T-p} \end{pmatrix} \in \mathbb{R}^{(T-p)\times(p+1)}.$$

In modern times such a problem can be solved very efficiently by, for example, the open source software HiGHS, which is also available as Python package.

From Equation (2.4) it follows that

$$\hat{\phi} = \arg\min_{\phi}\max_{t}\left|x_t - \sum_{j=1}^{p}\phi_j x_{t-j}\right|$$
$$\hat{h} = \max_{t}\left|x_t - \sum_{j=1}^{p}\hat{\phi}_j x_{t-j}\right|$$

Thus, conditional maximum likelihood under uniform density leads to L^∞-regression analogous to how conditional ML under Gaussian density leads to L^2-regression.

2.3 Order selection methods

There are several established methods for choosing an order $\hat{p}$ given a realization $x = (x_1, \dots, x_T)$ that is assumed to come from an AR(p) process. All of them require to obtain estimates $\hat{\phi}_1^{(k)}, \dots, \hat{\phi}_k^{(k)}$ for each candidate order $k = 1, \dots, K$. This can be done efficiently using the Durbin-Levinson algorithm.

Information criteria selectors

Denote by $l(\hat{\theta})$ the log-likelihood function evaluated at its maximizer $\hat{\theta}$. The class of information criteria selectors has the form

$$\hat{p}_{\mathrm{IC}} = \arg\min_{1 \le k \le K} \mathrm{IC}_T(k), \quad \mathrm{IC}_T(k) = -\frac{2}{T-K} l(\hat{\theta}) + \frac{k c_T}{T-K}.$$

For example, $\hat{\theta} = (\hat{\phi}, \hat{\sigma}^2)$ and $l(\hat{\phi}, \hat{\sigma}_k^2) = -\frac{T-K}{2}(\log(2\pi\hat{\sigma}_k^2) + 1)$ and therefore

$$\mathrm{IC}_T(k) = \log(\hat{\sigma}_k^2) + \frac{k c_T}{T-K}$$

where we have dropped all terms independent of k. Since $\hat{\sigma}_1^2 \ge \hat{\sigma}_2^2 \ge \cdots \ge \hat{\sigma}_K^2$ (the lack of fit decreases as the candidate order increases), it must be counteracted by a penalty term for the candidate order k.

Here, $c_T > 0$ is some penalty term sequence. Different choices of c_T lead to different AR order selectors that are known in the literature. Choosing $c_T = 2$ leads to the Akaike information criterion (AIC), $c_T = \log T$ to the Bayesian information criterion (BIC), and $c_T = 2 \log\log T$ to the Hannan-Quinn information criterion (HQIC). All of these criteria can be motivated in their own sense and serve a different purpose. The AIC is asymptotically efficient, but not strongly consistent. BIC and HQIC are strongly consistent but not asymptotically efficient.

Partial autocorrelation cutoff

The partial autocorrelation function α is given by $\alpha(0) = 1$ and $\alpha(k) = \phi_{kk}$, where $\phi_k = (\phi_{1k}, \dots, \phi_{kk})$ is the coefficient vector obtained as a solution of the Yule-Walker Equations. The sample partial autocorrelation function $\hat{\alpha}$ is defined analogously through empirical analogs. For an AR(p) process we have $\alpha(p) = \phi_p$ and $\alpha(k) = 0$ for $k > p$. For causal AR(p) processes with iid innovations it is known

that $\sqrt{T}\hat{\alpha}(k) \xrightarrow{d} N(0,1)$ as $T \to +\infty$ for each fixed $k > p$. In particular, assuming T is large, we have $\mathbb{P}(\sqrt{T}\hat{\alpha}(k) \in [-b, b]) \approx 2\Phi(b) - 1$. Specifically, setting $b = \Phi^{-1}(0.975) \approx 1.96$, we obtain $\mathbb{P}(\hat{\alpha}(k) \in [-\frac{1.96}{\sqrt{T}}, \frac{1.96}{\sqrt{T}}]) \approx 0.95$. Therefore, we might choose as the AR order the last k that $\hat{\alpha}(k)$ is outside the bounds $\pm\frac{1.96}{\sqrt{T}}$.

2.4 Embedding of time series

The sliding window embedding is the conceptual bridge between time series analysis and other fields in mathematics, for example, Topological Data Analysis (TDA) or Convex Geometry. The time series is embedded into $\mathbb{R}^k$ for some embedding dimension $k \in \mathbb{N}$. In TDA the obtained sliding window point cloud is put into the persistent homology pipeline to look for interesting topological features. See the papers [30]. We will instead study the geometric properties of the sliding window point cloud in Section 2.5 below. Before that, we introduce the sliding window embedding. Let $X = (X_1, \ldots, X_T)$ be one realization of some univariate time series. Then we obtain a point cloud from X by defining:

$$Y_t^{(k)} = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-k+1} \end{pmatrix} \in \mathbb{R}^k, \quad \mathcal{P}_k = \left\{ Y_t^{(k)} : t = K, \ldots, T \right\} \subseteq \mathbb{R}^k$$

Here, $k \in \mathbb{N}$ is the dimensionality of the point cloud $\mathcal{P}_k$ and K a predetermined upper cap, that is, $k \in \{1, \ldots, K\}$. Note that we are discarding observations $X_1, \ldots, X_{K-k}$ in the construction of the point cloud $\mathcal{P}_k$. This is mainly for computational and notational convenience. We choose the "backward" embedding $Y_t^{(k)}$ over the "forward" embedding $\tilde{Y}_t^{(k)} = (X_t, X_{t+1}, \ldots, X_{t+k-1})^T$ for notational convenience throughout this chapter. Geometrically, the resulting point clouds are equivalent up to an orthogonal transformation and a time index shift, that is, $\tilde{Y}_t^{(k)} = R_k Y_{t+k-1}^{(k)}$, where $R_k = \operatorname{adiag}(1, \ldots, 1)$.

2.5 Geometric Approach

In this section, we study how the geometry of the point cloud $\mathcal{P}_k \subseteq \mathbb{R}^k$ changes with increasing embedding dimension k. For bounded innovation distributions, like uniform or beta, the point cloud $\mathcal{P}_k$ is characterized by a well known convex–geometric object. The question becomes if we can apply some convex–geometric functional f to these well known objects and detect a transition at $k = p + 1$, where p is the AR order of the embedded time series.

In order to be able to talk about the "geometry of a point cloud" in a rigorous way, we first need to introduce some concepts from probability theory. Let Y be a k-dimensional random vector with density f_Y. The *support* of Y is given by

$$\operatorname{supp}(Y) = \{x \in \mathbb{R}^k : \mathbb{P}(Y \in B_r(x)) > 0 \text{ for all } r > 0\}.$$

In the case where f_Y is continuous, it holds $\operatorname{supp}(Y) = \overline{\{y \in \mathbb{R}^k : f_Y(y) > 0\}}$.

Let $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ be the MA(∞)-representation of an AR(p) process, where $\{Z_t\}_{t\in\mathbb{Z}}$ are the independent and identically distributed innovations, and where the sequence $\{\psi_j\}_{j\in\mathbb{N}_0}$ is given recursively by $\psi_{-(p-1)} = \cdots = \psi_{-1} = 0$, $\psi_0 = 1$ and $\psi_j = \phi_1\psi_{j-1} + \cdots + \phi_p\psi_{j-p}$ for all $j \geq 1$. Now each point $Y_t^{(k)} \in \mathcal{P}_k$ can be written as

$$Y_t^{(k)} = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-k+1} \end{pmatrix} = \sum_{j=0}^{\infty} g_j^{(k)} Z_{t-j}, \tag{2.5}$$

where $g_j^{(k)} = (\psi_j, \psi_{j-1}, \ldots, \psi_{j-k+1})^T$. Due to stationarity of $\{X\}_{t\in\mathbb{Z}}$, the distribution of $Y_t^{(k)}$ does not depend on t and we write $Y_t^{(k)} \sim \mu_k$ which means $\mu_k(B) = \mathbb{P}(Y_t^{(k)} \in B)$ for all $B \in \mathcal{B}(\mathbb{R}^k)$.

Bounded innovation distributions lead to $Y_t^{(k)}$ having compact support. Thus, in our work, we focus on innovations Z_t that have support $[a, b]$, $a < b$. Possible ideas for extending our approach to innovation distributions with unbounded support are given in the outlook chapter 5. The most prominent example is the uniform distribution $\mathcal{U}[a, b]$. More generally, we can consider the Beta distribution. A random

variable X follows the Beta distribution Beta(α, β) if its density is given by

$$f_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}\mathbf{1}\{x \in (0,1)\}, \quad \alpha, \beta > 0,$$

where $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ and Γ is the Gamma function. For $\alpha = \beta = 1$ we have Beta$(\alpha,\beta) =$ Beta$(1,1) = \mathcal{U}[0,1]$.

2.5.1 Convex–Geometric Definitions and Connection to sliding-window support

We will start exploring the geometry of the sequence of point clouds $\{\mathcal{P}_k\}_{k=1}^K$ by considering the case where the innovations are uniformly distributed, that is, $Z_t \sim \mathcal{U}[-1,1]$. To this end, we first introduce some common definitions from convex geometry. A good survey on convex geometry is the book [18].

Definition 2 (Minkowski sum)**.** *Let A and B be two subsets of a vector space V. Their Minkowski sum is given by the set $A + B = \{a + b : a \in A, b \in B\}$.*

Definition 3 (Hausdorff distance)**.** *Let (M,d) be a metric space and X, Y two nonempty subsets of M. The Hausdorff distance d_H between X and Y is defined by*

$$d_H(X,Y) = \max\left\{\sup_{x\in X} d(x,Y), \sup_{y\in Y} d(X,y)\right\},$$

where $d(x,Y) = \inf\{d(x,y) : y \in Y\}$.

Definition 4 (Zonotope)**.** *A zonotope is the Minkowski sum of finitely many line segments. It is said to be generated by $w_1, \ldots, w_n \in \mathbb{R}^d$ if it has the form*

$$\{a_1 w_1 + \cdots + a_n w_n : 0 \le a_j \le 1 \text{ for all } j\}$$

Definition 5 (Zonoid)**.** *Let $w = (w_n)_{n\in\mathbb{N}_0}$ be a sequence of vectors in $\mathbb{R}^d$. A set $\mathcal{Z}$ is called a zonoid generated by w if $d_H(\mathcal{Z}, \mathcal{Z}_n) \to 0$ for $n \to \infty$, where $\mathcal{Z}_n$ is a zonotope generated by $w_0, w_1, \ldots, w_n$.*

Using Equation (2.5) we see that the support of $Y_t^{(k)}$ is given by a zonoid generated by $\{g_j^{(k)}\}_{j\in\mathbb{N}_0}$. We denote this zonoid by $\mathcal{Z}^{(k)}$. By Definition 5 the zonoid $\mathcal{Z}^{(k)}$ can be approximated by the zonotopes $Z_n^{(k)}$ generated by $g_0^{(k)}, \dots, g_n^{(k)}$.

Let $[a, b]$, $a < b$ be the support of the innovations $\{Z_t\}$. By writing $[a, b] = c + d[-1, 1]$, where $c = (a + b)/2$ and $d = (b - a)/2$, we can pull everything back to the special case where the support is just the interval $[-1, 1]$. We have

$$\operatorname{supp}(Y_t^{(k)}) = \sum_{j=0}^{\infty} g_j^{(k)} \operatorname{supp}(Z_{t-j}) = \underbrace{c \sum_{j=0}^{\infty} g_j^{(k)}}_{=:C_k} + \underbrace{d \sum_{j=0}^{\infty} g_j^{(k)}[-1, 1]}_{=\mathcal{Z}^{(k)}},$$

that is, we have translated and scaled the original zonoid $\mathcal{Z}^{(k)}$.

In fact, we can derive the following convergence speed result. It tells us that the approximating zonotope converges to the zonoid faster if the underlying AR process is “more causal”.

Proposition 1. *Let $\mathcal{Z}^{(k)}$ and $\mathcal{Z}_n^{(k)}$ be like above coming from an causal autoregressive process where the impulse sequence $\{\psi_j\}$ fulfills $|\psi_j| \leq C\rho^j$ for some constants $C > 0$ and $\rho \in (0, 1)$. Then*

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}) \leq C \frac{\sqrt{k}}{1-\rho} \frac{1}{\rho^{k-2}} \rho^n.$$

In particular, $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}^{(k)}) = \mathcal{O}(\rho^n)$. That is, $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}^{(k)}) \to 0$ is exponentially fast.

Proof. See Appendix Section 6.1.1. □

A convex body is a compact convex set with a nonempty interior. As a convex geometric functional on a convex body K we use its minimum width. For a rigorous notion of width, we introduce the support function.

Definition 6 (Support function)**.** *Let $K \subseteq \mathbb{R}^d$ be a convex nonempty set. The support function of K is $h_K : \mathbb{R}^d \to \mathbb{R}$, $h_K(u) = \sup\{\langle x, u\rangle : x \in K\}$.*

We can express the Hausdorff distance between convex bodies using their support functions.

Lemma 1. *Let $K, L \subseteq \mathbb{R}^k$ be two convex bodies. Then*

$$d_H(K,L) = \sup_{u \in \mathbb{S}^{k-1}} |h_K(u) - h_L(u)| = \|h_K - h_L\|_{\infty, \mathbb{S}^{k-1}}.$$

Proof. See [16, Section 4]. □

The width of a convex body $K \subseteq \mathbb{R}^d$ in direction u is given by $w_K(u) = h_K(u) + h_K(-u)$ and if K is centrally symmetric, then $w_K(u) = 2h_K(u)$. The minimum width is $m(K) = \min\{w_K(u) : u \in \mathbb{S}^{d-1}\}$. We write $w_k(u) = w_{\mathcal{Z}^{(k)}}(u)$ and $m_k = m(\mathcal{Z}^{(k)})$ for the width in direction u and the minimum width of the zonoid $\mathcal{Z}^{(k)}$, respectively.

The support function of a zonoid $\mathcal{Z}^{(k)}$ generated by $\{g_j^{(k)}\}_{j \in \mathbb{N}_0}$ is given by

$$h_{\mathcal{Z}^{(k)}}(u) = \sum_{j=0}^{\infty} |\langle u, g_j^{(k)} \rangle|, \quad u \in \mathbb{S}^{k-1}.$$

Next, we formulate a result that states that the relation $k \mapsto m_k$ has an elbow point at $k = p + 1$.

Proposition 2. *Let $\mathcal{P}_k$ be the sliding window embedding of a causal AR(p) process with coefficients $\phi_1, \ldots, \phi_p$, $\phi_p \neq 0$ and $\mathcal{Z}^{(k)}$ the corresponding support zonoid. In the case $k \geq p + 1$ it holds*

$$m_k = \min_{u \in \mathbb{S}^{k-1}} w_k(u) = \frac{2}{\sqrt{1 + \phi_1^2 + \cdots + \phi_p^2}},$$

where the minimum is attained at the finite set of points

$$\left\{ \pm \frac{q_r^{(k)}}{\|q_\phi\|_2} : r = 0, \ldots, k - (p+1) \right\} \tag{2.6}$$

with $q_\phi = (1, -\phi_1, \ldots, -\phi_p)$ and

$$q_r^{(k)} = (\underbrace{0, \ldots, 0}_{r}, 1, -\phi_1, \ldots, -\phi_p, 0, \ldots, 0) \in \mathbb{R}^k.$$

In the case $k < p+1$ it holds

$$m_{k-1} \geq m_k > \frac{2}{\sqrt{1+\phi_1^2+\cdots+\phi_p^2}}, \quad \forall k = 2, \ldots, p.$$

Proof. See Appendix Section 6.1.2. □

Remark 1 (about Proposition 2)**.** *For $k = p+1$ the minimizing direction is $u = (u_0, u_1, \ldots, u_p) = q_\phi/\|q_\phi\|$. We can recover the AR coefficients by $\phi_j = -(q_\phi)_j = -\|q_\phi\|u_j = -u_j/u_0$, $j = 1, \ldots, p$. If $\hat{u}$ is a practically obtained minimizer (for example, via gradient descent, see below), we get coefficient estimates $\hat{\phi}_j = -\hat{u}_j/\hat{u}_0$.*

2.5.2 The empirical minimum width and its consistency

An empirical version of the width $w_k(u)$ is obtained by inserting $\mathrm{conv}(\mathcal{P}_k)$ in place of $\mathcal{Z}^{(k)}$. Consequently, we obtain

$$\begin{aligned}\hat{w}_{k,T}(u) &= h_{\mathrm{conv}(\mathcal{P}_k)}(u) + h_{\mathrm{conv}(\mathcal{P}_k)}(-u) \\ &= \max_{x\in\mathcal{P}_k}\langle x, u\rangle - \min_{x\in\mathcal{P}_k}\langle x, u\rangle \\ &= \max_t \langle Y_t^{(k)}, u\rangle - \min_t \langle Y_t^{(k)}, u\rangle\end{aligned}$$

and $\hat{m}_{k,T} = \min\{\hat{w}_k(u) : u \in \mathbb{S}^{k-1}\}$.

The next results establish the consistency of the empirical minimum width, that is, $\hat{m}_{k,T} \to m_k$ for $T \to \infty$. In the next Lemma we show that the convex hull of a point cloud converges to its support $\mathcal{Z}$ almost surely in Hausdorff distance given that the point cloud is dense in $\mathcal{Z}$ with probability 1.

Lemma 2. *Let $\{Y_t\}_{t\geq 1}$ be a $\mathbb{R}^k$ valued process and $\mathcal{P}_T = \{Y_1, \ldots, Y_T\}$. Let $\mathcal{Z} \subseteq \mathbb{R}^k$ be nonempty, compact and convex. Assume*

$$\mathbb{P}\left(\overline{\{Y_t : t \geq 1\}} = \mathcal{Z}\right) = 1$$

Then

$$d_H(\mathrm{conv}(\mathcal{P}_T), \mathcal{Z}) \overset{T\to+\infty}{\to} 0, \quad \textit{almost surely}$$

Proof. See Appendix Section 6.1.3. □

There exists work on the convergence speed of the sequence $d_H(\text{conv}(\mathcal{P}_T), \mathcal{Z})$ for $T \to +\infty$, but only under certain assumptions. Brunel [8] proves $\mathbb{E}[d_H(\mathcal{P}_T, \mathcal{Z})] = \mathbf{O}((\frac{\log n}{n})^{1/\alpha})$ under the assumption that "enough probability mass is put at the caps of $\mathcal{Z}$". He also assumes that $Y_1, \ldots, Y_T$ are independent and identically distributed. Louhichi [27] instead considers stationary points $Y_1, \ldots, Y_T$ but assumes a stronger "a-b-condition" in addition with other specific conditions. She then proceeds to show the same kind of convergence speed as Brunel.

We state the "enough probability mass" condition from Brunel and show that the sliding window point cloud fails this condition. The ϵ-cap of a convex body $\mathcal{Z} \subseteq \mathbb{R}^k$ in direction $u \in \mathbb{S}^{k-1}$ is the set $C_{\mathcal{Z}}(u, \epsilon) = \{x \in K : \langle x, u \rangle \geq h_{\mathcal{Z}}(u) - \epsilon\}$. Brunel's condition is that there exist $\alpha, L > 0$ and $0 < \epsilon_0 \leq 1$ such that $\mu_k(C_{\mathcal{Z}}(u, \epsilon)) \geq L\epsilon^\alpha$ for all $u \in \mathbb{S}^{k-1}$ and $\epsilon \in [0, \epsilon_0]$.

Louhichi's "a-b condition" is given as follows. There exist $a, b, \epsilon_0 > 0$ such that $\inf_{y \in \mathcal{Z}^{(k)}} \mathbb{P}(Y_t^{(k)} \in B(y, \epsilon)) \geq a\epsilon^b$ for all $0 < \epsilon \leq \epsilon_0$. This is a stronger condition as Brunel's condition, because $Y_t^{(k)} \in B(y, \epsilon)$ implies $Y_t^{(k)} \in C_{\mathcal{Z}}(u, \epsilon)$.

The next Proposition shows that in our setting, Brunel's condition fails and therefore also Louchichi's condition.

Proposition 3. *Let $\mathcal{Z}^{(k)}$ be the zonoid from above. Then we have*

$$\lim_{\epsilon \to 0} \frac{\mu_k(C_{\mathcal{Z}^{(k)}}(e_1, \epsilon))}{\epsilon^\alpha} = 0$$

for all $\alpha > 0$. In particular, Brunel's condition is not fulfilled.

Proof. See Appendix Section 6.1.5. □

decrease overlap to increase convergence speed?

The next Lemma now shows that the support of a stationary and ergodic process is in fact dense in $\mathcal{Z}$ with probability 1.

Lemma 3. *Let $\{Y_t\}$ be stationary and ergodic. Furthermore, let $\mathcal{Z} = \text{supp}(Y_t)$ be compact and convex. Then $\{Y_t : t \geq 1\}$ is dense in $\mathcal{Z}$ almost surely.*

Proof. See Appendix Section 6.1.4. □

Remark 2. *A causal AR process has the representation $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$. Such processes are stationary and ergodic given that the innovations $\{Z_t\}$ are stationary and ergodic. Furthermore, the embedded process $t \mapsto Y_t^{(k)}$ inherits the stationarity and ergodicity.*

Corollary 1. *We have $\hat{m}_{k,T} \to m_k$ for $T \to \infty$ almost surely.*

Proof. For once we have

$$|\hat{m}_{k,T} - m_k| = |\min_u \hat{w}_{k,T}(u) - \min_u w_k(u)| \leq \sup_u |\hat{w}_{k,T}(u) - w_k(u)|$$

By Lemma 1 we have

$$\sup_{u \in \mathbb{S}^{k-1}} |\hat{w}_{k,T}(u) - w_k(u)| \leq 2d_H(\mathrm{conv}(\mathcal{P}_k), \mathcal{Z}^{(k)})$$

Finally, applying Lemmas 2 and 3 and Remark 2 proves the statement. □

The proof of Corollary 1 in particular establishes uniform almost sure convergence of $\hat{w}_{k,T}$ to w_k on $\mathbb{S}^{k-1}$. This allows us to establish an argmin consistency result.

Corollary 2. *Let $\mathcal{U}_k$ be the set of minimizing directions of w_k and $\hat{\mathcal{U}}_{k,T}$ the set of minimizing directions of $\hat{w}_k$. Then we have*

$$\sup_{\hat{u} \in \hat{\mathcal{U}}_{k,T}} \mathrm{dist}(\hat{u}, \mathcal{U}_k) \to 0, \quad T \to +\infty, \quad \textit{almost surely},$$

that is, every empirical minimizer becomes arbitrarily close to the set of the population minimizing directions.

Proof. Let $\hat{u} \in \hat{\mathcal{U}}_{k,T}$. For every $\epsilon > 0$ define the set $A_\epsilon := \{u \in \mathbb{S}^{k-1} : \mathrm{dist}(u, \mathcal{U}_k) \geq \epsilon\}$. Note that A_ϵ is a compact set. Now fix $\epsilon > 0$ and assume that A_ϵ is nonempty, that is, there exists $u_0 \in \mathbb{S}^{k-1}$ such that $\|u_0 - \tilde{u}\| \geq \epsilon$ for all $\tilde{u} \in \mathcal{U}_k$. By continuity of w_k there exists $u_\epsilon \in A_\epsilon$ with $w_k(u_\epsilon) = \min_{u_0 \in A_\epsilon} w_k(u_0)$. Define $\delta_\epsilon := w_k(u_\epsilon) - m_k \geq 0$. Since $\delta_\epsilon = 0$ would be a contradiction, we have $\delta_\epsilon > 0$.

We already know that $0 \leq w_k(\hat{u}_{k,T}) - m_k \to 0$ for $T \to \infty$. In particular, $w_k(\hat{u}_{k,T}) - m_k < \delta_\epsilon$ for all sufficiently large T (all $T \geq T_0$ for some T_0).

Now $\hat{u}_{k,T} \in A_\epsilon$ would imply $w_k(\hat{u}_{k,T}) - m_k \geq \delta_\epsilon$, which would be a contradiction. Therefore, $\hat{u}_{k,T} \notin A_\epsilon$ for all $T \geq T_0$. This means $\operatorname{dist}(\hat{u}_{k,T}, \mathcal{U}_k) < \epsilon$. Since $\epsilon > 0$ was arbitrarily chosen, this proves $\operatorname{dist}(\hat{u}_{k,T}, \mathcal{U}_k) \to 0$ for $T \to \infty$. Finally, we get $\sup_{\hat{u} \in \hat{\mathcal{U}}_{k,T}} \operatorname{dist}(\hat{u}, \mathcal{U}_k) \to 0$ for $T \to \infty$. □

We can specify this result in the case where $k = p + 1$. Then we know from Proposition 2 that $\mathcal{U}_{p+1} = \{\pm \|q_\phi\|_2^{-1} q_\phi\}$. Consequently, we would get

$$\sup_{\hat{u} \in \hat{\mathcal{U}}_{p+1,T}} \min_{s \in \{-1,1\}} \|\hat{u} - s\|q_\phi\|_2^{-1} q_\phi\|_2 \to 0, \quad T \to \infty.$$

The first coordinate of $\|q_\phi\|_2^{-1} q_\phi$ is $(1 + \phi_1^2 + \cdots + \phi_p^2)^{-1/2}$ (nonzero), and by the convergence result above, the first coordinate of $\hat{u}_{k,T}$ is also eventually nonzero almost surely. The map $g : (u_0, u_1, \ldots, u_p) \mapsto (-u_1/u_0, \ldots, -u_p/u_0)$ is continuous on $\mathbb{S}^p \setminus \{u = (u_0, \ldots, u_p) : u_0 = 0\}$ and even. By Remark 1 we get

$$\sup_{\hat{u} \in \hat{\mathcal{U}}_{k,T}} \|g(\hat{u}) - \phi\|_2 \to 0, \quad T \to \infty.$$

Thus, we have shown that a continuous transformation of every empirical minimizer is a strongly consistent estimator for the AR coefficients ϕ.

cite classical argmin consistency theorem

2.5.3 Minimization via Projected sub-gradient descent

The idea now is to minimize $\hat{w}_{k,T}(u)$ over all $u \in \mathbb{S}^{k-1}$ and examine the relation $k \mapsto \hat{m}_{k,T}$. The hope is that the theoretically established elbow point that occurs at $k = p + 1$ will also be detectable at the empirical level.

Before discussing the minimization of $\hat{w}_k = \hat{w}_{k,T}$ we collect some properties of this function.

Proposition 4. *1. $\hat{w}_k$ is Lipschitz continuous on S^{k-1} with Lipschitz constant* $\operatorname{diam}(\mathcal{P}_k) = \max_{s,t} \|Y_s^{(k)} - Y_t^{(k)}\|$

2. $\hat{w}_k$ is convex on $\mathbb{R}^k$

3. The set of minimizers $\mathcal{X}_k = \arg\min\{\hat{w}_k(u) : u \in \mathbb{S}^{k-1}\}$ is nonempty and compact.

4. Denote by $\hat{m}_k$ the minimum of $\hat{w}_k$ on $\mathbb{S}^{k-1}$. If $T \geq K + k$, then $\hat{m}_k > 0$ almost surely.

5. $\hat{w}_k$ is piecewise linear and differentiable almost everywhere

6. The set of non-differentiable points is given by

$$\mathcal{N} = \bigcup_{(s,t)\neq(p,q)} \{u : \langle Y_s^{(k)} - Y_t^{(k)} + Y_q^{(k)} - Y_p^{(k)}, u\rangle = 0\}$$

7. The gradient of $\hat{w}_k$ in direction $u \notin \mathcal{N}$ is given by

$$\nabla_u \hat{w}_k = Y_{s_0(u)}^{(k)} - Y_{t_0(u)}^{(k)}, \quad (s_0(u), t_0(u)) = \arg\max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u\rangle$$

Proof. See Appendix Section 6.1.6. □

A classical minimization approach is the projected sub-gradient descent; see Algorithm 1 and Figure 2.1 which visualizes one step of the projected sub-gradient descent from Algorithm 1. [26] provide convergence guaranties. We show that our setting fulfills the assumptions stated in [26, Theorem 4] needed for the linear convergence of Algorithm 1 if the initial point u_0 is close to $\mathcal{X}_k$. But first, we state this result in our notation.

Theorem 1 ([26, Theorem 4])**.** *Consider the minimization problem $\min f(x)$, $x \in \mathbb{S}^{k-1}$, where f is convex on $\mathbb{R}^k$ and Lipschitz-continuous with constant $L > 0$. Let $\{u_k\}_{k\geq 0}$ be the sequence obtained from Algorithm 1. Let $\mathcal{X} = \arg\min\{f(x) : x \in \mathbb{S}^{k-1}\}$ be the set of minimizing directions of f and let f^* be the minimum of f over $\mathbb{S}^{k-1}$. Assume that the weak sharp minima condition holds, that is,*

$$\exists \alpha, \rho > 0 \text{ such that } \forall u \in \mathbb{S}^{k-1}, \operatorname{dist}(u, \mathcal{X}) < \rho : f(u) - f^* \geq \alpha \operatorname{dist}(u, \mathcal{X}). \quad (2.7)$$

Algorithm 1 Projected gradient descent

Require: $P = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$, $u_0 \in \mathbb{S}^{d-1}$, $\{\eta_t\}_{t \geq 0}$
Ensure: $\hat{u}$

```
Initialize ŵ ← ŵ_k(u_0), û ← u_0
for t = 0, 1, ... do
    (i_t, j_t) ← arg max_(i,j) ⟨x_i − x_j, u⟩
    g_t ← x_{i_t} − x_{j_t}
    ξ_t ← (I − u_t u_t^T) g_t
    u_{t+1} ← (u_t − η_t ξ_t) / ‖u_t − η_t ξ_t‖
    if ŵ_k(u_{t+1}) < ŵ, set û ← u_{t+1}
end for
return û
```

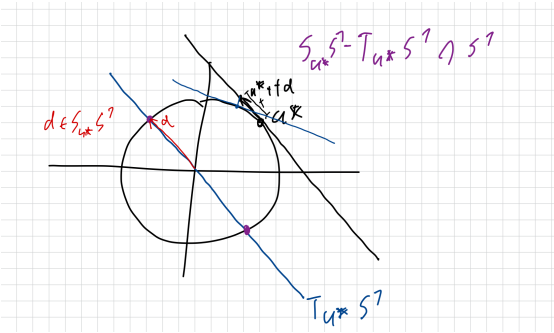


Figure 2.1: One step of the projected sub-gradient descent from Algorithm 1

Define the sequence of step sizes by $\gamma_t = \beta^t \gamma_0$, $t = 0, 1, \dots$, *where* β *and* γ_0 *satisfy certain conditions. If* $u_0 \in \mathbb{S}^{k-1}$ *fulfills* $\operatorname{dist}(u_0, \mathcal{X}) < \min\{\alpha/L, \rho\}$, *then* $\operatorname{dist}(u_t, \mathcal{X}) \leq \beta^t e_0$ *for all* $t \geq 0$.

We already established the convexity and Lipschitz-continuity of $f = \hat{w}_k$ in Proposition 4. It turns out that the weak sharp minima condition from Equation (2.7) is also fulfilled in our setting. We record this fact in the following Lemma.

Lemma 4. *There exist* $\alpha > 0$ *and* $\rho > 0$ *such that* $\hat{w}_k(u) - \hat{m}_k \geq \alpha \operatorname{dist}(u, \mathcal{X}_k)$ *for all* $u \in \mathbb{S}^{k-1}$ *with* $\operatorname{dist}(u, \mathcal{X}_k) < \rho$.

Proof. See Appendix Section 6.1.7. □

In practice, the parameters γ_0 and β need to be tuned as there is no chance of obtaining explicit values for α, ρ in general.

2.5.4 Elbow point selection

rework completely

The last step is to employ an elbow point selector with the property that it would always pick the correct kink in the population case. That is, $\hat{k} = p + 1$ is chosen given that $m_k = m^*$ for all $k \geq p + 1$ and $m_k > m*$ for all $k \leq p$.

fit two affine lines

Define $S_q(a, b) = \sum_{l=1}^{q}(al + b - \hat{m}_l)^2$ and $T_q(c, d) = \sum_{l=q}^{T}(cl + d - \hat{m}_l)^2$ for $q = 2, \ldots, T - 1$.

Two affine lines are fitted through $(1, \hat{m}_1), \ldots, (q, \hat{m}_q)$ and $(q, \hat{m}_q), \ldots, (K, \hat{m}_K)$ for all $q = 2, \ldots, K - 1$ and we select the q with the best fit.

$$S_{AA}(q) = \min_{a,b,c,d} \left(\sum_{l=1}^{q}(al + b - \hat{m}_l)^2 + \sum_{l=q}^{K}(cl + d - \hat{m}_l)^2 \right), \quad \hat{q}_{AA} = \arg \min_{2 \leq q \leq K-1} S_{AA}(q)$$

fit one affine line and one constant line

Since we know that the true minimum width m_k has a breaking point at $k = p + 1$, we can use a modified L-method, that is, fit an affine curve through the points $(1, \hat{m}_1), \ldots, (q, \hat{m}_q)$ and a constant line through $(q, \hat{m}_q), \ldots, (K, \hat{m}_K)$ for all $q = 2, \ldots, K - 1$ and select the q with the best fit.

$$S_{AC}(q) = \min_{a,b,d} \left(\sum_{l=1}^{q}(al + b - \hat{m}_l)^2 + \sum_{l=q}^{K}(d - \hat{m}_l)^2 \right), \quad \hat{q}_{AC} = \arg \min_{2 \leq q \leq K-1} S_{AC}(q)$$

2.6 Numerical Results

We test our newly developed geometric estimators on simulated and real data. Regarding the experiments with simulated data we will investigate different scenarios: first we will simulate AR(p) processes, "forget" what the order p is and try to recover it using our geometric approach coupled with an elbow method. We will compare it against the classical established methods BIC and PACF cutoff, that were presented in Section 2.3. The second scenario that we consider is, that we generate

AR(p) processes and keep the knowledge of the true AR(p) order. We then compare our estimation method (projected sub-gradient descent, see Remark 1) with the classical coefficient estimation methods presented in Section 2.2.

For our experiments, we want to sample the coefficients $\phi_1, \dots, \phi_p$ in a theoretically grounded way. Jones [25] gives one such approach. There, an algorithm is given which samples $\phi = (\phi_1, \dots, \phi_p)$ uniformly from

$$C_p = \{\phi \in \mathbb{R}^p : 1 - \phi_1 z - \cdots - \phi_p z^p \neq 0 \text{ for all } z \in \mathbb{C}, |z| \leq 1\}$$

This allows for fair and principled numerical experiments.

Explain Jones algorithm better

2.6.1 Simulated Data

First exemplary experiment

Figures 2.2, 2.3, and 2.4 illustrate the proposed empirical minimum-width order selection for representative realizations of AR(p) processes with $p \in \{1, 2, 3\}$. The coefficients used are $\phi = 0.6$ for the AR(1) process, $\phi_1 = 0.6, \phi_2 = -0.3$ for the AR(2) process, and $\phi_1 = 0.6, \phi_2 = -0.3, \phi_3 = 0.3$ for the AR(3) process. The observed time series had length $T = 2000$.

The blue lines show the minimum width obtained by the projected sub-gradient descent, that is, $\hat{w}_k(\hat{u}_k)$ with $\hat{u}_k$ being the output of Algorithm 1. For reference, the dashed black line shows the theoretical minimum width m_k, which is equal to $2(1 + \phi_1^2 + \cdots + \phi_p^2)^{-1/2}$ for $k \geq p + 1$.

The Figures also show the population width evaluated at the minimizer $\hat{u}_k$ obtained by Algorithm 1, that is, $w_k(\hat{u}_k)$ as a dashed orange line. The difference $w_k(\hat{u}_k) - \hat{w}_k(\hat{u}_k)$ is always positive, due to the monotonicity of the support function with respect to set inclusion. Since we do not know the true support $\mathcal{Z}^{(k)}$, because we do not know the AR coefficients, we replace it with the convex hull of the sliding-window point cloud $\mathcal{P}_k$. Thus, the difference between the blue and orange lines can be interpreted as a "point filling error", that is, the error we make by replacing $\mathcal{Z}^{(k)}$ with $\mathrm{conv}(\mathcal{P}_k)$.

This error goes to zero for $T \to +\infty$ because

$$0 \leq w_k(\hat{u}_k) - \hat{w}_k(\hat{u}_k) \leq 2d_H(\mathrm{conv}(\mathcal{P}_{k,T}), \mathcal{Z}^{(k)}) \to 0, \quad T \to +\infty$$

as we have seen in the proof of Corollary 1.

The closeness between m_k (black) and $\hat{w}_k(\hat{u}_k)$ (blue) does not indicate that the minimization algorithm worked well. The reason for this can be demonstrated by the following error decomposition:

$$\hat{w}_{k,T}(\hat{u}_{k,T}) - w_k(u_k) = \left(\underbrace{w_k(\hat{u}_{k,T}) - w_k(u_k)}_{\geq 0} \right) - \left(\underbrace{w_k(\hat{u}_{k,T}) - \hat{w}_k(\hat{u}_{k,T})}_{=:F_{k,T}(\hat{u}_{k,T}) \geq 0} \right)$$

The second difference $F_{k,T}(\hat{u}_{k,T})$ is the "point filling error" in direction $\hat{u}_{k,T}$ that goes to zero for $T \to \infty$.

The first difference can be further decomposed:

$$0 \leq w_k(\hat{u}_k) - w_k(u_k) = \underbrace{w_k(\tilde{u}_{k,T}) - w_k(u_k)}_{=:E_{k,T} \geq 0} + \underbrace{w_k(\hat{u}_{k,T}) - w_k(\tilde{u}_{k,T})}_{=:R_{k,T}}.$$

Here $\tilde{u}_k$ is the true minimizer of the empirical width $\hat{w}_k$. By our considerations in Section 2.5.2 any minimizer $\tilde{u}_k$ of the empirical width converges towards the set of minimizing directions of the population width, thus the first difference $E_{k,T}$ goes to zero for $T \to \infty$.

The second difference $R_{k,T}$ is not guaranteed to be nonnegative as well. This term measures the difference in population width between two directions.

It can be written as $R_{k,T} = \epsilon_{k,T} + F_{k,T}(\hat{u}_{k,T}) - F_{k,T}(\tilde{u}_{k,T})$, where $\epsilon_{k,T} := \hat{w}_{k,T}(\hat{u}_{k,T}) - \hat{w}_{k,T}(\tilde{u}_{k,T})$ is the optimization error.

Note that by (the proof of) Corollary 1 we have

$$F_{k,T}(\hat{u}_{k,T}) = |w_k(\hat{u}_{k,T}) - \hat{w}_{k,T}(\hat{u}_{k,T})| \leq \sup_{u \in \mathcal{S}^{k-1}} |w_k(u) - w_{k,T}(u)| =: \delta_{k,T} \to 0, \quad T \to \infty.$$

It follows $|R_{k,T} - \epsilon_{k,T}| \leq \delta_{k,T} \to 0$ for $T \to \infty$. The interpretation is that $R_{k,T}$ differs from the optimization error $\epsilon_{k,T}$ by a quantity that vanishes. The optimization error

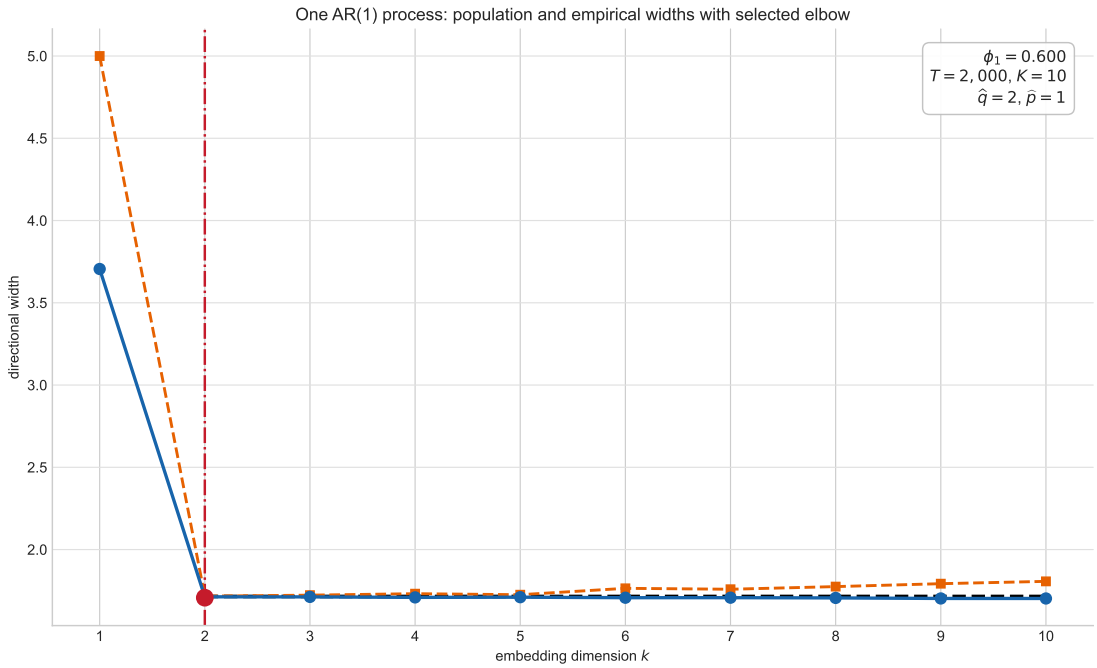


Figure 2.2: Blue: $\hat{w}_k(\hat{u}_k)$, Orange: $w_k(\hat{u}_k)$ (population width w_k evaluated at empirical minimizer $\hat{u}_k$), Black: m_k (only for $k \geq 1$)

$\epsilon_{k,T}$ goes to zero for $T \to \infty$ under certain conditions by Theorem 1.

The vertical red dashed line indicates the selected order using $\hat{p}_{AC} = \hat{q}_{AC} - 1$. For these three exemplary processes, the correct order was selected. Using Remark 1 the following coefficient estimates are obtained from the minimizing directions:

- AR(1) process: $\hat{\phi} = 0.599652$
- AR(2) process: $\hat{\phi}_1 = 0.605058, \hat{\phi}_2 = -0.308672$
- AR(3) process: $\hat{\phi}_1 = 0.602940, \hat{\phi}_2 = -0.302585, \hat{\phi}_3 = 0.302667$

Order selection experiments

Table 2.1 summarizes the results from 50 simulations for each true order $p \in \{1, 2, 3\}$. We compare the classical order selectors BIC and PACF cutoff with our minimum width approach, where we test the different methods of detecting the elbow point from Section 2.5.4.

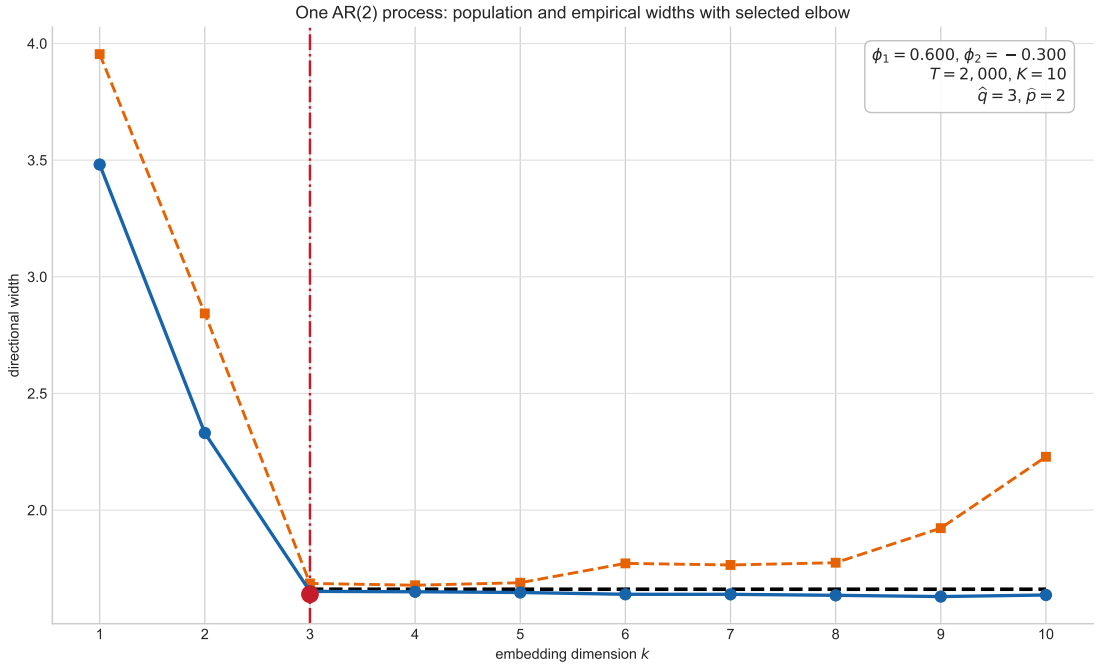


Figure 2.3: Blue: $\hat{w}_k(\hat{u}_k)$, Orange: $w_k(\hat{u}_k)$ (population width w_k evaluated at empirical minimizer $\hat{u}_k$), Black: m_k (only for $k \geq 2$)

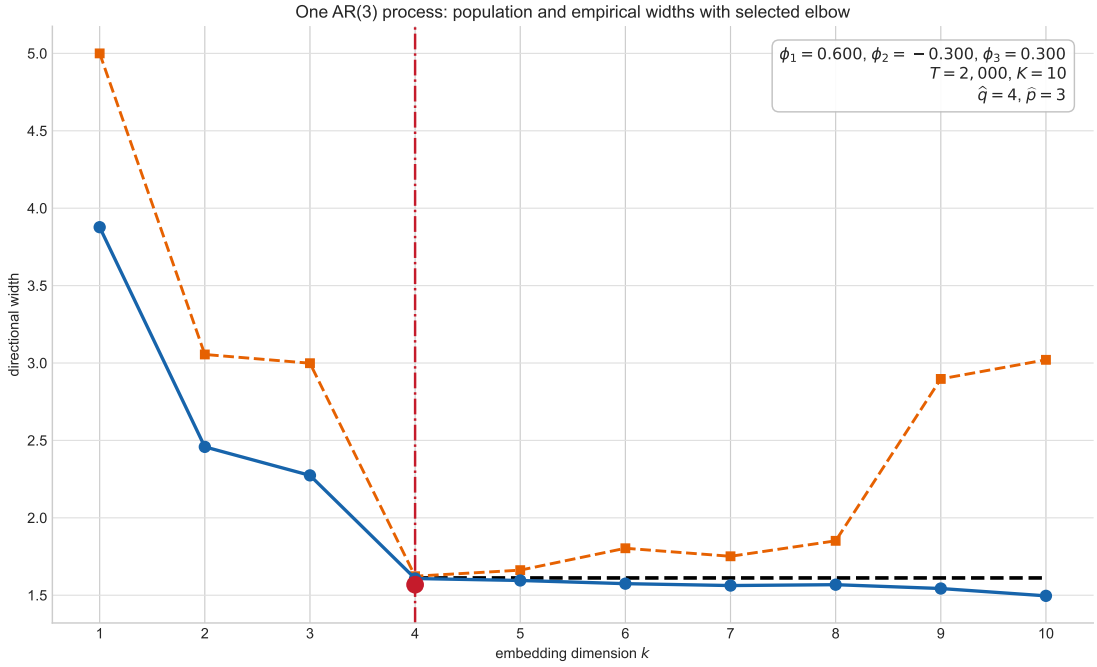


Figure 2.4: Blue: $\hat{w}_k(\hat{u}_k)$, Orange: $w_k(\hat{u}_k)$ (population width w_k evaluated at empirical minimizer $\hat{u}_k$), Black: m_k (only for $k \geq 3$)

Table 2.1: AR-order selections for 50 processes per true order. Each entry reports the number of exact selections, followed by (underselections/overselections).

Selector	AR(1)	AR(2)	AR(3)
Empirical minimum width (L-method)	**49/50** (0/1)	36/50 (14/0)	14/50 (29/7)
BIC	47/50 (3/0)	**46/50** (4/0)	**43/50** (7/0)
PACF cutoff	31/50 (1/18)	34/50 (1/15)	33/50 (3/14)

Coefficient estimation experiments

Here we compare the four presented coefficient estimation methods presented in Section 2.2, namely, OLS (L_2), Yule-Walker, Horváth's L_∞ linear program and our minimum-width approach.

Our experiment looks like this: We sample $L = 100$ causal AR(p)-processes for $p \in \{1, 2, 3, 4, 5\}$. We assume that we know the true order p and compute the estimates $\hat{\phi}_1^{(l)}, \ldots, \hat{\phi}_p^{(l)}$, $l = 1, \ldots, L$. As a measure of error we use $\mathrm{MAE} = L^{-1} \sum_{l=1}^{L} \|\hat{\phi}^{(l)} - \phi^{(l)}\|_1$ (mean absolute error) and median absolute error.

Table 2.2 shows the results. We see that the L^∞-regression approach works best across all orders and that for all methods except the Minimum width approach, the error increases for larger orders p. See Figure 2.5. Strikingly the minimum width approach fails for orders $p \geq 3$. This leads us to an investigation of why this is the case. From the table we also see that the median absolute errors of the Minimum width are much less of a failure. In fact, they beat OLS and Yule-Walker, but are still below L^∞-regression. This indicates that there are some specific runs where the Minimum width approach failed badly. Figure 2.6 shows a distribution of how often each method was best.

2.7 Literature positioning and novelty assessment

15.10: Is there literature of how to determine the minimum width of a point cloud in $\mathbb{R}^k$?

Table 2.2: Coefficient estimation errors for 100 simulated AR(p) processes per true order. Each entry reports the mean ℓ_1 coefficient error, with the median per-process error in parentheses below. The lowest values are shown in bold.

Estimator	AR(1)	AR(2)	AR(3)	AR(4)	AR(5)
OLS (L_2)	0.012681 (0.011231)	0.027507 (0.021934)	0.050458 (0.045155)	0.068460 (0.062511)	0.100692 (0.080998)
Yule–Walker	0.012586 (0.010970)	0.028103 (0.022924)	0.053082 (0.048355)	0.123852 (0.066709)	0.235747 (0.094575)
Horváth (L_∞)	**0.001073** (**0.000493**)	**0.002199** (**0.001458**)	**0.004234** (**0.003029**)	**0.007227** (**0.006078**)	**0.012418** (**0.011283**)
Minimum width	0.001742 (0.001217)	0.005930 (0.003376)	4.384671 (0.008823)	18.644876 (0.017886)	6.546884 (0.044592)

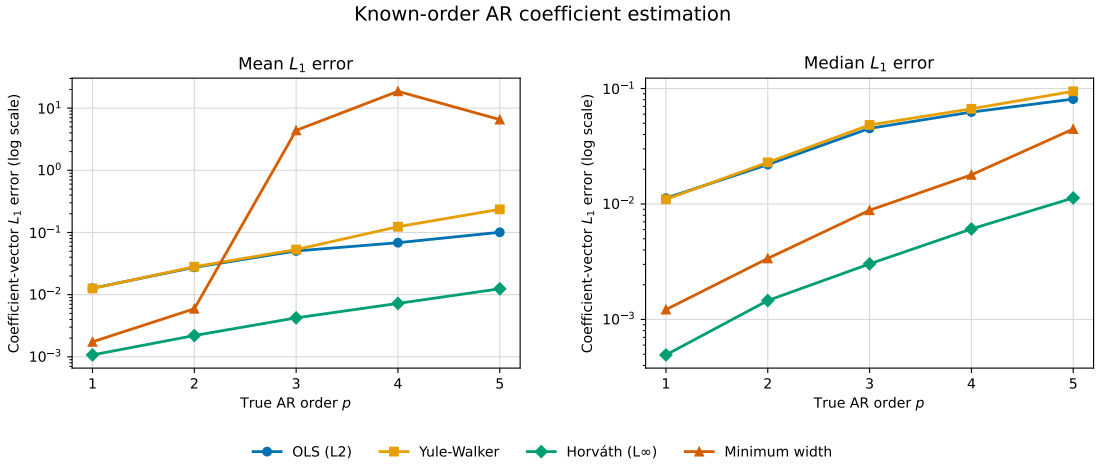


Figure 2.5

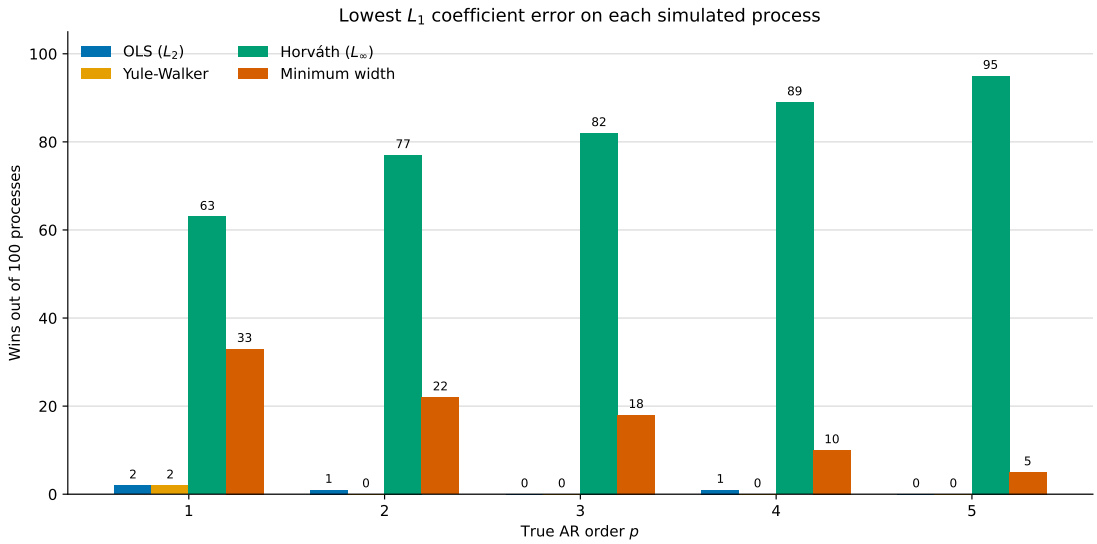


Figure 2.6

The conditional maximum likelihood approach with innovations assumed to be from $U[-h,h]$ leads to L^∞-regression:

$$\min_{\phi\in\mathbb{R}^p}\max_t\left|\underbrace{X_t-\sum_{j=1}^p\phi_jX_{t-j}}_{=\langle Y_t^{(p+1)},q_\phi\rangle}\right|$$

Geometric interpretation of L^∞-regression?

Introducing an intercept term b we get

$$\min_{\phi,b}\max_t\left|\langle Y_t^{(p+1)},q_\phi\rangle-b\right|=\frac{1}{2}\min_\phi\left(\max_t\langle Y_t^{(p+1)},q_\phi\rangle-\min_t\langle Y_t^{p+1},q_\phi\rangle\right)$$

3 Spectral Scale Selection for Kernel Density Estimation

In this chapter, we apply the scale selection methodology introduced in [4] to a sequence of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$.

Goal: Develop spectral bandwidth ranking pipeline, explain why the spectral construction selects what it selects

working hypothesis

Let $h_D^* = \arg\min \mathbb{E}[D(f, \hat{f}_h)]$ and let $S = \exp(-D/\tau)$. Does the peak of the Perron vector of S relate to h_D^*?

Literature positioning: Lindeberg, Goldenshluger (2011), SiZer (?)

energy functional precedence: physics, kde bandwidth

3.1 Preliminaries on Kernel Density Estimation

We have data $X_1, \ldots . X_n \in \mathbb{R}$ sampled from an (unknown) distribution with density f. The kernel density function $\hat{f}_h$ estimates this unknown density. It is defined by

$$\hat{f}_h(x) \coloneqq \frac{1}{nh} \sum_{j=1}^{n} K\left(\frac{x - X_j}{h}\right), \quad x \in \mathbb{R}.$$

Using the shorthand notation $K_h(y) := h^{-1}K(h^{-1}y)$ we can also write $\hat{f}_h(x) = n^{-1}\sum_{j=1}^n K_h(x - X_j)$. The kernel density estimate $\hat{f}_h(x)$ is a Monte Carlo estimate of $(f * K_h)(x) = \mathbb{E}_{X\sim f}[K_h(x - X)]$. Writing $P_n = n^{-1}\sum_{j=1}^n \delta_{X_j}$ for the empirical

measure, we see that $(P_n * K_h)(x) = \int P_n(y)K_h(x-y)\,dy = n^{-1}\sum_{j=1}^n K_h(x - X_j) = \hat{f}_h(x)$.

Two common choices for the kernel K are:

- the Epanechnikov kernel $K(u) = \mathbf{1}\{-1 \le u \le 1\}3(1-u^2)/4$, $u \in \mathbb{R}$
- the Gaussian kernel $K(u) = (2\pi)^{-1/2}\exp(-u^2/2)$, $u \in \mathbb{R}$

From now on, we always work with the Gaussian kernel, unless indicated otherwise.

The question of how to choose the bandwidth h is widely known in the literature. See reviews []. We present some of the most popular methods below, which we also use to compare with our newly developed spectral selector. Following the review [20], bandwidth selectors can be divided into several classes, including plug-in and cross-validation methods. The methods of both classes minimize some approximation of the (mean) integrated squared error.

Definition 7. *Let f be a density and $\hat{f}_h$ its kernel density estimate. The integrated squared error (ISE) between f and $\hat{f}_h$ is given by*

$$\mathrm{ISE}(h) \coloneqq \int \left(f(x) - \hat{f}_h(x)\right)^2 dx.$$

The mean integrated squared error (MISE) is $\mathrm{MISE}(h) \coloneqq \mathbb{E}[\mathrm{ISE}(h)]$.

Cross-validation methods

The ISE can be written as

$$\mathrm{ISE}(h) = \int \hat{f}_h(x)^2\,dx - 2\mathbb{E}_{X\sim f}\left[\hat{f}_h(X)\right] + \int f(x)^2\,dx.$$

The first term $\int \hat{f}_h^2$ can be computed from the data. The third term $\int f^2$ does not depend on the bandwidth and can be ignored when minimizing. We are left to find an (unbiased) estimator for $\mathbb{E}_{X\sim f}[\hat{f}_h(X)]$. Such an estimator is given by $n^{-1}\sum_{i=1}^n \hat{f}_{h,-i}(X_i)$, where $\hat{f}_{h,-i}$ is the leave-one-out kernel density estimate given

by

$$\hat{f}_{h,-i}(x) = \frac{1}{(n-1)h} \sum_{j \neq i} K\left(\frac{x - X_j}{h}\right), \quad x \in \mathbb{R}.$$

The Ordinary Least Squares Cross-Validation bandwidth is

$$\hat{h}_{\mathrm{CV}} = \arg\min_{h>0} \mathrm{CV}(h), \quad \mathrm{CV}(h) = \int \hat{f}_h(x)^2 \, dx - \frac{2}{n} \sum_{i=1}^{n} \hat{f}_{h,-i}(X_i).$$

It is known that (add source!!) $\mathrm{ISE}(\hat{h}_{\mathrm{CV}}) - \mathrm{ISE}(\hat{h}_0) = \mathbf{O}(n^{-1})$, that is, the integrated squared error evaluated at the cross-validation selected bandwidth converges to the integrated squared error evaluated at its true minimizer $\hat{h}_0$ to zero at rate n^{-1}.

Plug-in methods

The bias-variance decomposition of the MISE is

$$\mathrm{MISE}(h) = \int \mathrm{Var}[\hat{f}_h(x)] \, dx + \int \mathrm{Bias}(\hat{f}_h(x))^2 \, dx$$

with

$$\mathrm{IV}(h) := \int \mathrm{Var}[\hat{f}_h(x)] \, dx = \frac{\|K\|_2^2}{nh} - \frac{\|K_h * f\|_2^2}{n}, \quad \text{and}$$
$$\mathrm{ISB}(h) := \int \mathrm{Bias}(\hat{f}_h(x))^2 \, dx = \|K_h * f - f\|_2^2.$$

This decomposition gives conditions on $h = h_n$ so that the kernel density estimate $\hat{f}_h$ is consistent. In order for the bias to vanish, it must hold $h_n \to 0$, because then $K_h * f \to f$. For the variance to vanish, we must require $nh_n \to \infty$. An example of such a sequence is $h_n = cn^{-\alpha}$ for some $c > 0$ and $\alpha \in (0, 1)$.

It is important to be convinced that MISE actually has a global minimum. To this end, we refer to Chacón et al. [12, Theorem 1] which tells us that a global minimum always exists if K is a nonnegative kernel that peaks at 0.

Asymptotic expansion of the integrated variance

Note that $\|K_h\|_1 = \int K_h(x)\,dx = \int K(x)\,dx = 1$ for every $h > 0$. By Young's convolution inequality $\|K_{h_n} * f\| \leq \|K_{h_n}\|_1 \|f\|_2 = \|f\|_2$. Therefore,

$$\frac{\frac{\|K_{h_n} * f\|_2^2}{n}}{\frac{\|K\|_2^2}{nh_n}} = \frac{h_n \|K_{h_n} * f\|_2^2}{\|K\|_2^2} \leq h_n \frac{\|f\|_2^2}{\|K\|_2^2} \to 0$$

and thus $-n^{-1}\|K_{h_n} * f\|_2^2 = \mathbf{o}((nh_n)^{-1}\|K\|_2^2)$. We therefore may omit the term $-n^{-1}\|K_{h_n} * f\|_2^2$ as it is asymptotically negligible.

Asymptotic expansion of the integrated squared bias

For the $\mathrm{ISB}(h)$ term we get

$$\mathrm{ISB}(h) = \frac{h_n^4}{4}\mu_2(K)^2 R(f'') + \mathbf{o}(h_n^4)$$

Asymptotic expansion of $\mathrm{MISE}(h)$

The first non-vanishing asymptotic approximation of the MISE is

$$\mathrm{MISE}(h_n) = \underbrace{\frac{\|K\|_2^2}{nh_n} + \frac{h_n^4}{4}\mu_2(K)^2\|f''\|_2^2}_{=\mathrm{AMISE}(h_n)} + o\left(\frac{1}{nh_n} + h_n^4\right).$$

We have found scaling sequences a_n and b_n such that $a_n \,\mathrm{MISE}(b_n t) \to G(t) := at^{-1} + bt^4$ with constants $a = \|K\|_2^2$ and $b = \mu_2(K)^2\|f''\|_2^2/4$, namely

$$n^{4/5}\,\mathrm{MISE}(n^{-1/5}t) \to \frac{\|K\|_2^2}{t} + \frac{\mu_2(K)^2\|f''\|_2^2}{4}t^4, \quad n \to \infty$$

Minimizing the AMISE leads to

$$h_{\mathrm{AMISE}}(n) = \left(\frac{\|K\|_2^2}{n\mu_2(K)^2\|f''\|_2^2}\right)^{1/5} \tag{3.1}$$

We have $h_{\text{oracle}} = h_{\mathrm{AMISE}} + \mathbf{o}(n^{-1/5})$ (see again [12]), which implies that h_{oracle} and h_{AMISE} are asymptotically equivalent.The optimal bandwidth thus behaves like $n^{-1/5}$.

Plug-in methods use Equation (3.1) as a starting point and try to find estimators

$\hat{R}(f'')$ for $R(f'') = \|f''\|_2^2$. Silverman's rule of thumb assumes that f is the density of a normal distribution $N(0, \sigma^2)$. In this case, $\|f^{''}\|_2^2$ can simply be computed and plugged into Equation (3.1):

$$\|f^{''}\|_2^2 = \frac{3}{8\sqrt{\pi}\sigma^5}, \quad h_{\text{AMISE}} = \left(\frac{4}{3}\right)^{1/5} \sigma n^{-1/5} \approx 1.06\sigma n^{-1/5}$$

Among the many other more sophisticated plug in methods, we present the (improved) Sheather and Jones' plug in selector, since it is used fairly commonly in practice; see for example the documentation of the kde packages of the statistics sowftware R and SAS.

They approximate $R(f'') = \|f''\|_2^2$ by $R(\hat{f}_g'') = \|\hat{f}_g''\|_2^2$, where $\hat{f}_g''$ is the second derivative of the kernel density estimate with pilot bandwidth g, whose choice is a major problem. They choose it so that the leading bias term of the mean squared error $\mathbb{E}[(R(f'')-R(\hat{f}_g''))^2]$ cancels. This choice leads to $g^7 = R(K'')\mu_2(K)^{-1}R(f''')^{-1}n^{-1}$. They then make this depend on h_{AMISE} by solving Equation (3.1) for n and plugging it into the formula for g^7, which results in $g(h) = C_K(R(f'')/R(f'''))^{1/7}h^{5/7}$.

Of course, the ratio $R(f'')/R(f''')$ has to be estimated separately, which is done by $R(\hat{f}_a'')/R(\hat{f}_b''')$ with a and b chosen following Silverman's rule of thumb. The resulting equation $\hat{g}(h) = C_K(R(\hat{f}_a'')/R(\hat{f}_b'''))^{1/7}h^{5/7}$ is plugged into $R(f_g'')$, which in turn is plugged into Equation (3.1). The Equation

$$h = \left(\frac{\|K\|_2^2}{n\mu_2(K)^2\|\hat{f}_{\hat{g}(h)}''\|_2^2}\right)^{1/5}$$

is solved numerically for h. There exist consistency results for this plug in selector. It converges to the oracle bandwidth h_{oracle} at a rate $n^{-5/14}$.

For illustration, the next Figure [] shows 50 points sampled from $N(0,1)$ together with the true density f and the kernel density estimates $\hat{f}_h$ with $h \in \{\hat{h}_{CV}, h_{\text{oracle}}, h_{\text{AMISE}}, h_{\text{Sil}}, h_{\text{SJ}}\}$.

Exact MISE representation for Gaussian mixture model

Exact computations are generally not possible since the true density f of the sampled data $X_1, \ldots, X_n$ is not known. In simulation experiments, of course, we simulate the data ourselves and thus know the density f. For certain choices of the true density f, explicit formulas for the MISE exist.

A particularly broad and well studied class is that of Gaussian mixture models. They are characterized by densities f of the form $f = \sum_{l=1}^{L} w_l \phi_{m_l,\sigma_l^2}$, where $w_l > 0$, $\sum_l w_l = 1$, $m_l \in \mathbb{R}$, and $\sigma_l^2 > 0$.

Marron and Wand [28] derive an explicit formula for the MISE between a density f of the Gaussian mixture model and its kernel density estimate with Gaussian kernel of order $r \in \mathbb{N}$. Here we state the formula for the special case $r = 1$, that is, for the standard Gaussian kernel.

Proposition 5. *The mean integrated squared error between a kernel density estimate $\hat{f}_h$ with Gaussian kernel and the density function f of a Gaussian mixture model with weights w_l, variances s_l^2 and means m_l is given by*

$$\mathrm{MISE}(h) = \frac{1}{2\sqrt{\pi}}\frac{1}{nh} + \left(1 - \frac{1}{n}\right)\mathcal{K}(h,2) - 2\mathcal{K}(h,1) + \mathcal{K}(h,0),$$
$$\mathcal{K}(h,q) = \sum_{l,l'=1}^{L} \frac{w_l w_{l'}}{\sqrt{s_l^2 + s_{l'}^2 + qh^2}} \phi\left(\frac{m_l - m_{l'}}{\sqrt{s_l^2 + s_{l'}^2 + qh^2}}\right), \quad q \in \{0,1,2\}.$$

Minimizing $\mathrm{MISE}(h)$ yields the "oracle" bandwidth h_{oracle}. We will use the oracle bandwidth as an ultimate benchmark in all settings where MISE has an explicit representation and thus can be minimized using standard methods.

We already established above that a minimizer of MISE always exists. The question of uniqueness is a more complicated problem. For a general Gaussian mixture model we cannot assume the uniqueness of the minimizer, but we are able to do so in the simple case where $f \sim N(0, \sigma^2)$.

First, notice that $h \mapsto M_n(h) = (2\sqrt{\pi})^{-1}((nh)^{-1} + (1-1/n)/(\sqrt{\sigma^2 + h^2}) + \sigma^{-1}) - 2(\sqrt{2\pi}\sqrt{2\sigma^2 + h^2})^{-1}$ is continuous on $(0, \infty)$. We set $Q_n(h) = 2\sqrt{\pi}h^2 M_n'(h)$ and see that $Q_n(h) \to 2\sqrt{2} - 1 > 0$ for $h \to \infty$ and $Q_n(h) \to -1/n < 0$ for $h \to 0$. Also, we have $Q_n'(h) > 0$ for all $h > 0$, which implies that Q_n only has one zero, and, in turn, the minimizer of M_n is unique.

As a function of σ, the second derivative of M_n at the unique minimizer behaves like σ^{-3}, which means that, for increasing variance, the valley in which the unique minimizer lies becomes flatter.

The integrated squared error between $\hat{f}_h$ and $f \sim N(0, \sigma^2)$ is given by

$$\mathrm{ISE}(h) = \frac{1}{n^2} \sum_{i,j=1}^{n} \phi_{\sqrt{2}h}(X_i - X_j) - \frac{2}{n} \sum_{i=1}^{n} \phi_{\sqrt{h^2+\sigma^2}}(X_i) + \phi_{\sqrt{2}\sigma}(0)$$

The connection between the minimum of ISE and the minimum of MISE is as follows. First, $\min_h \mathrm{ISE}(h) \leq \mathrm{ISE}(h_0)$ for all $h_0 > 0$. Therefore $\mathbb{E}[\min_h \mathrm{ISE}(h)] \leq \mathbb{E}[\mathrm{ISE}(h_0)]$ and since this is true for all $h_0 > 0$, it follows $\mathbb{E}[\min_h \mathrm{ISE}(h)] \leq \min_h[\mathrm{MISE}(h)]$.

3.2 Divergences between distributions

For information on information theory and divergences, see the book [32] or the survey [29].

Let $L^1(\mathbb{R})$ be the set of all absolutely integrable functions $f : \mathbb{R} \to \mathbb{R}$ and $\mathcal{P} \subseteq L^1$ be the set of all absolutely continuous probability distributions. That is,

$$\mathcal{P} = \left\{ f \in L^1 : f \geq 0, \int f(x)\, dx = 1 \right\}$$

Note that $\mathcal{P} \subseteq L^1$ is convex. A divergence is a map $D : \mathcal{P} \times \mathcal{P} \to \mathbb{R}_{\geq 0}$ such that $D(p, q) = 0$ if and only if $p = q$.

The two most important classes of divergences are f-divergences and Bregman divergences.

Computing divergences between densities amounts to evaluating integrals over $\mathbb{R}$. In general, these integrals cannot be computed explicitly, and numerical procedures have to be applied. For a practical guide on numerical integration, see [14, Chapter 9].

3.2.1 f-divergences

We take material on f-divergences mainly from [32, Chapter 7]. There, divergences are defined between two probability distributions P and Q. For our purposes, it

is enough to define them directly between probability density functions (Lebesgue integral).

Definition 8. *Let $f : \mathbb{R}_{>0} \to \mathbb{R}$ be a convex function with $f(1) = 0$. The f-divergence is defined as*

$$D_f(p\|q) \coloneqq \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx.$$

The f-divergence has the following invariance property: it holds $D_f(p\|q) = D_g(p\|q)$ whenever $f(x) - g(x) = c(x-1)$ for some constant $c \in \mathbb{R}$. We will write $f \sim g$ in short.

Since we will use divergences to construct scale similarity matrices, it is important to know under which conditions on f the f-divergence and thus the scale similarity matrix are symmetric. Proposition 6 yields the answer.

Proposition 6. *The f-divergence is symmetric iff $f \sim \tilde{f}$, where $\tilde{f}$ is defined by $\tilde{f}(x) = xf(1/x)$.*

Proof. We make use of [32, Proposition 7.2]. First, assume that $f(x) = xf(x^{-1}) + c(x-1)$. Then $D_f(p\|q) = \int qf(p/q) = \int q((p/q)f(q/p)+c((p/q)-1)) = \int pf(q/p)+c\int(p-q) = D_f(q\|p)$. Now assume that the f-divergence is symmetric on $\mathcal{P}$. We have $D_f(p\|q) = D_f(q\|p) = \int pf(q/p) = \int q(p/q)f(1/(p/q)) = \int q\tilde{f}(p/q) = D_{\tilde{f}}(p\|q)$. Now, let $x, y \in \mathbb{R}$ with $x < 1 < y$. Define $q_1 = (y-1)/(y-x)$ and $q_2 = (1-x)/(y-x)$ and let $g(z) = \delta_{q_1}(z) + \delta_{q_2}(z)$ and $h(z) = \delta_{xq_1}(z) + \delta_{yq_2}$. We get

$$D_f(h\|g) = \int gf(h/g) = q_1 f(x) + q_2 f(y)$$
$$D_{\tilde{f}}(h\|g) = \int g\tilde{f}(h/g) = q_1 \tilde{f}(x) + q_2 \tilde{f}(y)$$

and therefore $(y-1)(f(x) - \tilde{f}(x)) = (1-x)(\tilde{f}(y) - f(y))$ which is equivalent to $(f(x) - \tilde{f}(x))/(x-1) = (f(y) - \tilde{f}(y))/(y-1)$. It follows that $(f(x) - \tilde{f}(x))/(x-1)$ is constant on $\mathbb{R} \setminus \{1\}$ and thus $f(x) - \tilde{f}(x) = c(x-1)$ for all $x \in \mathbb{R}$ (it trivially holds for $x = 1$ too). □

In the following we introduce a few classes of parametrized f-divergences. Each of those parametrized classes has exactly one symmetric divergence among them. The numerical experiments below will make use of just the symmetric divergences since such divergences lead to symmetric scale similarity matrices of which we analyze the Perron vector. A possible extension of our scale selection principle to asymmetric scale similarity matrices is discussed in the Outlook 5. The reason we still introduce the divergences in such generality is to that the ideas in the Outlook can be better appreciated.

Amari alpha divergence

A prominent parametrized family of f-divergences is the so called α-divergence (see [13]) where the function $f = f_\alpha$ is given by

$$f_\alpha(t) = \frac{1}{\alpha(\alpha-1)}\left(t^\alpha - \alpha t - (1-\alpha)\right)$$

The actual divergence is then given by

$$D_\alpha(p\|q) = \frac{1}{\alpha(\alpha-1)}\left(\int p(x)^\alpha q(x)^{1-\alpha}\,dx - 1\right) \tag{3.2}$$

Taking the limit $\alpha \to 1$ we recover the Kullback-Leibler divergence, whereas taking the limit $\alpha \to 0$ recovers the reverse Kullback-Leibler divergence. At $\alpha = 1/2$ we obtain $D_{1/2}(p\|q) = 4D_H(p,q)$, where $D_H(p,q) = (1/2)\int(\sqrt{f}-\sqrt{g})^2 = 1 - \int\sqrt{pq}$ is the squared Hellinger distance.

By Hölders inequality $0 \le C_\alpha(p,q) \le 1$, where $C_\alpha(p,q) = \int p^\alpha q^{1-\alpha}$ and thus $0 \le D_\alpha(p\|q) \le \alpha^{-1}(1-\alpha)^{-1}$ for all $\alpha \in (0,1)$.

In particular, by Proposition 6, the α-divergence is symmetric iff $\alpha = 1/2$ (Hellinger distance).

For completeness, we mention that in the literature there exists a second parameterization, where $\alpha = (1-\alpha_A)/2$ is plugged into the formula in Equation (3.2) to obtain

$$D_{\alpha_A}(p\|q) = \frac{4}{1-\alpha_A^2}\left(1 - \int p(x)^{\frac{1-\alpha_A}{2}} q(x)^{\frac{1+\alpha_A}{2}}\right).$$

χ^α-divergences (total variation)

Another class of divergences are the χ^α-divergences (see []). They are determined by $f_\alpha(t) = (1/2)|t-1|^\alpha$, where $\alpha \geq 1$. Setting $\alpha = 1$ recovers the well known total variation distance: $d_{\mathsf{TV}}(p,q) = (1/2)\int |p-q|$.

The only symmetric χ^α-divergence is the total variation distance.

Total variation and squared Hellinger are related by the following chain of inequalities:

$$\begin{aligned} D_H(p,q) &= \frac{1}{2}\int (\sqrt{p}-\sqrt{q})^2 \leq \frac{1}{2}\int |p-q| = d_{\mathsf{TV}}(p,q) \\ &= \frac{1}{2}\int |p-q| = \frac{1}{2}\int (\sqrt{p}+\sqrt{q})|\sqrt{p}-\sqrt{q}| \\ &\leq \sqrt{\int (\sqrt{p}+\sqrt{q})^2}\sqrt{\int (\sqrt{p}-\sqrt{q})^2} \\ &= \sqrt{\left(2+2\int\sqrt{pq}\right)\left(2-2\int\sqrt{pq}\right)} \\ &= \sqrt{4-4\left(\int\sqrt{pq}\right)} = 2\sqrt{1-\int\sqrt{pq}} = 2\sqrt{D_H(p,q)} \end{aligned}$$

Numerical Approximation

We decide to use the classical quadrature formula

$$\int_a^b f(x)\,dx \approx Q_M(f) = (b-a)\sum_{m=0}^{M} w_m f(x_m),$$

where w_m are the quadrature weights and x_m the (equidistant) quadrature nodes. As quadrature weights we choose $w_m = 1/(2M)$ for $m = 0, M$ and $w_m = 1/M$ for $m = 1, \ldots, M-1$. This choice corresponds to the composite trapezoidal rule.

Let $A = [a,b]$ and $A^c = \mathbb{R} \setminus A$. We first aim to bound the truncation error $E_{\mathsf{trunc}} = |\int_{A^c} f(p(x),q(x))\,dx|$, where f is any of the functions that appear in a divergence,

and $p = \hat{f}_h$ and $q = \hat{f}_g$ are kernel density estimates with a Gaussian kernel. It holds

$$\tau_h := \int_{A^c} \hat{f}_h = \frac{1}{n} \sum_{j=1}^{n} \underbrace{\int_{\mathbb{R}\setminus(a,b)} \frac{1}{h} K\left(\frac{x - X_j}{h}\right) dx}_{=\Phi((a-X_j)/h)+\Phi((X_j-b)/h)} .$$

Setting $a = X_{\min} - Lh_{\max}$ and $b = X_{\max} + Lh_{\max}$, where $X_{\min} = \min X_j$, $X_{\max} = \max X_j$ and $h_{\max} = \max h_i$, we obtain

$$\Phi((a - X_j)/h) + \Phi((X_j - b)/h) \leq 2\Phi(-L) =: \delta_L$$

and thus $\tau_h \leq 2\Phi(-L)$ for any $h \leq h_{\max}$. Using the upper bound for the above truncation error, we choose L such that $\delta_L \leq \epsilon$. This can be achieved by setting $L \geq \Phi^{-1}(1 - \epsilon/2)$.

Replacing $\int p^\alpha q^{1-\alpha}$ in Equation (3.2) with the truncated integral $\int_a^b p^\alpha q^{1-\alpha}$ leads to the truncation error

$$E_{\text{trunc}} = \frac{1}{\alpha(1-\alpha)} \int_{A^c} p^\alpha q^{1-\alpha} \overset{\text{Hölder}}{\leq} \frac{1}{\alpha(1-\alpha)} \left(\int_{A^c} p\right)^\alpha \left(\int_{A^c} q\right)^{1-\alpha} \leq \frac{\delta_L}{\alpha(1-\alpha)}$$

for the α-divergence and truncating the total variation yields

$$\frac{1}{2} \int_{A^c} |p - q| \leq \frac{1}{2} \int_{A^c} p + \frac{1}{2} \int_{A^c} q \leq \delta_L.$$

As a consequence, we see that approximating α-divergence and total variation using the composite trapezoidal rule allows clean error control.

JS divergence

Another important divergence that belongs to the class of f-divergences is the Jensen Shannon divergence. The function f is set to be

$$f(t) = \frac{1}{2}\left(t \log t - (t+1) \log \frac{t+1}{2}\right)$$

The divergence is then given by

$$\mathrm{JS}(f,g) = H\left(\frac{f+g}{2}\right) - \frac{1}{2}H(f) - \frac{1}{2}H(g),$$

where $H(f) = -\int f \log f$ is the Shannon entropy.

The JS-divergence is also symmetric. In fact, the JS-divergence is a symmetrization of the Kullback-Leibler divergence, as it can be written like

$$\mathrm{JS}(p,q) = \frac{1}{2}\left(\mathrm{KL}(p\|\frac{p+q}{2}) + \mathrm{KL}(q\|\frac{p+q}{2})\right), \tag{3.3}$$

that is, the JS-divergence is the arithmetic mean of the KL-divergences of p and q to their equally weighted mixture.

Numerical Approximation

Gauss-Hermite seems slow, just use composite trapezoidal again but derive truncation error bound

For KL and JS divergence it is natural to use the Gauss-Hermite quadrature, which is a quadrature rule that approximates integrals of the form $\int_{-\infty}^{\infty} e^{-x^2} f(x)\,dx$ by $\sum_{m=1}^{M} w_m f(x_m)$ with the nodes x_m being the M-th Hermite Polynomial H_M and weights $w_m = (2^{M-1}M!\sqrt{\pi})/(M^2 H_{M-1}^2(x_m))$. For reference on Gauss-Hermite quadrature, see [1, Section 25.4.46]. We can write (and consequently approximate) the KL-divergence between two kernel density estimates $\hat{f}_h$ and $\hat{f}_g$ as

$$\begin{aligned} D_{\mathrm{KL}}(\hat{f}_h\|\hat{f}_g) &= \frac{1}{\sqrt{\pi}}\frac{1}{n}\sum_{i=1}^{n}\int_{-\infty}^{+\infty} e^{-x^2}\log\left(\hat{f}_h(X_j + h\sqrt{2}x) - \hat{f}_g(X_j + h\sqrt{2}x)\right)dx \\ &\approx \frac{1}{\sqrt{\pi}}\frac{1}{n}\sum_{i=1}^{n}\sum_{m=1}^{M} w_m \log\left(\hat{f}_h(X_j + h\sqrt{2}x_m) - \hat{f}_g(X_j + h\sqrt{2}x_m)\right). \end{aligned}$$

Using the formula in Equation (3.3) we can also approximate the JS-divergence between $\hat{f}_h$ and $\hat{f}_g$.

One advantage of the Gauss-Hermite quadrature is that we do not need to care about the truncation error because there is none.

3.2.2 Bregman divergences

The second major class are Bregman divergences. We give a general definition below to show that this concept of distance applies not just to probability densities. The Bregman divergence is a generalization of the L^2 distance and is applied in a variety of instances where the L^2-distance is also used, for example, clustering [3].

Definition 9. *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be a Hilbert space and $C \subseteq \mathcal{H}$ nonempty, open and convex and $F : C \to \mathbb{R}$ strictly convex, and Fréchet differentiable on C. The Bregman divergence between $x \in C$ and $y \in C$ is defined by*

$$D_F(x\|y) := F(x) - F(y) - \langle x - y, \nabla F(y)\rangle_{\mathcal{H}}. \tag{3.4}$$

The first order Taylor approximation of F around y evaluated at x is $F(y)+DF(y)[x-y]$, where $DF(x) : \mathcal{H} \to \mathbb{R}$ is a bounded linear operator. By the Riesz representation theorem, $DF(y)[h] = \langle \nabla F(y), h\rangle$ for all $h \in \mathcal{H}$ and in particular $DF(y)[x-y] = \langle x - y, \nabla F(y)\rangle_{\mathcal{H}}$. The Bregman divergence $D_F(x\|y)$ as defined in Equation (3.4) is therefore the difference between F evaluated at x and the first order Taylor approximation of F around y evaluated at x.

Similarly to the f-divergence, we have invariance $D_F = D_G$ if and only if $F - G$ is an affine functional, that is, $F(x) - G(x) = \langle a, x\rangle_{\mathcal{H}} + b$ for some $a \in \mathcal{H}$ and $b \in \mathbb{R}$ and all $x \in C$.

If F is twice continuously Fréchet differentiable, there is a symmetry characterization for Bregman divergences similar to the one for f-divergences stated in Proposition 6. Theorem 12 in [35] states that in this case D_F is symmetric if and only if

$$F(p) = \frac{1}{2}\langle p, Tp\rangle_{\mathcal{H}} + \langle b, p\rangle_{\mathcal{H}} + c, \quad p \in \operatorname{int}(\operatorname{dom}(F)), \tag{3.5}$$

where T is some bounded, self-adjoint and strictly positive operator, $b \in \mathcal{H}$ and $c \in \mathbb{R}$. That is, the Hessian $\nabla^2 F(g)$ of F at $g \in \mathcal{H}$ is T and, in particular, independent of g.

Subsequently, we will consider the inner product space $(L^2, \langle\cdot,\cdot\rangle_2)$ with functions F of the form $F_\phi(p) = \int \phi(p(x))\,dx$, where $\phi : \mathbb{R} \to \mathbb{R}$ is continuously differentiable,

strictly convex, ϕ' globally Lipschitz-continuous and $\phi(0) = \phi'(0) = 0$. In this setting, since $\nabla F_\phi(q) = \phi' \circ q$, the Bregman divergence becomes

$$D_\phi(p\|q) := D_{F_\phi}(p\|q) = \int (\phi(p(x)) - \phi(q(x)) - (p(x) - q(x))\phi'(q(x))) \, dx.$$

We have $D_\phi = D_{\tilde{\phi}}$ if and only if $\phi - \tilde{\phi}$ is an affine function. To see the one direction, we know that this is equivalent to $F_\phi(p) - F_{\tilde{\phi}}(p) = \langle a, p\rangle_2 + b$ for all $p \in L^2$. In particular for $p_t(x) = t\mathbf{1}\{x \in [0,1]\}$. Plugging this particular choice of p_t into the equation yields $\phi(x) - \tilde{\phi}(x) = t\int_0^1 a(x)\,dx + b$, that is, $\phi - \tilde{\phi}$ is affine. The other direction follows from a simple calculation.

We also have that D_ϕ is symmetric if and only if ϕ is a quadratic function given that we additionally assume $\phi \in C^2(\mathbb{R})$. If D_ϕ is symmetric, then $F_\phi(p)$ has the representation in Equation (3.5) for all $p \in L^2$. Plugging in the particular choice $p(x) = x\mathbf{1}\{x \in [0,1]\}$ into Equation (3.5) and recalling the condition $\phi(0) = \phi'(0) = 0$, we arrive at ϕ having the form $\phi(x) = (\kappa/2)x^2$ with $\kappa = \langle \mathbf{1}\{\cdot \in [0,1]\}T\mathbf{1}\{\cdot \in [0,1]\}\rangle$. The converse direction is again a simple calculation.

An example of such a function ϕ is the class of β-divergences (see [13, Equation (46)]) determined by

$$\phi_\beta(t) = \frac{t^{1+\beta}}{\beta(1+\beta)}, \quad \beta > 0.$$

The β-divergence $D_\beta := D_{\phi_\beta}$ is thus given by

$$D_\beta(p\|q) = \frac{1}{\beta(1+\beta)}\left(\int \left(p(x)^{1+\beta} - (1+\beta)p(x)q(x)^\beta + \beta q(x)^{1+\beta}\right) dx\right).$$

By the above symmetry characterization, the β-divergence is symmetric only for $\beta = 1$. In this case, we get

$$D_1(p\|q) = \frac{1}{2}\int (p(x) - q(x))^2\,dx = \frac{1}{2}\|p - q\|_2^2 = \frac{1}{2}\langle p, p\rangle_{L^2} + \frac{1}{2}\langle q, q\rangle_{L^2} - \langle p, q\rangle_{L^2}.$$

If $p(x) = \hat{f}_h(x) = n^{-1}\sum_j K_h(x - X_j)$ and $q(x) = \hat{f}_s(x) = n^{-1}\sum_j K_s(x - X_j)$ are kernel density estimates with a Gaussian kernel K, we can obtain an explicit

formula for $D_1(\hat{f}_h, \hat{f}_s)$. We have

$$D_1(\hat{f}_h, \hat{f}_s) = \frac{1}{2}G_{hh} + \frac{1}{2}G_{ss} - G_{hs}, \tag{3.6}$$

where

$$G_{hs} = \langle \hat{f}_h, \hat{f}_s \rangle_{L^2} = \frac{1}{n^2} \sum_{i,j=1}^{n} \phi_{\sqrt{h^2+s^2}} (X_i - X_j) \tag{3.7}$$

The derivation of the formula uses mainly the Gaussian convolution identity, that is, the fact that for Gaussian densities $f_1 = \phi_{\mu_1,\sigma_1^2}$ and $f_2 = \phi_{\mu_2,\sigma_2^2}$ their convolution is $f_1 * f_2 = \phi_{\mu_1+\mu_2,\sigma_1^2+\sigma_2^2}$.

Numerical Approximation

The L^2 distance between two Gaussian kernel density estimates can be computed extremely fast using the Fourier transform. Although an exact formula is given in Equations (3.6) and (3.7) the computational efficiency is of the order $\mathcal{O}(N^2 n^2)$, where N is the size of the bandwidth grid and n the sample size.

In the following, we will briefly describe the approach following the book [31]. For any function $f \in L^1(\mathbb{R})$ the Fourier transform is defined as $\mathcal{F} : f \mapsto \mathcal{F}[f]$ where $\mathcal{F}[f](\omega) = \int f(x) \exp(-ix\omega)\, dx$. The Fourier transform of the Gaussian density ϕ_σ is famously given by $\mathcal{F}[\phi_\sigma](\omega) = \exp(-\sigma^2\omega^2/2)$. Using the linearity and translation property of the Fourier transform, we obtain the Fourier transform of a kernel density function with a Gaussian kernel:

$$\mathcal{F}[\hat{f}_h](\omega) = \underbrace{\left(\frac{1}{n} \sum_{j=1}^{n} \exp(-iX_j\omega) \right)}_{=P_n(\omega)} \exp(-h^2\omega^2)/2.$$

Next, we use Parseval's identity to get

$$\begin{aligned} G_{r,s} = \int \hat{f}_{h_r}(x)\hat{f}_{h_s}(x)\, dx &= \frac{1}{2\pi} \int \mathcal{F}[\hat{f}_h](\omega)\overline{\mathcal{F}[\hat{f}_s](\omega)}\, d\omega \\ &= \frac{1}{2\pi} \int |P_n(\omega)|^2 \exp(-(h^2+s^2)\omega^2/2)\, d\omega \\ &= \frac{1}{\pi} \int_0^\infty |P_n(\omega)|^2 \exp(-(h^2+s^2)\omega^2/2)\, d\omega. \end{aligned}$$

Finally, we approximate G_{rs} by truncating the integral and applying the composite trapezoidal rule:

$$G_{rs} \approx \frac{1}{2\pi}\frac{\Omega}{K}\sum_{k=0}^{K} c_k |P_n(\omega_k)|^2 \exp\left(-(h_r^2 + h_s^2)\omega_k^2/2\right).$$

Compare exact, quadrature and Fourier in terms of speed and accuracy OR find source that already does this like I do

3.3 Assembly of Scale Similarity Matrices

3.3.1 Properties of Divergence matrices and its Perron vector

In this subsection, we give structural results on the divergence matrix D built from the L^2-divergence. In particular we show that D is a so called Robinson matrix. We also show that it fulfills the acute chain condition, which leads to the Perron vector of D peaking at either of the bordering scales. This further motivates the above similarity transformation, because else our scale selection procedure will choose either the first or last scale.

Definition 10 (Robinson matrix)**.** *A matrix D with entries $D_{ij} = D(x_i, x_j)$ is called a (two way) Robinson matrix (cf. [10]) if there exists a permutation π on $\{1, \ldots, N\}$ such that for $D^\pi_{ij} = D(x_{\pi(i)}, x_{\pi(j)})$ it holds*

$$D^\pi_{ik} \geq \max\{D^\pi_{ij}, D^\pi_{jk}\} \text{ and } D^\pi_{ki} \geq \max\{D^\pi_{ji}, D^\pi_{kj}\} \tag{3.8}$$

whenever $i < j < k$.

Robinson matrices are generalizations of Min matrices, that is, every Min matrix is a Robinson similarity. Note that for a Robinson similarity, the inequalities in Equation (3.8) are reversed. It is always beneficial to know the kind of matrix one works with to potentially be able to draw from established results about these kinds of matrices. Certain choices of divergences from Section 3.2 lead to Robinson matrices, see Proposition 7 below.

The reason why we do not want to work with the matrix D and instead transform to a similarity matrix $\phi(D)$ is due to the following Lemma. An entry v_i is large if the scale i has a large distance to other scales j. The Perron vector thus gives scales that are far apart from many other scales larger weight. These scales are usually the bordering scales. The next Lemma makes this intuition precise.

Lemma 5. *Let H be a Hilbert space, and let $x_1, \ldots, x_N$ be distinct points in $\mathcal{H}$ that form an acute chain, that is,*

$$\langle x_{k+1} - x_k, x_{l+1} - x_l \rangle \geq 0 \quad \textit{for all } k, l = 1, \ldots, N-1. \tag{3.9}$$

Let $D = (D_{ij}) \in \mathbb{R}^{N \times N}$, $D_{ij} = \|x_i - x_j\|^2$ be the pairwise distance matrix and $v = (v_j) \in \mathbb{R}^N$ its Perron vector. Then v has its largest entries at the border, that is,

$$\arg\max_{1 \leq j \leq N} v_j \subseteq \{1, N\}.$$

Proof. See Appendix Section 6.2.1. □

The "acute chain condition" has precedence in the theory of so called Monge matrices, for reference see, for example [23]. Monge matrices have a special structural property that allows for certain problems to have faster algorithms when assuming this structure. A matrix D is a Monge matrix if $D_{ij} + D_{kl} \leq D_{il} + D_{kj}$ whenever $i \leq k$ and $j \leq l$. In fact, a matrix D with entries $D_{ij} = \|x_i - x_j\|^2$ is Monge if and only if the acute chain condition holds for $x_1, \ldots, x_N$. All symmetric Monge matrices with zero diagonal are Robinson matrices. The acute chain condition has a geometric interpretation: the angle between all the differences of two consecutive points is at most 90 degrees.

We now show that when choosing the Gaussian kernel, the family of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$ satisfies the acute chain condition and, as a consequence, the matrix $D = (D_{ij})$, $D_{ij} = (1/2)\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2$ is a Robinson matrix.

Proposition 7. *The following two properties hold.*

(a) The family of kernel density estimates $\{\hat{f}_{h_i}\}_{i=1}^N$ satisfies the acute chain condition stated in Equation (3.9)*.*

(b) The divergence matrix $D = (D_{ij})$ with entries given by $D_{ij} = (1/2)\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2$ is a Robinson matrix.

Proof. The Fourier transform $\mathcal{F}[\hat{f}_h]$ of a kernel density estimate $\hat{f}_h$ with a Gaussian kernel is given by $\mathcal{F}[\hat{f}_h](\omega) = P_n(\omega)\exp(-h^2\omega^2/2)$. We set $\delta_k = \hat{f}_{h_{k+1}} - \hat{f}_{h_k}$. By linearity of the Fourier transform, $\mathcal{F}[\delta_k] = P_n(\omega)b_k(\omega)$, where we have set $b_k(\omega) = \exp(-h_{k+1}^2\omega^2/2) - \exp(-h_k^2\omega^2/2)$. Using Parseval's identity, we write

$$\langle \delta_k, \delta_l \rangle = \int \delta_k \delta_l = \frac{1}{2\pi} \int_{\mathbb{R}} |P_n(\omega)|^2 b_k(\omega) b_l(\omega)\, d\omega.$$

Since $h_k < h_{k+1}$, we have $b_k(\omega) < 0$ for all $\omega \neq 0$. Also, notice that $\omega \mapsto P_n(\omega)$ and $P_n(0) = 1$. Thus $P_n(\omega) \neq 0$ for all $\omega \in B_\epsilon(0)$ for some $\epsilon > 0$. It follows that the function $\omega \mapsto |P_n(\omega)|^2 \delta_k(\omega)\delta_l(\omega)$ is strictly positive on $B_\epsilon(0) \setminus \{0\}$. We have found a set with positive Lebesgue measure on which the integrand is positive, which implies that the integral $\langle \delta_k, \delta_l \rangle$ itself is positive, that is, $\langle \delta_k, \delta_l \rangle > 0$.

Now, let $i < j < k$. We have

$$\begin{aligned}
\|\hat{f}_{h_k} - \hat{f}_{h_i}\|_2^2 &= \|\hat{f}_{h_k} - \hat{f}_{h_j} + \hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 \\
&= \|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2 + \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 + 2\langle \hat{f}_{h_k} - \hat{f}_{h_j}, \hat{f}_{h_j} - \hat{f}_{h_i} \rangle \\
&> \|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2 + \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2 \\
&\geq \max\{\|\hat{f}_{h_k} - \hat{f}_{h_j}\|_2^2, \|\hat{f}_{h_j} - \hat{f}_{h_i}\|_2^2\},
\end{aligned}$$

because

$$\begin{aligned}
\langle \hat{f}_{h_k} - \hat{f}_{h_j}, \hat{f}_{h_j} - \hat{f}_{h_i} \rangle &= \langle \sum_{r=j}^{k-1} \underbrace{(\hat{f}_{h_{r+1}} - \hat{f}_{h_r})}_{=\delta_r}, \sum_{s=i}^{j-1} \underbrace{(\hat{f}_{h_{s+1}} - \hat{f}_{h_s})}_{=\delta_s} \rangle \\
&= \sum_{r=j}^{k-1} \sum_{s=i}^{j-1} \langle \delta_r, \delta_s \rangle > 0,
\end{aligned}$$

by the first part of the proof. □

3.3.2 Similarity transformations

A matrix $A \in \mathbb{R}^{n\times n}$ is positive definite if $x^T Ax > 0$ for all $x \in \mathbb{R}^n$, $x \neq 0$. A symmetric map $D : X \times X \to \mathbb{R}$ is called conditionally negative definite if

$$\sum_{i,j=1}^{n} a_i a_j D(x_i, x_j) \leq 0$$

for all $x_1, \ldots, x_n \in X$ and $a_1, \ldots, a_n \in \mathbb{R}$ with $\sum_i a_i = 0$. A function $\phi : (0.\infty) \to \mathbb{R}$ is called completely monotone if $\phi \in C^\infty(0, \infty)$ and $(-1)^n \phi^{(n)}(t) \geq 0$ for all $n \in \mathbb{N}_0$ and $t > 0$. The most prominent example of a completely monotone function is $\phi(t) = e^{-t/\tau}$, $\tau > 0$.

The following Theorem connects the concepts presented above and is due to Schoenberg; see [5].

Theorem 2 (Schoenberg)**.** *The following statements are equivalent.*

(a) $D : X \times X \to \mathbb{R}$ *is conditionally negative definite with* $D(x, x) = 0$ *for all* $x \in X$

(b) For all $x_1, \ldots, x_n \in X$ *the matrix* $S \in \mathbb{R}^{n\times n}$ *defined by the entries* $S_{ij} = \phi(D(x_i, x_j))$ *is positive semi-definite for all completely monotone functions* $\phi : \mathbb{R}_{\geq 0} \to \mathbb{R}$*.*

(c) There exists some Hilbert space embedding $\Phi : X \to \mathcal{H}$ *such that* $D(x, y) = \|\Phi(x) - \Phi(y)\|^2_{\mathcal{H}}$*.*

The equivalence $(a) \iff (b)$ *can be strengthened to strict positive definiteness by additionally assuming that* $D(x, y) > 0$ *for all* $x \neq y$ *(which is the case for divergences) and requiring* ϕ *to be non-constant and the points* $x_1, \ldots, x_n$ *to be pairwise distinct.*

Proof. Let D be conditionally negative. By [5, Theorem 2.2] this is equivalent to $\tilde{S}_t = (\exp(-tD(x_i, x_j)))_{i,j=1}^n$ being positive semi-definite for all finite choices of points $x_1, \ldots, x_n \in X$. Let ϕ be a completely monotone function. Then by [5, Corollary 6.14], there exists a measure μ on $\mathcal{B}((0, \infty))$ such that $\phi(z) = \int_0^\infty e^{-tz}\, d\mu(t)$.

Consequently, we write

$$\sum_{i,j=1}^{n} a_i a_j \phi(D(x_i, x_j)) = \sum_{i,j=1}^{n} a_i a_j \int_0^\infty \exp\left(-tD(x_i, x_j)\right) d\mu(t)$$
$$= \int_0^\infty \underbrace{\sum_{i,j=1}^{n} a_i a_j \exp\left(-tD(x_i, x_j)\right)}_{\geq 0 \text{ by assumption}} d\mu(t) \geq 0$$

for all $a_1, \ldots, a_n \in \mathbb{R}$. This shows that the matrix $S = (\phi(D(x_i, x_j)))_{i,j=1}^n$ is positive semi-definite. If (b) holds, then (a) follows directly by the equivalence stated in [5, Theorem 2.2] since $\exp(-t\cdot)$ is a completely monotone function for all $t > 0$.[5, Proposition 3.2] gives the implication $(a) \implies (c)$, while the reverse direction can easily be checked by us. It holds

$$\sum_{i,j=1}^{n} c_i c_j \|\Phi(x_i) - \Phi(x_j)\|_{\mathcal{H}}^2 = -2 \sum_{i,j=1}^{n} c_i c_j \langle \Phi(x_i), \Phi(x_j) \rangle = -2 \left\| \sum_{j=1}^{n} c_j \Phi(x_j) \right\|_{\mathcal{H}}^2 \leq 0.$$

For $i \neq j$ we have $0 < D(x_i, x_j) = \|\Phi(x_i) - \Phi(x_j)\|_{\mathcal{H}}^2$. Thus, the points $z_i = \Phi(x_i)$ are pairwise distinct. Invoking [19, Theorem 3.1], we see that the matrix $\tilde{S}_t$ is strictly positive definite, that is, $a^T \tilde{S}_t a > 0$ for all $a \in \mathbb{R}^n$, $a \neq 0$. Since ϕ is assumed to be non-constant, this forces $\mu((0, \infty)) > 0$. Thus, $a^T S a = \int_0^\infty a^T \tilde{S}_t a \, d\mu(t) > 0$, i.e., S is strictly positive definite. □

Fuglede and Topsøe [15] show that the Jensen-Shannon divergence is conditionally negative definite. In fact, using Schoenberg's theorem above, we can also include L^2-divergence, squared Hellinger, and Total Variation in the class of conditionally negative definite divergences. We can use the equivalence $(a) \iff (c)$ from Theorem 2 to show that L^2-divergence, squared Hellinger and Total Variation are conditionally negative definite by giving the isometric Hilbert space embeddings:

- L^2-divergence: $\mathcal{H} = L^2(\mathbb{R})$, $\Phi(p) = p/\sqrt{2}$, $\|\Phi(p) - \Phi(q)\|_{L^2}^2 = (1/2)\|p - q\|_2^2$
- squared Hellinger: $\mathcal{H} = L^2(\mathbb{R})$, $\Phi(p) = \sqrt{p/2}$, $\|\Phi(p) - \Phi(q)\|_{L^2}^2 = (1/2)\|\sqrt{p} - \sqrt{q}\|_2^2$
- Total Variation: $\mathcal{H} = L^2(\mathbb{R}, (0, \infty))$, $\Phi(p)(x, t) = (1/\sqrt{2})\mathbf{1}\{0 < t < p(x)\}$, we have $\|\Phi(p) - \Phi(q)\|_{L^2(\mathbb{R}, (0,\infty))}^2 = (1/2) \int |p - q|$

- Jensen Shannon: $\mathcal{H} = L^2(\mathbb{R}, (0,\infty))$, $\Phi(p)(x,t) = (1 - e^{-tp(x)/2})/(\sqrt{2}t)$, $\|\Phi(p) - \Phi(q)\|^2_{L^2(\mathbb{R},(0,\infty))} = \mathrm{JS}(p,q)$. This can be seen easily:

$$\begin{aligned}
\|\Phi(p) - \Phi(q)\|^2_{L^2(\mathbb{R},(0,\infty))} &= \int_{-\infty}^{\infty}\int_{0}^{\infty} \left(\Phi(p)(x,t) - \Phi(q)(x,t)\right)^2 \, dt\, dx \\
&= \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{\left(e^{-tq(x)/2} - e^{-tp(x)/2}\right)^2}{2t^2} \, dt\, dx \\
&\overset{(*)}{=} \int_{-\infty}^{\infty} \frac{1}{2} \log\left(\frac{q(x)^{q(x)} p(x)^{p(x)}}{(q(x)/2 + p(x)/2)^{q(x)+p(x)}}\right) dx \\
&= \int_{-\infty}^{\infty} j(p(x), q(x))\, dx = \mathsf{JS}(p,q),
\end{aligned}$$

  where in step $(*)$ we have use a well known integral identity, see, for example [37].

In the context of our scale selection methodology, Theorem 2 gives a general construction principle for a scale similarity matrix $\Sigma^{(N)} = (\Sigma^{(N)}_{ij})$ that is guaranteed to be positive (semi-) definite. Namely, we set $\Sigma^{(N)}_{ij} = \phi(D(\hat{f}_{h_i}, \hat{f}_{h_j}))$ where D is some conditionally negative definite divergence and ϕ a completely monotone function.

Additionally, the transformation $S_{ij} = \exp(-D_{ij}/\tau)$ can be motivated as follows. Let $H(p) = -\sum_j p_j \log p_j$ be the entropy of a probability vector $p \in \mathbb{R}^N$. It can be interpreted as an uncertainty or concentration measure. $H(p)$ is large if p is close to the discrete uniform distribution and maximal for $p = (N^{-1}, \ldots, N^{-1})$ (maximal uncertainty and minimal concentration). It is small if p is concentrated on some entry and minimal for $p = e_j$ for some $j = 1, \ldots, N$, where e_j is the j-th unit vector in $\mathbb{R}^N$ (minimal uncertainty and maximal concentration).

The quantity

$$\sum_{j=1}^{N} p_j D_{ij} - \tau H(p) = \sum_{j=1}^{N} p_j D_{ij} + \tau \sum_{j=1}^{N} p_j \log p_j$$

is minimized by $\hat{p}_i = (\hat{p}_{i1}, \ldots, \hat{p}_{iN})$, where $\hat{p}_{ij} = S_{ij} / \sum_{k=1}^{N} S_{ik}$. See [2] for reference. We can interpret $\hat{p}_i$ as a probability vector that describes how strongly the scale h_i is connected to any other scales.

The minimization of the first term $D_i p = \sum_j p_j D_{ij}$ ($D_i \in \mathbb{R}^{1\times N}$ is the i-th row

of D) leads to probability vectors that concentrate on the i-th component since $\min_j D_{ij} = D_{ii} = 0$ (the largest probability weights is put on a bandwidth that is closest to bandwidth h_i with respect to the divergence D). The penalty term $\sum_j p_j \log p_j$ counteracts this concentration by penalizing it, where τ controls the amount of penalization. In that sense, τ can be interpreted as a penalty weight.

This is another instance of the Donsker–Varadhan variational principle.

Taking the limit $\tau \to +\infty$ yields $S = 11^T$, which corresponds to maximal penalization on concentration. Taking the limit $\tau \to 0$ gives $S = I_N$, which corresponds to no penalization on concentration.

Yet another advantage of the exponential transformation is the obvious fact that the entries of the resulting similarity matrix S are all positive and therefore by the Perron-Frobenius theorem, S has a real valued eigenvalue $\lambda_{\max}$ which is strictly larger than the absolute value of all other eigenvalues. The corresponding eigenvector $v_{\max}$ (called the Perron vector) has all positive entries and is unique up to scaling.

3.3.3 Bandwidth grid

By properly scaling the data it we can take $h_{\max} = 1$ without loss of generality.

grid must scale with n? Otherwise spectrally selected bandwidth will not be consistent

The choice of the correct bandwidth grid is essential. Set

$$\mathcal{H}_N = \{h_1, \dots, h_N\}, \quad 0 < h_{\min} := h_1 < \cdots < h_N =: h_{\max}.$$

The next definition gives a general construction principle for such a bandwidth grid.

Definition 11. *Let $0 < r \leq 1$. A grid $\mathcal{H}_N$ is said to be r-linear if*

$$\frac{1}{r}h_j^r = \frac{1}{r}h_1^r + \underbrace{\frac{h_N^r - h_1^r}{r(N-1)}}_{=:\Delta_{r,N}}(j-1), \quad j = 1, \dots, N. \tag{3.10}$$

Remark 3. *The construction in Equation* (3.10) *is just the Box-Cox transformation of a linear grid $\mathcal{H}_N$, compare with [6, Equation (1)]. Setting $\psi_r(h) = r^{-1}(h^r - 1)$*

we see that a grid $\mathcal{H}_N$ is r-linear if $\{\psi_r(h_j)\}_{j=1}^N$ is a linear grid. Taking the limit $r \to 0$ we obtain $\psi_0(h) := \lim_{r\to 0} \psi_r(h) = \log h$. A 0-linear grid can therefore be seen as a geometric grid. Another interpretation is the following: h_j is the weighted power mean between the two endpoints h_1 and h_N with exponent r and weights $w_1 = 1 - (j-1)/(N-1)$ and $w_2 = (j-1)/(N-1)$. For reference, see [9, Chapter 3.1].

Definition 11 smoothly connects a linear and a geometric grid. Setting $r = 1$ gives a linear grid, that is,

$$h_j = h_{\min} + \Delta_{1,N}(j-1), \quad \Delta_{1,N} = \frac{h_{\max} - h_{\min}}{N-1}.$$

Interpreting $r = 0$ as limit we obtain a geometric grid, that is,

$$\log h_j = \log h_{\min} + \Delta_{0,N}(j-1), \quad \Delta_{0,N} = \frac{\log h_{\max} - \log h_{\min}}{N-1}.$$

The geometric grid can equivalently be written as

$$h_j = h_{\min} q_N^{j-1}, \quad q_N = \left(\frac{h_{\max}}{h_{\min}}\right)^{1/(N-1)}.$$

Taking the limit $N \to +\infty$ increases the fineness of the grid.

Next we formulate a Lemma that gives the first order asymptotic approximation for the spacing $h_{j_N} - h_{i_N}$ of a r-linear grid, where (i_N) and (j_N) are appropriately chosen index sequences. It tells us that for large N the difference between the two scales h_{j_N} and h_{i_N} is approximately $h_{i_N}^{1-r}\delta_{ij}^{r,N}$. In other words, we express the difference between two untransformed bandwidths $h_{j_N} - h_{i_N}$ using the difference between two transformed bandwidths $\psi_r(h_{j_N}) - \psi_r(h_{i_N})$ approximately when N is large and $j_N - i_N = \mathbf{o}(N)$.

Lemma 6. *Let $\mathcal{H}_N = \{0 < h_1 < \cdots < h_N\}$ be a r-linear grid with fixed endpoints $h_1 = h_{\min}$ and $h_N = h_{\max}$, and let $i = (i_N)_{N\in\mathbb{N}}$ and $j = (j_N)_{N\in\mathbb{N}}$ be two index sequences that satisfy $1 \le i_N, j_N \le N$ for all $N \in \mathbb{N}$. Set $\delta_{ij}^{r,N} := (j_N - i_N)\Delta_{r,N}$ (the length of the gap between $\psi_r(h_{i_N})$ and $\psi_r(h_{j_N})$). Assume $|j_N - i_N| = \mathbf{o}(N)$.*

Then

$$h_{j_N} - h_{i_N} = h_{i_N}^{1-r}\delta_{ij}^{r,N} + \mathbf{O}((\delta_{ij}^{r,N})^2).$$

Proof. We have $h_{j_N}^r - h_{i_N}^r = r(j_N - i_N)\Delta_{r,N}$ from which follows

$$h_{j_N} = \left(h_{i_N}^r + r\delta_{ij}^{r,N}\right)^{1/r} = h_{i_N}\left(1 + r\underbrace{\frac{\delta_{ij}^{r,N}}{h_{i_N}^r}}_{=:c_N}\right)^{1/r}.$$

Note that $c_N = \mathbf{O}(\delta_{ij}^{r,N})$ because $h_{i_N}^{-r}$ is bounded above by $h_{\min}^{-r}$. In particular, since $|j_N - i_N| = \mathbf{o}(N)$ by assumption, c_N is a null sequence. The first order Taylor approximation of $(1+x)^{1/r}$ around 0 is $(1+x)^{1/r} = 1 + x/r + \mathbf{O}(x^2)$. We get

$$\begin{aligned} h_{j_N} - h_{i_N} &= h_{i_N}\left((1 + rc_N)^{1/r} - 1\right) = h_{i_N}\left(c_N + \mathbf{O}((\delta_{ij}^{r,N})^2)\right) \\ &= h_{i_N}^{1-r}\delta_{ij}^{r,N} + \mathbf{O}((\delta_{ij}^{r,N})^2) \end{aligned}$$

by properties of the $\mathbf{O}$-notation. □

The stronger condition $j_N - i_N = \mathbf{O}(1)$ (the difference between the index sequences stays asymptotically bounded) would lead to the remainder $\mathbf{O}(N^{-2})$.

Motivation for why a geometric grid is the natural choice

The transformation $K \mapsto K_h$ stretches K horizontally while keeping the area under the curve the same. We can measure the width of a kernel K, for example, by $s(K) := \sqrt{\mu_2(K)} = (\int x^2 K(x)\,dx)^{1/2}$. The width of the stretched kernel is $s(K_h) = hs(K)$. Increasing h by the factor a increases $s(K_h)$ by the factor a. Choosing a geometric grid means that going from h_j to h_{j+1} increases $s(K_{h_j})$ by the same factor q_N. Thus, each step stretches the kernel by the same factor.

3.4 Numerical Experiments

How does software compute eigenvectors

stable Perron vector computation

Power method (compute Perron vector through Power iteration)

In our numerical experiments, we restrict ourselves to these four divergences:

- L_2 divergence (β-divergence, $\beta = 1$)
- Total variation: $1 - \mathrm{TV}(p, q) = \int \min\{p, q\}$ TBD
- squared Hellinger: $1 - H^2(p, q) = \int \sqrt{pq}$ TBD
- Jensen-Shannon $D_{\mathrm{JS}}(p, q)$ TBD

add spectral bandwidth choices for total variation, squared Hellinger and Jensen Shannon

We will start our explorations using the L_2 divergence as it is closely related to the MISE criterion for optimal bandwidth selection and it can be computed efficiently using the Fourier transform. We use the similarity transformation $S = \exp(-\lambda D)$ and vary the penalty parameter $\lambda = \tau^{-1}$. Specifically, we will look at the limit $\lambda \to \infty$, which corresponds to a decreasing penalization on the concentration of the Perron vector. Later, we will explore the other symmetric divergences: Total variation, squared Hellinger, and Jensen-Shannon. We leave the asymmetric parametrized divergences as future work; see Outlook 5.

In our experiments, we focus first on the case $f \sim N(0, 1)$, that is, normally distributed data. This is the simplest case in which there is only one "optimal" bandwidth (the one minimizing a certain error criterion). We start with this simple case to see if there is even a connection between our spectral pipeline and the notion of "optimal" bandwidth selection. Experimentation with other data distributions is future work; see the Outlook chapter 5. Especially Gaussian mixture models are interesting because in such cases there might not exist a unique optimal bandwidth and the Perron vector could serve as a ranking procedure for multiple candidate bandwidths.

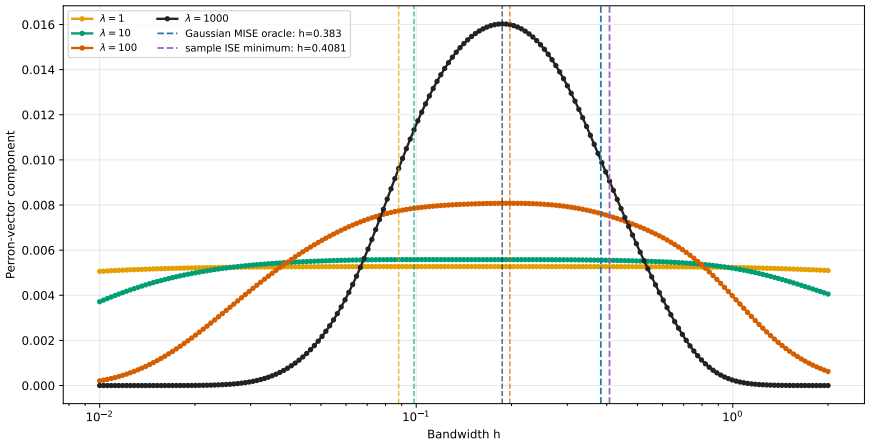


Figure 3.1: $X_i \sim N(0,1)$, $n = 200$, $N = 191$, $h_{\min} = 0.01$, $h_{\max} = 2$, $\lambda \in \{10^k : k = 0, 1, 2, 3\}$, geometric bandwidth grid; the blue line indicates $h_{\text{oracle}} = 0.38304$

3.4.1 Fixed sample experiments

For the first set of experiments, we draw one fixed sample $X = (X_1, \ldots, X_n) \sim f$, $n = 200$ and study the influence of the grid fineness N and the concentration penalty $\lambda = \lambda_N$.

Figure 3.1 shows the relation $i \mapsto v_i$, where v is the Perron vector of $S^{(\lambda)} = (S_{ij}^{(\lambda)})$, $S_{ij}^{(\lambda)} = \exp\left(-\frac{\lambda}{2}\|\hat{f}_{h_i} - \hat{f}_{h_j}\|_2^2\right)$, $\lambda = 1$.

We see that the Perron vector has a flat plateau for $\lambda = 1$ (yellow) and $\lambda = 10$ (green) which means that many bandwidths are considered central. Increasing λ leads to the Perron vector having a more pronounced peak, which is consistent with the interpretation of $\tau = \lambda^{-1}$ being a penalty for how much the Perron vector concentrates.

Of course, a single such image tells us not much and does not give us hints about a possible connection between the oracle bandwidth and the spectrally chosen bandwidth.

Figure 3.2 shows the stability with respect to the grid fineness N. The sample $X = (X_1, \ldots, X_n)$ was kept fixed. We observe a stabilization of the chosen bandwidth for increasing grid fineness.

It can be seen that the grid stability curves for $\lambda = 1$ (yellow) and $\lambda = 10$ (green)

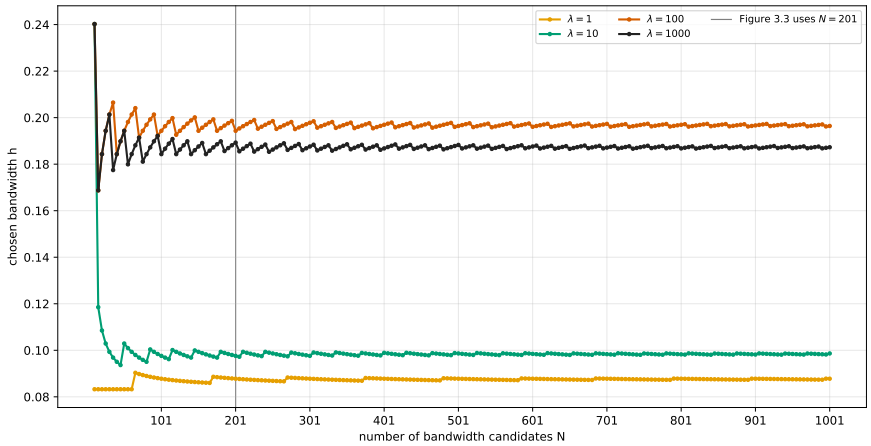


Figure 3.2: For a fixed sample $X = (X_1, \ldots, X_{200})$ the spectral bandwidth selection pipeline was repeated for $N = 11, 16, 21, \ldots, 1001$; $\lambda = 1$

converge towards a smaller bandwidth scale than the curves for $\lambda = 100$ (red) and $\lambda = 1000$ (black). This indicates that some kind of switch is happening between $\lambda = 10$ and $\lambda = 100$. Interestingly, the grid stability plots exhibit a sawtooth pattern that smoothens out.

In Section 3.3.2 we established the interpretation of τ as a penalty weight. Henceforth, we set $\lambda = 1/\tau$ and study $\lambda \to \infty$. This corresponds to a decrease in concentration penalization. Figure 3.3 shows the chosen bandwidth for one fixed sample as λ increases. The path does not approach the oracle bandwidth over the displayed range. We will explain this behavior in Section 3.5 below. Also note that the curve sharply jumps around $\lambda \approx 17.87375$.

The observation regarding the sharp jumps motivates to ask more questions: does it happen under repeated sampling? At which λ does the switch happen each time? Is it an interesting property about this method or just a numerical artifact?

After having examined the fixed λ increasing N behavior in Figure 3.2 and the fixed N increasing λ behavior in Figure 3.3, we would like next to study the joint limit $(N, \lambda_N) \to (\infty, \infty)$. We study the following regimes: $\lambda_N = N^\alpha$, where $\alpha \in \{0, 0.5, 1, 1.5, 2, 3\}$. Figure 3.4 summarizes the results. For $\alpha > 1/2$ the chosen bandwidth seems to converge towards some bandwidth $h_\infty = 0.1764$. In Section 3.5 we explain that this is the minimizer of $\hat{Q}_{200,2}(h) = (h^2/2)\|\partial_h \hat{f}_h\|_2^2$.

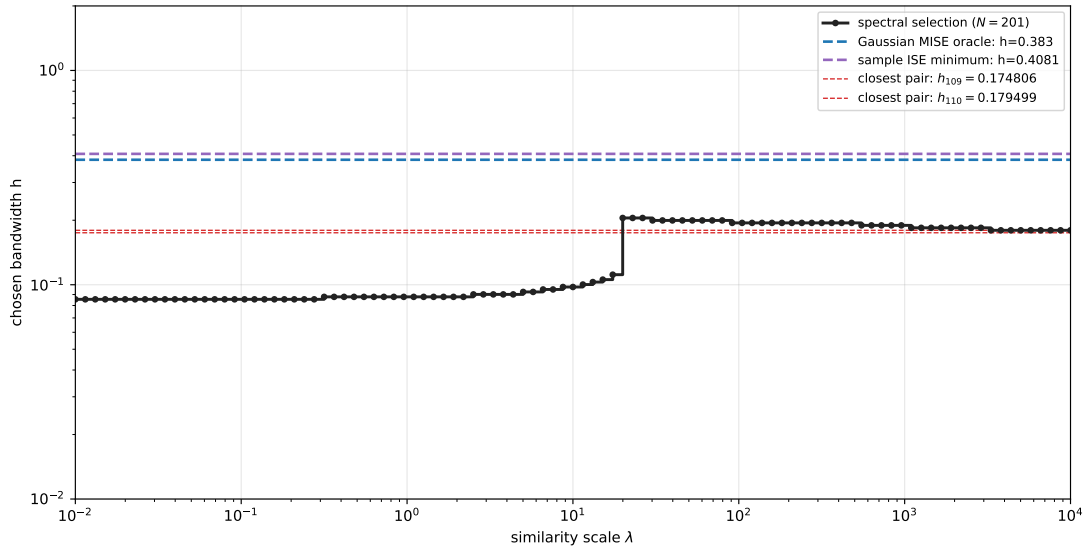


Figure 3.3: Spectrally chosen bandwidth for one fixed $N(0,1)$ sample ($n = 200$) as λ increases; $N = 201$, $h_{\min} = 0.01$, and $h_{\max} = 2$.

explain the sharp jump?

3.4.2 Repeated sampling experiments

Now we draw repeated samples $(X_1^{(i)}, \ldots, X_n^{(i)})$, $i = 1, \ldots, M$ and study the distribution of chosen bandwidths. Figure 3.5 shows the results of studying the distribution of the chosen bandwidths under repeatedly drawn samples $(X_1^{(i)}, \ldots, X_n^{(i)})$, $i = 1, \ldots, M$.

We see a large discrepancy between the distributions of the spectrally chosen bandwidth and the distribution of the minimizer of the energy functional and the minimizer of the ISE.

3.4.3 Increasing sample size experiments

In this section, we want to see what happens when the sample size increases, that is, $n \to \infty$.

It is interesting to ask what happens to this distribution for increased sample size n. The answer is shown in Figure 3.6.

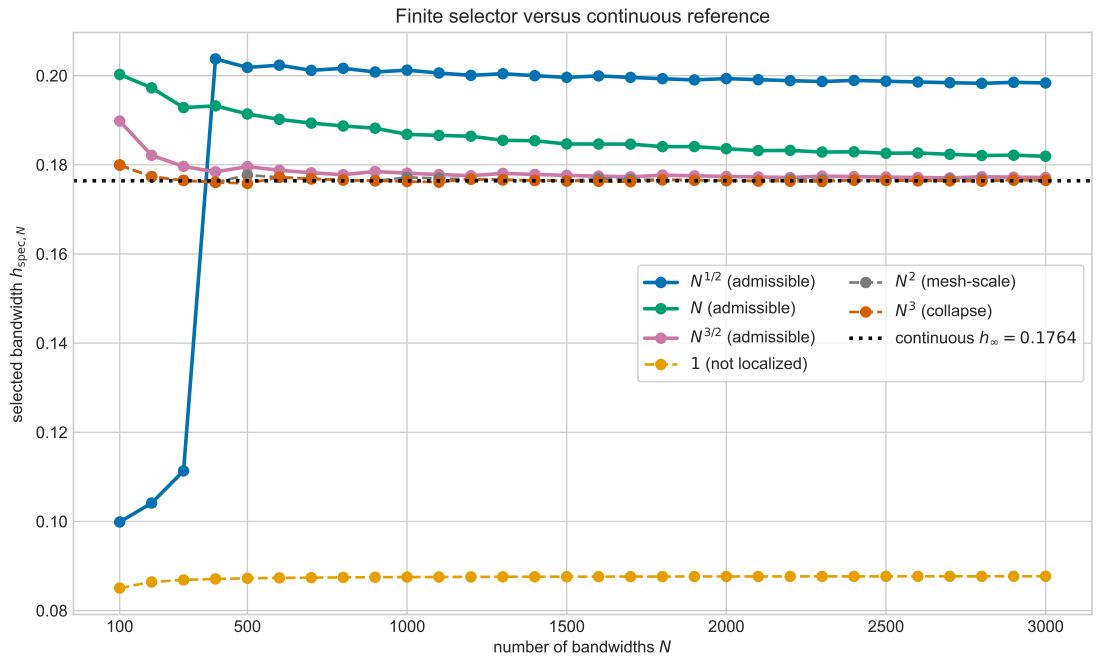
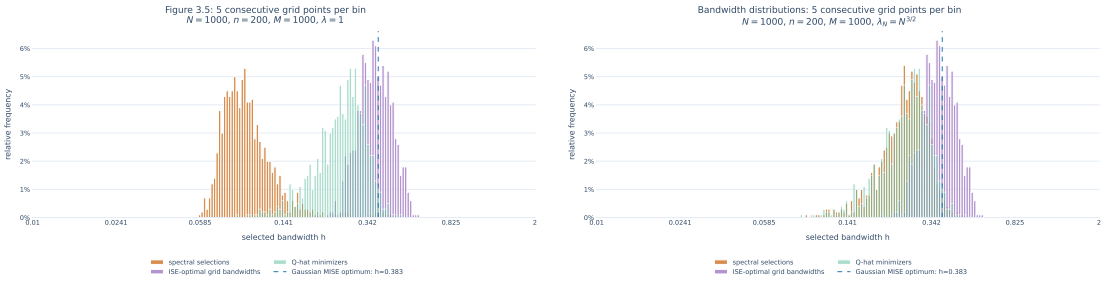


Figure 3.4: For $N \to \infty$ the chosen bandwidth approaches a limit $h_\infty = 0.1764$, for all $\alpha > 1/2$. The convergence is slow for $\alpha = 1$ and faster for $\alpha = 3/2$

explain N^2 and N^3 behavior

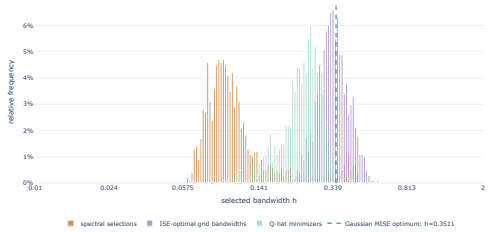
add experiments that show failure for exponents $\alpha > 2$

repeat the experiments for different r and under repeated sampling

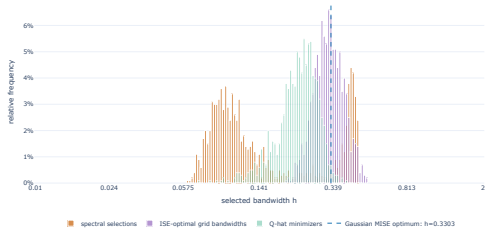


(a) $\lambda = 1$ (b) $\lambda_N = N^{3/2} \approx 31622.78$

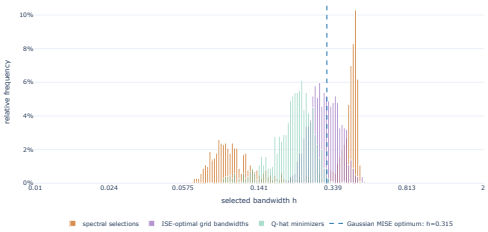
Figure 3.5: Distributions of chosen bandwidths for $N = 1000$, $n = 200$, and $M = 1000$ repetitions, combining five consecutive grid positions per bin.



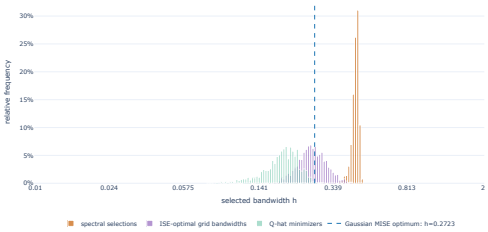
(a) $n = 300$



(b) $n = 400$



(c) $n = 500$



(d) $n = 1000$

Figure 3.6: Distribution of chosen bandwidths changes for increasing sample size n. A clear transition between two regimes is visible.

Can the two peaks be described?

n_{critical} (finite sample mode switching), gap between two local Perron maxima??

3.5 Explanation of the Results

3.5.1 Convergence to closest adjacent pair

Here we explain the results of the fixed N and $\lambda \to \infty$ experiment.

Let $\Sigma^{(\lambda)}$ be the scale similarity matrix with entries $\Sigma_{ij}^{(\lambda)} = \exp(-\lambda D_{ij})$. Since the diagonal entries are equal to 1, we can write $\Sigma^{(\lambda)} = I_N + A^{(\lambda)}$. The matrices $\Sigma^{(\lambda)}$ and $A^{(\lambda)}$ have the same Perron vector. The Perron vector also stays the same if we multiply the matrix by a positive scalar. We therefore define $\tilde{A}^{(\lambda)} = e^{\lambda d_*} A^{(\lambda)}$, where $d_* = \min_{i \neq j} D_{ij}$ is the smallest distance between two distinct scales h_i and h_j. The off diagonal entries of $\tilde{A}^{(\lambda)}$ are given by $\tilde{A}_{ij}^{(\lambda)} = e^{\lambda(d_* - D_{ij})}$, which all lie between 0 and 1. One particular advantage of this construction is that large λ become numerically manageable while the Perron vector does not change.

Taking the limit $\lambda \to \infty$ we get

$$\lim_{\lambda \to \infty} \tilde{A}_{ij}^{(\lambda)} = \begin{cases} 1 & , D_{ij} = d_* \\ 0 & , D_{ij} > d_*. \end{cases}$$

In the limit, this is an adjacency matrix of a graph on $\{h_1, \ldots, h_N\}$ with edges between h_i and h_j if and only if D_{ij} is minimal. In fact if D is a Robinson matrix (for example if we take the L^2-divergence) this forces $d_* = d_{j,j+1}$ for some $j = 1, \ldots, N-1$.

Assuming that $d_* = d_{j,j+1}$ for some unique $j = 1, \ldots, N-1$, we can show that the Perron vector $v^{(\lambda)}$ of $\tilde{A}^{(\lambda)}$ converges to $(1/2)e_j + (1/2)e_{j+1}$, that is, it concentrates at $\arg\min_{(j,j+1)} D_{j,j+1}$. We write $B = \lim_{\lambda \to \infty} \tilde{A}^{(\lambda)}$ (where the limit is understood entry-wise). The matrix B can be written as $B = e_j^T e_{j+1} + e_{j+1} e_j^T$. Direct calculations show that $e_j + e_{j+1}$ is an eigenvector with eigenvalue 1, $e_j - e_{j+1}$ is an eigenvector with eigenvalue -1 and e_i, $i \neq j, j+1$ are eigenvectors with eigenvalues 0. Therefore, $v = (1/2)e_j + (1/2)e_{j+1}$ is the sum normalized Perron vector of B.

This explains the results shown in Figure 3.3, as well as the yellow line in Figure 3.4. In Figure 3.3 the chosen bandwidth approaches the scales h_{j_*} and h_{j_*+1} with the smallest consecutive jump $D_{j,j+1}$, that is, $j_* \in \arg\min_j D_{j,j+1}$.

3.5.2 Perron vector concentrates at minimizer of normalized curvature-energy functional

To explain the results, we need the concept of a Fréchet-derivative. A good book is Zeidler [36] and Section 4.2 therein.

We introduce the general definition of a Fréchet-derivative. Let X and Y be Banach spaces and $U \subseteq X$ an open nonempty subset. Let $\Psi : U \to Y$ be a map. A linear and bounded operator $d\Psi(u) : X \to Y$ is called Fréchet-derivative of Ψ at the point $u \in U$ if

$$\Psi(u+h) = \Psi(u) + d\Psi(u)[h] + r_u(h), \quad \forall h \in X : u+h \in U,$$

where $r_u(h) : \{h \in X : u+h \in U\} \to Y$ is a remainder that fulfills $\|r_u(h)\|_Y / \|h\|_X \to 0$ for $h \to 0$. Ψ is called continuously differentiable on U if $d\Psi(u)$ exists for all $u \in U$ and if $d\Psi : U \mapsto \mathcal{L}(X,Y)$ is continuous. Here, $\mathcal{L}(X,Y)$ is the set of all linear and bounded operators from X to Y. We write $\Psi \in C^1(U)$.

Assume that the line $\{u+th : t \in [0,1]\}$ is inside U and define $\gamma : [0,1] \to Y$, $\gamma(t) = \Psi(u+th)$. We have $\gamma'(t) = d\Psi(u+th)[h]$. Now, if $\Psi \in C^1(U)$ then γ is continuously differentiable and by the fundamental theorem of calculus $\Psi(u+h) - \Psi(u) = \int_0^1 d\Psi(u+th)[h]\,dt$. We can use this to bound the norm of the remainder:

$$\begin{aligned}
\|r_u(h)\|_Y &\leq \int_0^1 \|(d\Psi(u+th) - d\Psi(u))[h]\|_Y \, dt \\
&\leq \|h\|_X \int_0^1 \|d\Psi(u+th) - d\Psi(u)\|_{\mathsf{op}} \, dt \\
&\leq \|h\|_X \sup_{t\in[0,1]} \|d\Psi(u+th) - d\Psi(u)\|_{\mathsf{op}}
\end{aligned}$$

In our setting, $X = \mathbb{R}$, $U = \mathbb{R}_{>0}$, and $Y = L^2(\mathbb{R})$. The map is $\Psi(h) = \hat{f}_h$. The next Lemma gives the Fréchet-derivative of $h \mapsto \hat{f}_h$.

Lemma 7. *Let K be differentiable on $\mathbb{R}$ and square integrable. Let L be defined as $L(u) = K(u) + uK'(u)$. We must assume a certain decay assumption on L, namely $|L(v)| \leq C(1+|v|)^{-\alpha}$ for some $C > 0$, $\alpha > 1/2$ and all $v \in \mathbb{R}$. The Fréchet-derivative of $h \mapsto \hat{f}_h$ at h is an operator $d\Psi(h) : \mathbb{R} \to L^2(\mathbb{R})$ given by*

$d\Psi(h)[\tau] = \tau\partial_h \hat{f}_h$, *where*

$$\partial_h \hat{f}_h(x) = -\frac{1}{nh^2}\sum_{j=1}^{n} L\left(\frac{x - X_j}{h}\right), \quad x \in \mathbb{R}.$$

Additionally, the norm of the remainder term $r_h(\tau)$ *can be bounded above by:*

$$\|r_h(\tau)\|_2 \leq |\tau|\mu_J(|\tau|)$$

if $h, h + \tau \in J := [a, b] \subseteq (0, \infty)$. *Also,* $\mu_J \geq 0$ *and* $\mu_J(s) \to 0$ *as* $s \downarrow 0$.

Proof. See Appendix Section 6.2.2. □

Next, we state the main theorem of this thesis.

Proposition 8. *Let* $D : \mathcal{P}_2 \times \mathcal{P}_2 \to \mathbb{R}_{\geq 0}$ *be the* L^2*-divergence on* $\mathcal{P}_2 := \mathcal{P} \cap L^2(\mathbb{R})$. *Let* $\mathcal{H}_N = \{0 < h_1 < \cdots < h_N\}$ *be a* r*-linear grid for some* $0 < r \leq 1$ *with fixed endpoints* $h_1 = h_{\min}$ *and* $h_N = h_{\max}$. *Let* λ_N *be a sequence such that* $\lambda_N \to +\infty$ *and* $\lambda_N/N^2 \to 0$. *Define the scale similarity matrix* $S^{(N)} = (S_{ij}^{(N)})_{i,j=1}^{N}$ *with entries* $S_{ij}^{(N)} = \exp\left(-\lambda_N D(\hat{f}_{h_i}, \hat{f}_{h_j})\right)$. *Here* $\hat{f}_h$ *is the kernel density estimate with Gaussian kernel* K *and fixed data* $X_1, \ldots, X_n \sim f$.

Let i_N *be an index sequence such that* $h_{i_N} \to h_0 \in (h_{\min}, h_{\max})$. *Let* $R_{i_N}^{(N)} = \sum_{j=1}^{N} S_{i_N j}^{(N)}$ *be the* i_N*-th row sum of* $S^{(N)}$. *Then*

$$\sqrt{\lambda_N}\Delta_{r,N} R_{i_N}^{(N)} \to \sqrt{\frac{\pi}{\hat{Q}_r(h_0)}}, \quad N \to +\infty,$$

where $\hat{Q}_r(h) = 2^{-1}h^{2(1-r)}\|\partial_s \hat{f}_s|_{s=h}\|_2^2$ *is an energy functional.*

Proof. See Appendix Section 6.2.3. □

Let $v^{(N)}$ be the Perron vector of $S^{(N)}$, that is,

$$\sum_{j=1}^{N} S_{ij}^{(N)} v_j^{(N)} = \underbrace{\theta_{\max}^{(N)}}_{>0} v_i^{(N)}, \quad \text{for all } i = 1, \ldots, N.$$

The Perron vector $v^{(N)}$ has large entries $v_i^{(N)}$ if the weighted row sum $\sum_{j=1}^N S_{ij}^{(N)} v_j^{(N)}$ is large. Using the normalization $\|v^{(N)}\| = 1$ we can see the entries $v_i^{(N)}$ as weights, which allows us to say that the Perron vector gives a bandwidth h_i a higher weight if $\hat{f}_{h_i}$ is similar to $\hat{f}_{h_j}$ (in terms of L^2-distance) where the bandwidths h_j also have a large weight.

The Proposition 8 establishes that for a large N (finer and finer bandwidth grid) the unweighted row sum $\sum_{j=1}^N S_{i_N j}^{(N)}$ is approximately equal to $\lambda_N^{-1/2} \Delta_{r,N}^{-1} \pi^{1/2} \hat{Q}_r^{-1/2}(h_{i_N})$ (a properly scaled energy functional). This means that large row sums correspond to a low value of the energy functional $\hat{Q}_r$ and since the energy functional $\hat{Q}_r$ is the squared L^2 norm of the Fréchet derivative of the kernel density estimate, its minimizer can be interpreted as the point at which the L^2-curve $h \mapsto \hat{f}_h$ changes most slowly. This provides an informal explanation for why the Perron vector peaks at the minimizer of the energy functional $\hat{Q}_r$.

The spectral machinery is asymptotically redundant: the spectral selector is asymptotically equivalent to a normalized curvature-energy criterion.

Laplace method!!

For the Gaussian Kernel, the energy functional $\hat{Q}_r$ can be written down explicitly:

$$\hat{Q}_r(h) = \frac{h^{-5-2r}}{32\sqrt{\pi} n^2} \sum_{i,j=1}^{n} e^{-d_{ij}^2/(4h^2)} \left(d_{ij}^4 - 12h^2 d_{ij}^2 + 12h^4 \right),$$

where $d_{ij} = X_i - X_j$. Using standard optimization methods, we can minimize $\hat{Q}_r$ on the interval $[h_{\min}, h_{\max}]$. The black dotted horizontal line in Figure 3.4 represents $h_* = \arg\min_{h \in [h_{\min}, h_{\max}]} \hat{Q}_2(h)$. The green histograms inside Figures 3.5 and 3.6 also represent the minimizer of the energy functional $\hat{Q}_2$.

3.5.3 Asymptotic minimizer of expected energy functional is asymptotically equivalent to mean integrated squared error minimizer

The next result connects the asymptotic minimizer of $\mathbb{E}[\hat{Q}_r(h)]$ to the asymptotic minimizer of $\mathrm{MISE}(h)$.

The expectation of the energy functional is

$$\mathbb{E}[\hat{Q}_r] = \frac{1}{2} h^{2(1-r)} \left(\frac{1}{n} \|\partial_h K_h\|_2^2 + \left(1 - \frac{1}{n}\right) \|\partial_h (K_h * f)\|_2^2 \right)$$

Similarly to the AMISE we do a leading term approximation:

$$\mathbb{E}[\hat{Q}_r] = \underbrace{\frac{1}{2n} h^{-(1+2r)} \|L\|_2^2 + \frac{n-1}{2n} h^{4-2r} \mu_2(K)^2 \|f''\|_2^2}_{=:\mathrm{AMQ}_r(h)} + \mathbf{o}(h^{4-2r})$$

The minimizer of $\mathrm{AMQ}_r(h)$ is

$$h_{\mathrm{AMQ},r}(n) = \left(\frac{1+2r}{4-2r} \frac{\|L\|_2^2}{(n-1)\mu_2(K)^2 \|f''\|_2^2} \right)^{1/5}$$

We recall that the asymptotic minimizer of the mean integrated squared error is given in Equation (3.1) and we get

$$\frac{h_{\mathrm{AMQ},r}(n)}{h_{\mathrm{AMISE}}(n)} \to \frac{1+2r}{4-2r} \left(\frac{\|L\|_2^2}{\|K\|_2^2} \right)^{1/5}, \quad n \to \infty. \tag{3.11}$$

For $r = 13/14$ we thus see that $h_{\mathrm{AMQ},r}(n)$ is asymptotically equivalent. This is because if K is the Gaussian kernel, then $R(K) =?$ and $R(L) =?$ and the right and the limit in Equation (3.11) becomes 1.

Figure [] visualizes our theoretical considerations. It compares the distribution of $M = 1000$ selected bandwidths minimizing:

- ISE (violet)
- $\hat{Q}_0$ (blue)
- $\hat{Q}_1$ (orange)
- $\hat{Q}_{13/14}$ (green)

4 Conclusion

In the first part of this thesis we developed a new method for estimating the coefficients and determining the order of auto-regressive processes. It is based on the minimization of the width of the sliding window embedding of the AR process via projected sub-gradient descent. By establishing a connection to convex geometry, we were able to derive consistency results for this new method.

The second part of this thesis was concerned about applying the spectral scale selection methodology developed in the manuscript [4] to bandwidth selection in kernel density estimation. We developed a framework for how to construct pairwise similarity matrices using the notion of divergences between probability distributions. Our main result establishes an interesting connection between energy functional minimization and spectral scale selection.

5 Outlook

Geometric Estimation of Autoregressive processes

Is the minimum width in any way related to the Fourier transform

Big Question: How to generalize this approach for a wide class of innovation distributions? We only get the nice zonotope formulas because of uniform innovations

A vital question is how to generalize the approach developed above to a broader class of innovation distributions. The main difficulty to overcome is that $\operatorname{supp}(Y_t^{(k)})$ is generally not compact but usually unbounded. Thus, the minimum width is infinite.

Probability integral transform (PIT)

linear filter, Wold representation

The *minimum volume set* (compare [33]) is defined by

$$E_\alpha \in \arg\min\{\lambda_d(A) : \mathbb{P}(Y \in A) \geq \alpha, A \in \mathcal{B}(\mathbb{R}^d)\}.$$

The *highest density region* C_α, $\alpha \in (0,1)$ (see [22]) is defined by

$$\begin{aligned} C_\alpha &= C_{h_\alpha} \\ C_h &= \{x \in \mathbb{R}^d : f(x) \geq h\} \\ h_\alpha &= \sup\{h \in \mathbb{R} : \mathbb{P}(Y \in C_h) \geq 1 - \alpha\} \end{aligned}$$

connect these concepts with each other. For many distributions they are the same. Using minimum volume sets would remove dependence on having a smooth density

estimate HDR directly from the points $Y_t^{(k)}$; non parametric estimation, empirical HDR

Maybe find a class of distributions where the highest density region (or minimum volume set) has a nice geometric shape and compute the intrinsic volumes of that shape

log concave distributed innovations give convex highest density regions; VERY INTERESTING

A probability density is log concave if

$$f(x) = \frac{e^{-V(x)}}{\int e^{-V(u)}\, du}$$

for some convex function V.

Normally distributed innovations

Next, we turn our attention to Gaussian innovations, which is often seen as the default case, that is, $Z_t \sim N(0, \sigma^2)$ for some $\sigma > 0$. Using the MA(∞) representation, we see that $Y_t^{(k)} \sim \mathcal{N}(0, \Gamma_k)$ with covariance matrix

$$\Gamma_k = \sigma^2 \sum_{j=0}^{\infty} g_j^{(k)} (g_j^{(k)})^T$$

The covariance matrix plays an important role in describing the geometry of the point cloud $\mathcal{P}_k$. Let $E_c = \{y \in \mathbb{R}^k : y^T \Gamma_k^{-1} y \leq c\}$ be the ellipsoid of size $c > 0$. Then for all $\alpha \in (0, 1)$ we have

$$E_{c_\alpha} \in \arg\min \left\{ \mathrm{Vol}(A) : \mathbb{P}(Y \in A) \geq \alpha : A \subseteq \mathbb{R}^k \right\}$$

That is, the ellipsoid with size c_α is the set with minimum volume, such that the probability that Y is inside E_{c_α} is at least α.

find reference or prove

Other convex–geometric functionals

empirical version of intrinsic volumes? And also convergence to theoretical object

spherical harmonics: Q_0 (mean width), Q_2, Q_4, ...

numerical analogues $\hat{Q}_0$, $\hat{Q}_2$, $\hat{Q}_4$

Besides the minimum width m_k we can ask if other convex–geometric functionals can be used. A natural candidate would be the mean width, given by integrating $w_k(u)$ over $\mathbb{S}^{k-1}$:

$$\overline{w}_k = \int_{\mathbb{S}^{k-1}} w_k(u)\, d\sigma(u).$$

In practice one would estimate the integral by Monte Carlo simulation. The estimator would take the form

$$\hat{\overline{w}}_k = \int_{\mathbb{S}^{k-1}} \hat{w}_{k,T}(u)\, d\sigma(u) \approx \frac{1}{M} \sum_{m=1}^{M} \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, U_k \rangle,$$

where $U_k \sim \mathcal{U}(\mathbb{S}^{k-1})$.

Real-world Data

Talk about applications, argue where AR models with bounded innovations might be used

- Wind or solar power generation
- traffic occupancy
- industrial control, system identification

Spectral Scale Selection for Kernel Density Estimation

Rényi divergence

Rényi is not conditionally negative definite?

Rényi is unbounded?

Remark 4 (Rényi divergence)**.** *The class of Rényi divergences is defined as*

$$R_\alpha(p\|q) = \frac{1}{\alpha-1}\log\left(\int p(x)^\alpha q(x)^{1-\alpha}\,dx\right),$$

where $\alpha \in \mathbb{R}_{>0}\setminus\{1\}$ is the order. In case of $\alpha > 1$, we read $f^\alpha g^{1-\alpha} = f^\alpha/g^{\alpha-1}$. The limiting case $\alpha \to 1$ recovers the Kullback-Leibler divergence.

The Rényi divergence is not a f-divergence, but the Rényi and Amari divergences are both strictly decreasing transformations of the same integral $C_\alpha(p,q) = \int p^\alpha q^{1-\alpha}$. We have

$$\begin{aligned} R_\alpha(p\|q) &= g(C_\alpha(p,q)), \quad g(t) = \frac{1}{\alpha-1}\log t \\ D_\alpha(p\|q) &= h(C_\alpha(p,q)), \quad h(t) = \frac{1}{\alpha(\alpha-1)}(t-1) \end{aligned}$$

Thus, the Rényi and Amari divergences can be transformed into each other using a strictly increasing function.

Bounded Divergences

From [] we have the uniform bound $D_f(p,q) \le C_f := f(0) + f^*(0)$ over all $p, q \in \mathcal{P}$. Here, $f^*(0) = \lim_{t\downarrow 0} tf(t^{-1})$. It follows that a f-divergence is bounded if and only if $f(0) + f^*(0) < \infty$. In case f is symmetric this is just equivalent to $2f(0) < \infty$.

For bounded divergences, we can use a more straightforward transformation approach by setting $\Sigma_{ij}^{(N)} = C_f - D_f(\hat{f}_{h_i}, \hat{f}_{h_j})$ and normalizing $\tilde{\Sigma}_{ij}^{(N)} = C_f^{-1}\Sigma_{ij}^{(N)} = 1 - D_f(\hat{f}_{h_i}, \hat{f}_{h_j})/C_f$.

For example, the total variation divergence can be written as

$$\begin{aligned} d_{\mathsf{TV}}(p,q) &= \frac{1}{2}\int |p-q| = 1 - 1 + \frac{1}{2}\int |p-q| \\ &= 1 - \frac{1}{2}\int p - \frac{1}{2}\int q + \frac{1}{2}\int |p-q| \\ &= 1 - \int \frac{p+q-|p-q|}{2} \\ &= 1 - \int \min\{p,q\}. \end{aligned}$$

Thus, $d_{\mathsf{TV}}(p,q) \le 1$ and $1 - d_{\mathsf{TV}}(p,q) = \int \min\{p,q\}$ is a similarity measure between

p and q. Similarly for the squared Hellinger distance we can write

$$H^2(p,q) = \frac{1}{2}\int (\sqrt{p} - \sqrt{q})^2 = 1 - \int \sqrt{pq}$$

So again $H^2(p,q) \leq 1$ and $1 - H^2(p,q) = \int \sqrt{pq}$ is a similarity measure between p and q. For total variation, we use the $\min$ kernel and for squared Hellinger Bhattacharyya kernel (see [24]). The JS divergence can also is also bounded:

$$f(t) = \frac{1}{2} t \log \underbrace{\frac{2t}{1+t}}_{\leq 2} + \frac{1}{2}\log \underbrace{\frac{2}{1+t}}_{\leq 2} \leq \frac{1}{2}\log 2(t+1), \quad t \geq 0,$$

$$\mathrm{JS}(p,q) = \int q f(p/q) \leq \int q \frac{1}{2} \log 2 \left(\frac{p}{q} + 1\right) = \log 2$$

Asymmetric divergences

- Kullback-Leibler: $D_{\mathrm{KL}}(p,q) = \int p \log (p/q)$ (as a limit of α-divergence for $\alpha \to 1$)

Data coming from the Gaussian mixture model

Later we inspect data coming from a Gaussian mixture model. In such cases, there might not exist a unique "optimal" bandwidth, and our spectral approach could serve as a ranking procedure for the candidate bandwidths. We will be particularly interested in cases where the globally optimal bandwidth (with respect to some error criterion) would not be picked by the Perron vector, and instead another bandwidth is picked because it has a more central position.

Perron vector conjecture

Somehow establish connection to the Perron vector of S trough the row sums R_i

Conjecture 1. *Let $v^{(\lambda)}$ be the Perron vector of $S^{(\lambda)} = (S_{ij}^{(\lambda)})$, $S_{ij}^{(\lambda)} = \exp(-\lambda D(\hat{f}_{h_i}, \hat{f}_{h_j}))$, where D is the L^2-divergence. Let $\hat{Q}_{n,\beta}(h)$ be like in Equation (**??**). Let $i^*_{N,\lambda_N} = \arg\max_{1\leq i\leq N} v_i^{(\lambda)}$ be the index where the Perron vector has its largest entry. Then,*

$$\mathrm{dist}\left(h_{i^*_{N,\lambda_N}}, \operatorname*{arg\,min}_{h\in[h_{\min},h_{\max}]} \hat{Q}_{n,\beta}(h)\right) \to 0.$$

for $N \to \infty$, $\lambda_N \to \infty$, $\lambda_N / N^2 \to 0$.

Extension to asymmetric divergences

Stationary distribution construction for asymmetric scale similarity matrices

The fact that the similarity matrix S is not symmetric in general is not a problem because the Doob h transform also works in this case. The resulting stationary distribution π is given by

$$\pi_i = \frac{l_i r_i}{\sum_{j=1}^{N} l_j r_j}$$

where r and l are the right and left principal eigenvectors of S, that is, $Sr = \lambda_{\max} r$ and $S^T l = \lambda_{\max} l$.

Mention Davis Kahan theorem and spectral gap

Matrix pertubation theory

Davis-Kahan: Stability $\leftrightarrow$ informativeness

6 Appendix

6.1 Proofs Chapter 2

6.1.1 Proof of Proposition 1

Lemma 8. *There exist $C > 0$ and $\rho \in (0,1)$ such that $|\psi_j| \leq C\rho^j$. In particular, $\lim_{j\to\infty} \psi_j = 0$.*

Proof. Define $r_\phi := \min\{|z| : z \in \mathbb{C}, \phi(z) = 0\}$. Since $\{X_t\}$ is causal, $r_\phi > 1$. Let $R \in (1, r_\phi)$ be arbitrary. From complex analysis, it follows that $1/\phi$ is analytical on the disk $D_0(R) = \{z \in \mathbb{C} : |z| \leq R\}$. Let $1/\phi(z) = \sum_{j=0}^{\infty} \psi_j z^j$ be its representation. By Cauchy's integral inequality, see [34, Corollary 4.3], $|\psi_j| \leq R^{-j} \max_{|z|=R} |1/\phi(z)|$. Setting $C = \max_{|z|=R} |1/\phi(z)|$ and $\rho = R^{-1}$ yields the statement. □

Proof. First, notice that $\mathcal{Z}_n^{(k)} \subseteq \mathcal{Z}$ for all $n \in \mathbb{N}$ and therefore $d(y, \mathcal{Z}^{(k)}) = 0$ for all $y \in \mathcal{Z}_n^{(k)}$. By definition of the Hausdorff distance $d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}) = \sup\{d(\mathcal{Z}_n^{(k)}, z) : z \in \mathcal{Z}^{(k)}\}$. Let $y \in \mathcal{Z}_n^{(k)}$ and $z \in \mathcal{Z}^{(k)}$ be arbitrary. They have representations $\sum_{j=0}^{n} a_j g_j^{(k)}$ and $\sum_{j=0}^{\infty} b_j g_j^{(k)}$ respectively for some $a_j, b_j \in [-1, 1]$. Now we have

$$\begin{aligned} d(y,z) &= \left\| \sum_{j=0}^{n} a_j g_j^{(k)} - \sum_{j=0}^{\infty} b_j g_j^{(k)} \right\| = \left\| \sum_{j=n+1}^{\infty} b_j g_j^{(k)} + \sum_{j=0}^{n} (b_j - a_j) g_j^{(k)} \right\| \\ &\leq \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| + \sum_{j=0}^{n} |b_j - a_j| \|g_j^{(k)}\| \end{aligned}$$

and consequently

$$\begin{aligned} d(\mathcal{Z}_n^{(k)}, z) &= \inf_{a_0,\dots,a_n} d\left(\sum_{j=0}^{n} a_j g_j^{(k)}, z\right) \\ &\leq \inf_{a_0,\dots,a_n} \left(\sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| + \sum_{j=0}^{n} |b_j - a_j| \|g_j^{(k)}\|\right) \\ &= \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| \end{aligned}$$

Finally,

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}) = \sup_{b_0,b_1,\dots} d\left(\mathcal{Z}_{\setminus}^{(\|)}, \sum_{j=0}^{\infty} b_j g_j^{(k)}\right) \leq \sup_{b_0,b_1,\dots} \sum_{j=n+1}^{\infty} |b_j| \|g_j^{(k)}\| = \sum_{j=n+1}^{\infty} \|g_j^{(k)}\|$$

Using Lemma 8 we obtain

$$\|g_j^{(k)}\| = \sqrt{\sum_{r=0}^{k-1} |\psi_{j-r}|^2} \leq \sqrt{\sum_{r=0}^{k-1} C^2 \rho^{2j} \rho^{-2r}} = C\rho^{j-k+1} \sqrt{\frac{1-\rho^{2k}}{1-\rho^2}} \leq C\sqrt{k}\rho^{j-k+1}$$

In the end

$$d_H(\mathcal{Z}_n^{(k)}, \mathcal{Z}^{(k)}) \leq \sum_{j=n+1}^{\infty} \|g_j^{(k)}\| \leq C\sqrt{k} \frac{1}{\rho^{k-2}} \rho^n \sum_{j=0}^{\infty} \rho^j \leq C \frac{\sqrt{k}}{1-\rho} \frac{1}{\rho^{k-2}} \rho^n$$

□

6.1.2 Proof of Proposition 2

Lemma 9. *Let $a = (a_n)_{n\in\mathbb{Z}} \in l^2(\mathbb{Z}) \setminus \{0\}$ and $b = (b_n)_{n\in\mathbb{Z}} \in l^1(\mathbb{Z})$ be sequences. Then it holds $\|a * b\|_2 = \|a\|_2 \|b\|_1$ if and only if $b = 0$ or $b = c\delta_r$ for some $c \neq 0$ and $r \in \mathbb{Z}$.*

Proof. First, assume that the equality holds. We define the shifted sequence $\tau_m a$ by $(\tau_m a)[n] = a_{n-m}$. Then the convolution can be seen as an infinitely weighted sum of shifts, that is, $a * b = \sum_m b_m \tau_m a$. We set $x_m = b_m \tau_m a$. By assumption $\{x_m\}$ is

absolutely summable, which implies $\| \sum_{|m|\leq N} x_m - \sum_m x_m \| \to 0$ for $N \to \infty$ and by the reverse triangle inequality $|\| \sum_{|m|\leq N} x_m \|_2 - \| \sum_m x_m \|_2| \leq \| \sum_{|m|\leq N} x_m - \sum_m x_m \|_2$. Therefore,

$$\sum_{m,k} \langle x_m, x_k \rangle = \lim_{N\to\infty} \sum_{|m|\leq N} \sum_{|k|\leq N} \langle x_m, x_k \rangle = \lim_{N\to\infty} \left\| \sum_{|m|\leq N} x_m \right\|_2^2 = \left\| \sum_m x_m \right\|_2^2.$$

By assumption we also have

$$\sum_m \|x_m\|_2 = \|a\|_2 \|b\|_1 = \|a * b\|_2 = \left\| \sum_m x_m \right\|_2$$

Thus,

$$\sum_{m,k} \langle x_m, x_k \rangle = \left\| \sum_m x_m \right\|_2^2 = \sum_{m,k} \|x_m\|_2 \|x_k\|_2$$

By Cauchy Schwarz, $\langle x_m, x_k \rangle \leq \|x_m\|_2 \|x_k\|_2$ and because of the above equality it follows $\langle x_m, x_k \rangle = \|x_m\|_2 \|x_k\|_2$. This means that x_m and x_k are linearly dependent for all m, k with $b_m, b_k \neq 0$ and therefore $x_m = c_{m,k} x_k$ for some $c_{m,k} \neq 0$. It further follows that $|a_{n-m}| = |a_{n-k}|$ for all n. Now $m \neq k$ would lead to $a \notin l^2(\mathbb{Z})$. This shows that b must either be a null sequence or can only have one nonzero entry.

The other direction simply follows by plugging in and checking. □

Proof. Recall that $g_j^{(k)} = (\psi_j, \dots, \psi_{j-k+1})^T$ where $\{\psi_j\}_{j\in\mathbb{Z}}$ is a sequence given by

$$\psi_j = \begin{cases} \sum_{i=1}^p \phi_i \psi_{j-i} & , j \geq 1 \\ 1 & , j = 0 \\ 0 & , j < 0. \end{cases}$$

Let $u = (u_0, \dots, u_{k-1}) \in \mathbb{S}^{k-1}$ and consider the extension of u to a sequence $\{\tilde{u}_l\}_{l\in\mathbb{Z}}$ where $\tilde{u}_l = \mathbf{1}\{0 \leq l \leq k-1\} u_l$. Define the sequence $a = \{a_j\}_{j\in\mathbb{Z}}$ by

$$a_j = \langle g_j^{(k)}, u \rangle = \sum_{l=0}^{k-1} u_l \psi_{j-l} = \sum_{l\in\mathbb{Z}} \tilde{u}_l \psi_{j-l} = (\tilde{u} * \psi)[j].$$

By Definition of $\{\psi_j\}_{j\in\mathbb{Z}}$ we have $(q_\phi * \psi) = \delta_0$ and thus $q_\phi * a = q_\phi * (\tilde{u} * \psi) = (q_\phi * \psi) * \tilde{u} = \delta_0 * \tilde{u} = \tilde{u}$ by properties of the convolution. By Young's convolution inequality (see [17, Theorem 1.2.12])

$$1 = \|u\|_2 = \|q_\phi * a\|_2 \leq \|q_\phi\|_2 \|a\|_1. \tag{6.1}$$

Next, we use the simple formula for the support function of the zonoid $\mathcal{Z}^{(k)}$ to obtain

$$w_k(u) = 2h_{\mathcal{Z}^{(k)}}(u) = 2\sum_{j=0}^{\infty} |\langle g_j^{(k)}, u\rangle| = 2\|a\|_1.$$

Therefore, we obtain

$$w_k(u) = 2\|a\|_1 \geq \frac{2}{\|q_\phi\|_2}.$$

Plugging in the directions from the set in Equation (2.6) in w_k we easily see that the above lower bound is actually attained. It remains to show that there are no other minimizing directions. Inequality (6.1) becomes an equality iff a has exactly one nonzero entry (see Lemma 9 in the Appendix 6). This means $a = c\delta_r$ and it follows $\tilde{u} = c\tau_r q_\phi$, which forces u to have the form $\pm q_r^{(k)}/\|q_\phi\|_2$ for some $r = 0, \ldots, k-(p+1)$.

Now let $k < p+1$. We have shown above that $w_k(u)$ attains the lower bound $2/\|q_\phi\|_2$ if and only if u has the specific form above. But if $k < p+1$ there are no permitted r that would lead to the specific form. Since w_k is continuous it attains its minimum on $\mathbb{S}^{k-1}$ and therefore $m_k > 2/\|q_\phi\|_2$.

We are left to show $m_{k-1} \geq m_k$ for $k = 2, \ldots, p$. Let $v = (v_0, \ldots, v_{k-2}) \in \mathbb{S}^{k-2}$ and $u = (v_0, \ldots, v_{k-2}, 0) \in \mathbb{S}^{k-1}$ and thus

$$w_k(u) = 2\sum_j \left|\sum_{l=0}^{k-1} u_l \psi_{j-l}\right| = 2\sum_j \left|\sum_{l=0}^{k-2} v_l \psi_{j-l}\right| = w_{k-1}(v).$$

Let v be a minimizing direction. Then $m_{k-1} = w_{k-1}(v) = w_k(u) \geq \min_{u\in\mathbb{S}^{k-1}} w_k(u) = m_k$. □

6.1.3 Proof of Lemma 2

Lemma 10. *Let A, B, C be subsets of a metric space with $A \subseteq B \subseteq C$. Then $d_H(B, C) \leq d_H(A, C)$.*

Proof. Since $A \subseteq B$, we have $\inf_{b \in B} d(b, c) \leq \inf_{a \in A} d(a, c)$. Therefore,

$$d_H(B, C) = \sup_{c \in C} \inf_{b \in B} d(b, c) \leq \sup_{c \in C} \inf_{a \in A} d(a, c) = d_H(A, C).$$

□

Proof. Let $\epsilon > 0$ be arbitrary. Let $\Omega_0 = \{\omega \in \Omega : \overline{\{Y_t(\omega) : t \geq 1\}} = \mathcal{Z}\}$. By assumption, $\mathbb{P}(\Omega_0) = 1$. Fix $\omega \in \Omega_0$. In what follows, we abbreviate $Y_t = Y_t(\omega)$ and $\mathcal{P}_T = \{Y_1(\omega), \ldots, Y_T(\omega)\}$. Because $\{Y_t : t \geq 1\}$ is dense in $\mathcal{Z}$, it holds $\mathcal{Z} \subseteq \bigcup_{t \geq 1} B(Y_t, \epsilon)$ and since $\mathcal{Z}$ is compact, there exists a finite sub-cover $\mathcal{Z} \subseteq \bigcup_{j=1}^{m(\epsilon)} B(Y_{t_j(\epsilon)}, \epsilon)$. Set $T_\epsilon = \max\{t_1(\epsilon), \ldots, t_{m(\epsilon)}(\epsilon)\}$. Now, let $z \in \mathcal{Z}$ be arbitrary. It follows that $d(z, \mathcal{P}_T) = \inf\{\|z - Y_t\| : t = 1, \ldots, T\} \leq \epsilon$ for all $T \geq T_\epsilon$. Thus, $\sup_{z \in \mathcal{Z}} d(z, \mathcal{P}_T) \leq \epsilon$, since z was arbitrary. Since $\mathcal{P}_T \subseteq \mathcal{Z}$, we also have $\sup_{t=1,\ldots,T} d(Y_t, \mathcal{Z}) = 0$. By definition of the Hausdorff distance, we obtain $d_H(\mathcal{P}_T, \mathcal{Z}) = \sup_{z \in \mathcal{Z}} d(z, \mathcal{P}_T) \leq \epsilon$ and therefore $d_H(\mathcal{P}_T, \mathcal{Z}) \to 0$ for $T \to \infty$, since $\epsilon > 0$ was arbitrary. By Lemma 10 we also get $d_H(\mathrm{conv}(\mathcal{P}_T), \mathcal{Z}) \to 0$ for $T \to \infty$. Since we worked on Ω_0, this proves almost sure convergence. □

6.1.4 Proof of Lemma 3

Proof. It is well known that the family $\mathcal{B}$ of open balls with rational centers $q \in \mathbb{Q}^k$ and radii $r \in \mathbb{Q}_{>0}$ forms a countable base for $\mathbb{R}^k$. That is, $\mathcal{B}$ is countable and for any open set $U \subseteq \mathbb{R}^k$ and every $x \in U$ there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$. Define $\mathcal{B}_\mathcal{Z} = \{B \in \mathcal{B} : B \cap \mathcal{Z} \neq \emptyset\}$. Let $B \in \mathcal{B}_\mathcal{Z}$ and choose some $z \in B$. Because B is open, there exists $\delta > 0$ with $B(z, \delta) \subseteq B$. Since $z \in \mathcal{Z} = \mathrm{supp}(Y_t)$, by the definition of the support, we have $\mathbb{P}(Y_t \in B) \geq \mathbb{P}(Y_t \in B(z, \delta)) > 0$. By Birkhoff's ergodic theorem

$$\frac{1}{T} \sum_{t=1}^{T} 1\{Y_t \in B\} \overset{T \to \infty}{\to} \mathbb{P}(Y_t \in B) > 0$$

This implies that $Y_t \in B$ for infinitely many $t \geq 1$ almost surely. In particular, there exists $t \geq 1$ such that $Y_t \in B$. Let $z \in \mathcal{Z}$ and $\epsilon > 0$ be arbitrary. There exists $B \in \mathcal{B}$ such that $z \in B \subseteq B(z, \epsilon)$. From above we also know that there exists $t \geq 1$ such that $Y_t \in B$. Thus $\|Y_t - z\| \leq \epsilon$ □

6.1.5 Proof of Proposition 3

Extend to Beta distributed innovations

Proof. Let $\{Z_t\} \overset{\text{iid}}{\sim} U[-1,1]$ and write $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$. Then, $\operatorname{supp}(X_t) = [-H, H]$, where $H = \sum_{j=0}^{\infty} |\psi_j|$. Consequently

$$H - X_t = \sum_{j=0}^{\infty} |\psi_j| - \sum_{j=0}^{\infty} \psi_j Z_{t-j} = \sum_{j=0}^{\infty} |\psi_j| \underbrace{(1 - \operatorname{sgn}(\psi_j) Z_{t-j})}_{=D_{t-j} \in [0,2]}.$$

There exist infinitely many nonzero ψ_j. (Assume, for contradiction, that only finitely many ψ_j are nonzero. Then, the set $\{j \in \mathbb{N}_0 : \psi_j \neq 0\}$ is nonempty ($\psi_0 = 1$) and finite. Let J be its maximum. This would mean that $1/\phi(z) = \sum_{j=0}^{J} \psi_j z^j$ is a polynomial of order J. By assumption, the polynomial ϕ has degree p. But now we have $0 = \deg(\phi \cdot (1/\phi)) = \deg(\phi) + \deg(1/\phi) = p + J$. This would force $p = 0$ and $J = 0$, which contradicts our assumption that $p \geq 1$.) Thus, for each $\alpha > 0$ no matter how large we find distinct indices $j_1, \ldots, j_M$ where $M > \alpha$ such that $\psi_{j_m} \neq 0$ for all $m = 1, \ldots, M$. Let $\epsilon > 0$. Then $H - X_t \leq \epsilon$ implies $D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1}$ for all $m = 1, \ldots, M$. Now, it follows

$$\begin{aligned}
\mathbb{P}(H - X_t \leq \epsilon) &\leq \mathbb{P}\left(\bigcap_{m=1}^{M} \left\{ D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1} \right\} \right) \overset{(*)}{=} \prod_{m=1}^{M} \mathbb{P}(D_{t-j_m} \leq \epsilon |\psi_{j_m}|^{-1}) \\
&\overset{(**)}{=} \prod_{m=1}^{M} \min\left\{ \frac{\epsilon}{2|\psi_{j_m}|}, 1 \right\} \leq \prod_{m=1}^{M} \frac{\epsilon}{2|\psi_{j_m}|} = C_M \epsilon^M,
\end{aligned}$$

where in step $(*)$ we have used the independence of the innovations and in step $(**)$ that $D_{t-j_m} \sim \mathcal{U}[0,2]$. Because $H - X_t \leq \epsilon \iff Y_t^{(k)} \in C_{Z^{(k)}}(e_1, \epsilon)$, we

conclude

$$\frac{\mathbb{P}(Y_t^{(k)} \in C_{\mathcal{Z}^{(k)}}(e_1, \epsilon))}{\epsilon^\alpha} = \frac{\mathbb{P}(H - X_t \leq \epsilon)}{\epsilon^\alpha} \leq C_M \epsilon^{M-\alpha} \overset{\epsilon \to 0}{\to} 0.$$

□

6.1.6 Proof of Proposition 4

Proof. Let $u, v \in S^{k-1}$. Write $\hat{w}_k(u) = \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u \rangle$ and we obtain

$$\begin{aligned}
\hat{w}_k(u) - \hat{w}_k(v) &= \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u \rangle - \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, v \rangle \\
&\leq \max_{s,t} \left(\langle Y_s^{(k)} - Y_t^{(k)}, u \rangle - \langle Y_s^{(k)} - Y_t^{(k)}, v \rangle \right) \\
&= \max_{s,t} \langle Y_s^{(k)} - Y_t^{(k)}, u - v \rangle \\
&\leq \left(\max_{s,t} \| Y_s^{(k)} - Y_t^{(k)} \| \right) \| u - v \|
\end{aligned}$$

for all $u, v \in \mathbb{R}^k$.

Convexity of $\hat{w}_k$ follows from the convexity of linear functions and the pointwise maximum of convex functions. Piecewise linearity also follows directly from the standard fact that the pointwise maximum of linear functions is piecewise linear. Differentiability almost everywhere follows directly from Rademacher's theorem (Lipschitz-continuous functions are differentiable almost everywhere). The set of non-differentiable points is just the set of all points u where two distinct linear functions $\langle Y_s^{(k)} - Y_t^{(k)}, u \rangle$ and $\langle Y_q^{(k)} - Y_p^{(k)}, u \rangle$ coincide. The gradient of $\hat{w}_k$ in all directions $u \notin \mathcal{N}$ is obtained by straightforward differentiation.

By Weierstrass' extreme value theorem, $\hat{w}_k$ has a global minimum on S^{k-1}. The set of minimizers

$$\mathcal{X}_k = \arg\min \{ \hat{w}_k(u) : u \in \mathbb{S}^{k-1} \} \tag{6.2}$$

is not empty. We also note that $\mathcal{X}_k$ is compact, again due to the continuity of $\hat{w}_k$ and the compactness of $\mathbb{S}^{k-1}$.

Denote by $\hat{m}_{k,T}$ the minimum of $\hat{w}_k$ on $\mathbb{S}^{k-1}$. We have $\hat{m}_{k,T} > 0$ if and only if

the affine hull of $\mathcal{P}_k$ is the whole space $\mathbb{R}^k$. Let $\operatorname{aff}(\mathcal{P}_{k,T}) = \mathbb{R}^k$. Assume for a contradiction that $\hat{m}_{k,T} = 0$. Then there would exist a direction $u \in \mathbb{S}^{k-1}$ with $\langle Y_t^{(k)}, u\rangle = c_{k,u}$ constant for all observations t. Hence, all points $Y_t^{(k)}$ lie inside the affine hyperplane $H_u = \{x \in \mathbb{R}^k : \langle u, x\rangle = c_{k,u}\}$ and therefore $\operatorname{aff}(\mathcal{P}_{k,T}) \subseteq H_u$. But H_u is not the whole space, leading to a contradiction. If $\operatorname{aff}(\mathcal{P}_{k,T}) \neq \mathbb{R}^k$, then $V := \{Y_t^{(k)} - Y_K^{(k)} : t = K, \dots, T\}$ is a proper subspace since $Y_K^{(k)} + V = \operatorname{aff}(\mathcal{P}_{k,T})$. This means that we can find a direction $u' \in V^\perp, \|u'\| = 1$ and this direction fulfills $\langle Y_t^{(k)} - Y_K^{(k)}, u\rangle = 0$ for all $t = K, \dots, T$, implying that $\langle Y_t^{(k)}, u\rangle$ is constant for all t and thus $\hat{m}_{k,T} = 0$.

Take $Y_K^{(k)}, \dots, Y_{K+k}^{(k)}$ and let the matrix M be defined by having columns $Y_{K+1}^{(k)} - Y_K^{(k)}, \dots, Y_{K+k}^{(k)} - Y_K^{(k)}$. The determinant $D = \det M$ is a nonzero polynomial in the observations $W = (X_{K-k+1}, \dots, X_K, X_{K+1}, \dots, X_{K+k})$, implying that $\{x : D(x) = 0\}$ has Lebesgue measure zero; see [11]. Note that we need at least $T \geq K + k$ to form this vector W. The random vector W has a joint density if the innovations have a density. Therefore, it follows $\mathbb{P}(D(W) \neq 0) = 1$, that is, the determinant of M is almost surely nonzero. This immediately implies that the differences $Y_{K+1}^{(k)} - Y_K^{(k)}, \dots, Y_{K+k}^{(k)} - Y_K^{(k)}$ span the whole space $\mathbb{R}^k$ and thus $\operatorname{aff}(\mathcal{P}_{k,T}) = \mathbb{R}^k$. Combining the arguments, we find that $\hat{m}_{k,T} > 0$ is almost surely the case given that $T \geq K + k$. □

6.1.7 Proof of Lemma 4

Proof. The function $\hat{w}_k$ is a pointwise maximum over finitely many linear functions. In the following, we write $A = \{Y_s^{(k)} - Y_t^{(k)} : s, t\}$ and $f(u) = \max\{\langle a, u\rangle : a \in A\}$. Also, let $m = \min\{f(u) : u \in \mathbb{S}^{k-1}\}$ and $\mathcal{X} = \{u \in \mathbb{S}^{k-1} : f(u) = m\}$. Let $u^* \in \mathcal{X}$ and $T_{u^*}\mathbb{S}^{k-1}$ be the tangent space at u^*. Furthermore, let $A_* = \{a \in A : \langle a, u^*\rangle = m\}$ be the set of active vectors. Define $g(d) = \max\{\langle a, d\rangle : a \in A_*\}$ for all unit tangent directions $d \in T_{u^*}\mathbb{S}^{k-1} \cap \mathbb{S}^{k-1} =: S_{u^*}\mathbb{S}^{k-1}$.

Let $t > 0$. We can see that

$$f(u^* + td) = \max_{a \in A}\langle a, u^* + td\rangle = \underbrace{\max_{a \in A}\langle a, u^*\rangle}_{=m} + t \max_{a \in A}\langle a, d\rangle.$$

We want to assert that $f(u^* + td) = m + tg(d)$ for sufficiently small $t > 0$. In

general, $\max\{\langle a,d\rangle : a \in A\} \geq \max\{\langle a,d\rangle : a \in A_*\} = g(d)$ since $A_* \subseteq A$. In case of equality there is nothing to show. Therefore we assume $\max\{\langle a,d\rangle : a \in A\} > g(d)$. If this is the case, there must exist an inactive vector $b \in A \setminus A_*$ with $\langle b,d\rangle > g(d)$. We now write

$$\langle b, u^* + td\rangle = \langle b,u^*\rangle + t\langle b,d\rangle = m - \underbrace{(m - \langle b,u^*\rangle)}_{=:\delta_b>0} + t\langle b,d\rangle.$$

We see that the difference

$$m + tg(d) - \langle b, u^* + td\rangle = \delta_b - t\left(\langle b,d\rangle - g(d)\right)$$

stays positive for $t < \delta_b/(\langle b,d\rangle - g(d)) =: r_b > 0$. Set $r(u^*)$ to be the minimum over all $b \in A \setminus A_*$ with $\langle b,d\rangle > g(d)$. It follows that

$$f(u^* + td) = \max_{a\in A}\langle a, u^* + td\rangle = \max_{a\in A_*}\langle a, u^* + td\rangle = m + tg(d)$$

for all $t < r(u^*)$ and $d \in S_{u^*}\mathbb{S}^{k-1}$ and therefore also

$$f\left(\frac{u^* + td}{\sqrt{1+t^2}}\right) = \frac{m + tg(d)}{\sqrt{1+t^2}} \tag{6.3}$$

for all $d \in S_{u^*}\mathbb{S}^{k-1}$ and all $0 < t < r(u^*)$.

From this it follows that $g(d) > 0$ and therefore also $c^* = \min\{g(d) : d \in S_{u^*}\mathbb{S}^{k-1}\} > 0$ since g is a continuous function and the unit tangent sphere $S_{u^*}\mathbb{S}^{k-1}$ is compact. Now write $u_t(d) = (u^* + td)/\sqrt{1+t^2}$. For $t \in (0, r(u^*))$ we have

$$\begin{aligned} f(u_t(d)) - m &\overset{(6.3)}{=} \frac{m + tg(d)}{\sqrt{1+t^2}} - m \geq \frac{m + tc^*}{\sqrt{1+t^2}} - m \\ &= tc^* + (m + tc^*)\left(\frac{1}{\sqrt{1+t^2}} - 1\right) \\ &= tc^* - t^2 \underbrace{\frac{m + tc^*}{\sqrt{1+t^2}(1+\sqrt{1+t^2})}}_{\to m/2 \text{ for } t\downarrow 0}. \end{aligned}$$

Denote the remainder term by r_t. We have $|r_t|t^{-2} \to m/2$ for $t \downarrow 0$ and in particular there exist constants $C > 0$ and $r'_* > 0$ such that $|r_t| \leq Ct^2$ for all $t \in (0, r'_*)$.

We continue working on $t < \min\{r(u^*), r'_*, \sqrt{2}, c^*/(2C)\}$. Because $\|u_t(d) - u^*\|^2 = 2 - 2/\sqrt{1+t^2} \leq \min\{t^2, 2\} = t^2$ it follows

$$f(u_t(d)) - m \geq tc^* + r(t) \geq tc^* - Ct^2 = t(c^* - Ct) \geq \frac{c^*}{2}t \geq \frac{c^*}{2}\|u_t(d) - u^*\|, \tag{6.4}$$

which lets us see that $\mathcal{X}$ must be finite.

To see this, assume that $u_t(d) \in \mathcal{X}$ is another minimizer. It would follow from Equation (6.3) that $f(u_t(d)) - m \geq (c^*/2)\|u_t(d) - u^*\| > 0$, which is a contradiction to $u_t(d) \in \mathcal{X}$. Thus for each $u^* \in \mathcal{X}$ there exists a radius $s(u^*) = \min\{\dots\}$ such that $B(u^*, s(u^*)) \cap \mathcal{X} = \{u^*\}$. Obviously $\{B(u^*, s(u^*)) : u^* \in \mathcal{X}\}$ covers $\mathcal{X}$. Since $\mathcal{X}$ is compact, there exists a finite subcover. It follows that $\mathcal{X}$ must be finite and we denote by $u_1^*, \dots, u_q^* \in \mathcal{X}$ the finitely many minimizing directions. We can thus choose $\alpha = \min_{1 \leq i \leq q} c_i^*/2 > 0$ and $\rho = \min_{1 \leq i \leq q} s(u_i^*)$. For all $u \in \mathbb{S}^{k-1}$, $\operatorname{dist}(u, \mathcal{X}) < \rho$ now follows

$$f(u) - m \geq \|u - u_i^*\| \frac{c_i^*}{2} \geq \alpha \operatorname{dist}(u, \mathcal{X})$$

□

6.2 Proofs Chapter 3

6.2.1 Proof of Lemma 5

Proof. Since the points x_j are distinct, all entries of the matrix D are positive and by Perron-Frobenius, its Perron vector v has positive entries and is unique up to scaling. We fix v so that $v^T \mathbf{1} = 1$. We interpret v as a discrete probability distribution on $\{x_1, \dots, x_N\}$. Let X be a discrete random variable with $\mathbb{P}(X = x_j) = v_j$. Write the expected value $\mathbb{E}[X] = \mu = \sum_{j=1}^N v_j x_j$ and the variance $\operatorname{Var}[X] = \sigma^2 = \sum_{j=1}^N v_j \|x_j - \mu\|^2$. Now we have

$$(Dv)_i = \sum_{j=1}^N D_{ij} v_j = \sum_{j=1}^N \|x_i - x_j\|^2 v_j = \mathbb{E}[\|X - x_i\|^2] \overset{(*)}{=} \|\mu - x_i\|^2 + \sigma^2, \tag{6.5}$$

where at $(*)$ we have used the bias-variance decomposition. Since v is the Perron vector of D, the left hand side of Equation (6.5) is equal to ρv_i with $\rho > 0$ being the unique maximum eigenvalue of D.

Next, we write

$$\begin{aligned}
\mu &= \sum_{j=1}^{N} v_j x_j = v_1 x_1 + v_2 x_2 + \cdots + v_N x_N \\
&= x_1 - (v_2 + \cdots + v_N) x_1 \\
&+ v_2 x_2 + v_3 x_2 + \cdots + v_N x_2 - (v_3 x_2 + \cdots + v_N x_2) \\
&\vdots \\
&+ v_{N-2} x_{N-2} + v_{N-1} x_{N-2} + v_N x_{N-2} - (v_{N-1} x_{N-2} + v_N x_{N-2}) \\
&+ v_{N-1} x_{N-1} + v_N x_{N-1} - v_N x_{N-1} \\
&+ v_N x_N \\
&= x_1 + (v_2 + \cdots + v_N)(x_2 - x_1) + \cdots + (v_{N-1} + v_N)(x_{N-1} - x_{N-2}) + v_N (x_N - x_{N-1}) \\
&= x_1 + \sum_{k=1}^{N-1} \sum_{j=k+1}^{N} v_j (x_{k+1} - x_k) = x_1 + \sum_{k=1}^{N-1} a_k \delta_k,
\end{aligned}$$

where $a_k = \sum_{j=k+1}^{N} v_j$ and $\delta_k = x_{k+1} - x_k$. Note that because of the positivity of the Perron vector, it holds $1 > a_1 > a_2 > \cdots > a_{N-1} = v_N > 0$.

As a next step, write

$$\begin{aligned}
x_i - \mu = x_i - x_1 - (\mu - x_1) &= \sum_{k=1}^{i-1} \delta_k - \sum_{k=1}^{N-1} a_k \delta_k = \sum_{k=1}^{i-1} \delta_k - \sum_{k=1}^{i-1} a_k \delta_k - \sum_{k=i}^{N-1} a_k \delta_k \\
&= \underbrace{\sum_{k=1}^{i-1} (1 - a_k) \delta_k}_{=P_i} - \underbrace{\sum_{k=i}^{N-1} a_k \delta_k}_{=Q_i}
\end{aligned}$$

Consequently,

$$\begin{aligned}
\|x_i - \mu\|^2 = \|P_i - Q_i\|^2 &= \|P_i\|^2 + \|Q_i\|^2 - 2\langle P_i, Q_i \rangle \\
&\leq \|P_i\|^2 + \|Q_i\|^2 + 2\langle P_i, Q_i \rangle = \|P_i + Q_i\|^2
\end{aligned}$$

for all $i = 2, \dots, N-1$, where we have used

$$\langle P_i, Q_i \rangle = \sum_{1 \le k \le i-1} \sum_{i \le l \le N-1} \underbrace{(1-a_k)}_{>0} \underbrace{a_l}_{>0} \underbrace{\langle \delta_k, \delta_l \rangle}_{\ge 0} > 0.$$

In particular, this is the step where we use the acute chain condition (3.9).

Now we see that

$$\|P_i + Q_i\| \le \begin{cases} \|\sum_{k=1}^{N-1} a_k \delta_k\| = \|\mu - x_1\| & , a_{i-1} \ge \frac{1}{2} \\ \|\sum_{k=1}^{N-1} (1-a_k)\delta_k\| = \|x_N - \mu\| & , a_{i-1} < \frac{1}{2}. \end{cases} \tag{6.6}$$

Finally, by combining Equations (6.5) and (6.6) we arrive at

$$\begin{aligned} \rho v_i - \sigma^2 = \|\mu - x_i\|^2 \le \|P_i + Q_i\|^2 &\le \max\{\|\mu - x_1\|, \|x_N - \mu\|\} \\ &= \max\{\rho v_1 - \sigma^2, \rho v_N - \sigma^2\} \end{aligned}$$

and therefore $v_i \le \max\{v_1, v_N\}$ for all interior coordinates $i = 2, \dots, N-1$. □

6.2.2 Proof of Lemma 7

Proof. Let $h > 0$. For every $x \in \mathbb{R}$ we have

$$\begin{aligned} \lim_{\tau \to 0} \frac{1}{\tau} (\hat{f}_{h+\tau}(x) - \hat{f}_h(x)) &= \frac{1}{n} \sum_{j=1}^{n} \lim_{\tau \to 0} \frac{K_{h+\tau}(x - X_j) - K_h(x - X_j)}{\tau} \\ &= \frac{1}{n} \sum_{j=1}^{n} \partial_h K_h(x - X_j) \\ &= -\frac{1}{nh^2} \sum_{j=1}^{n} \left(K\left(\frac{x - X_j}{h}\right) + \frac{x - X_j}{h} K'\left(\frac{x - X_j}{h}\right) \right) \\ &=: \partial_h \hat{f}_h(x). \end{aligned}$$

Let $0 < |\tau| < h/2$. Using the mean value inequality, we bound

$$\begin{aligned}\left|\frac{K_{h+\tau}(u) - K_h(u)}{\tau}\right| &\leq \sup_{s\in(\min\{h,h+\tau\},\max\{h,h+\tau\})} |\partial_s K_s(u)| \\ &\leq \sup_{s\in(h/2,3h/2)} \left|\frac{1}{s^2} L\left(\frac{u}{s}\right)\right| \\ &\leq \frac{4}{h^2} \underbrace{\sup_{a\in(2/3,2)} |L(au/h)|}_{=:G(u/h)}\end{aligned}$$

and

$$\begin{aligned}\left|\frac{\hat{f}_{h+\tau}(x) - \hat{f}_h(x)}{\tau}\right| &\leq \frac{1}{n}\sum_{j=1}^n \left|\frac{K_{h+\tau}(x - X_j) - K_h(x - X_j)}{\tau}\right| \\ &\leq \frac{4}{nh^2}\sum_{j=1}^n G\left(\frac{x - X_j}{h}\right) =: M(x),\end{aligned}$$

where $M \in L^2(\mathbb{R})$ because by the Cauchy-Schwarz inequality $\|M\|_2^2 \leq 16h^{-3}\|G\|_2^2$ and by the decay assumption on L we get $\|G\|_2^2 \leq 3C^2/(2\alpha - 1) < +\infty$. By the dominated convergence theorem

$$\left\|\frac{\hat{f}_{h+\tau} - \hat{f}_h}{\tau} - \partial_h \hat{f}_h\right\|_2 \to 0, \quad \tau \to 0. \tag{6.7}$$

Set $d\Psi(h)[\tau] := \tau\partial_h \hat{f}_h$. Now it holds

$$\hat{f}_{h+\tau} - \hat{f}_h = \tau\partial_h \hat{f}_h + \underbrace{\left(\hat{f}_{h+\tau} - \hat{f}_h - \tau\partial_h \hat{f}_h\right)}_{=:r_h(\tau)},$$

where the remainder $r_h(\tau)$ fulfills $\|r_h(\tau)\|_2/|\tau| \to 0$ as $\tau \to 0$ as established in Equation (6.7).

We must now establish the continuity of $d\Psi : (0,\infty) \to \mathcal{L}(\mathbb{R}, L^2(\mathbb{R}))$.

Fix a closed nonempty interval $J := [a,b] \subseteq (0,\infty)$. Applying the bound from

above:

$$\begin{aligned}\|r_h(\tau)\|_2 &\leq |\tau| \sup_{t\in[0,1]} \|d\Psi(h+\tau t) - d\Psi(h)\|_{\mathsf{op}} \\ &\leq |\tau| \underbrace{\sup_{h,s\in J, |h-s|\leq|\tau|} \|d\Psi(s) - d\Psi(h)\|_{\mathsf{op}}}_{=:\mu_J(|\tau|)}\end{aligned}$$

for all $h \in [a,b]$ and $\tau \in \mathbb{R}$ with $h+\tau \in [a,b]$. $\mu_J(s) = \sup_{x,y\in J,|x-y|\leq s} \|d\Psi(x) - d\Psi(y)\|_{\mathsf{op}}$ is known in the literature as the modulus of continuity. Because $d\Psi$ is continuous, $d\Psi : J \to \mathcal{L}(\mathbb{R}, L^2(\mathbb{R}))$ is uniformly continuous, and in this case it is well known that $\mu_J(s) \to 0$ for $s \downarrow 0$. Also note that $\|d\Psi(s) - d\Psi(h)\|_{\mathsf{op}} = \|\partial_h \hat{f}_h|_{h=s} - \partial_h \hat{f}_h|_{h=h}\|_2$. Since we have established the continuity of $d\Psi$ before, the proof is finished. □

6.2.3 Proof of Proposition 8

Proof. Set $\epsilon_N := \sqrt{\lambda_N}\Delta_{r,N}$ and $t_{N,j} := \epsilon_N(j - i_N)$. Note that $\epsilon_N = t_{N,j+1} - t_{N,j}$ for all $j = 1, \ldots, N-1$. Fix $A > 0$ and set $I_N(A) := \{j = 1, \ldots, N : |t_{N,j}| \leq A\}$.

We have $\#I_N(A) \leq 2A\epsilon_N^{-1} + 1$ because $\#I_N(A) \leq \#\{j \in \mathbb{Z} : -A\epsilon_N^{-1} + i_N \leq j \leq A\epsilon_N^{-1} + i_N\} = \lfloor A\epsilon_N^{-1}\rfloor - \lceil -A\epsilon_N^{-1}\rceil + 1 = 2\lfloor A\epsilon_N^{-1}\rfloor + 1 \leq 2A\epsilon_N^{-1} + 1$.

We also set $\delta_{N,j} := (j - i_N)\Delta_{r,N} = \psi_r(h_j) - \psi_r(h_{i_N})$.

Split the row sum:

$$R_{i_N}^{(N)} = \sum_{j\in I_N(A)} S_{i_N j}^{(N)} + \underbrace{\sum_{j\notin I_N(A)} S_{i_N j}^{(N)}}_{=:T_N(A)}.$$

We have $D_{i_N j} \geq c(\delta_{N,j})^2$ for some constant $c > 0$. To see this, first write

$$\left| e^{-\frac{h_j^2\omega^2}{2}} - e^{-\frac{h_{i_N}^2\omega^2}{2}} \right| = \int_{\min\{h_j, h_{i_N}\}}^{\max\{h_j, h_{i_N}\}} t\omega^2 e^{-\frac{t^2\omega^2}{2}}\, dt \geq |h_{i_N} - h_j| h_{\min}\omega^2 e^{-h_{\max}^2\omega^2/2}$$

and, furthermore

$$\begin{aligned} D_{i_N j} &= \frac{1}{4\pi} \int |P_n(\omega)|^2 \left| e^{-\frac{h_j^2\omega^2}{2}} - e^{-\frac{h_{i_N}^2\omega^2}{2}} \right|^2 d\omega \\ &\geq |h_{i_N} - h_j|^2 \frac{h_{\min}^2}{4\pi} \int \underbrace{|P_n(\omega)|^2 \omega^4 e^{-h_{\max}^2\omega^2}}_{=:g(\omega)} \, d\omega. \end{aligned}$$

Now, $|\delta_{N,j}| = |h_j^r - h_{i_N}^r|/r = \int_{\min\{h_j,h_{i_N}\}}^{\max\{h_j,h_{i_N}\}} s^{r-1}\,ds \leq h_{\min}^{r-1}|h_j - h_{i_N}|$. Thus, it follows $D_{i_N j} \geq c(\delta_{N,j})^2$ with positive constant $c = h_{\min}^{4-2r}(4\pi)^{-1}\int g(\omega)\,d\omega$. It follows $S_{i_N j}^{(N)} = e^{-\lambda_N D_{i_N j}} \leq e^{-c\lambda_N(\delta_{N,j})^2} = e^{-ct_{N,j}^2}$. Note that the constant is, in fact, positive due to the continuity of the integrand $g(\omega)$ and because $g(\omega) \neq 0$ on a disk $\{\omega : |\omega| \leq \delta\} \setminus \{0\}$ for some $\delta > 0$.

Continuing

$$\begin{aligned} \epsilon_N T_N(A) &\leq \epsilon_N \sum_{1\leq j\leq N : \epsilon_N|j-i_N|>A} e^{-ct_{N,j}^2} = \epsilon_N \sum_{1-i_N\leq k\leq N-i_N : |k|>\epsilon_N^{-1}A} e^{-c\epsilon_N^2 k^2} \\ &\leq \epsilon_N \sum_{k\in\mathbb{Z}:|k|>\epsilon_N^{-1}A} e^{-c\epsilon_N^2 k^2} = 2\sum_{k=\lfloor\epsilon_N^{-1}A\rfloor+1}^{\infty} \epsilon_N e^{-c\epsilon_N^2 k^2} \\ &\leq 2\sum_{k=\lfloor\epsilon_N^{-1}A\rfloor+1}^{\infty} \int_{(k-1)\epsilon_N}^{k\epsilon_N} e^{-ct^2}\,dt \leq 2\int_{A-\epsilon_N}^{\infty} e^{-ct^2}\,dt \end{aligned}$$

We see that $\epsilon_N T_N(A)$ is asymptotically bounded and therefore

$$\limsup_{N\to\infty} \epsilon_N T_N(A) \leq 2\int_A^\infty e^{-ct^2}\,dt \leq \frac{2}{A}\int_A^\infty te^{-ct^2}\,dt = \frac{e^{-cA^2}}{cA} \to 0, \quad A\to\infty.$$

Now, let $j \in I_N(A)$, that is, $j - i_N \leq |j - i_N| \leq \epsilon_N^{-1}A = \mathbf{O}(N/\sqrt{\lambda_N}) = \mathbf{o}(N)$. Thus, by Lemma 6 we have $\tau_{N,j} := h_j - h_{i_N} = h_{i_N}^{1-r}\delta_{N,j} + \mathbf{O}(\delta_{N,j}^2)$ and by Lemma 7 follows

$$\begin{aligned} \hat{f}_{h_j} - \hat{f}_{h_{i_N}} &= \hat{f}_{h_{i_N}+\tau_{N,j}} - \hat{f}_{h_{i_N}} = \tau_{N,j}\partial_h \hat{f}_h|_{h=h_{i_N}} + u_{N,j} \\ &= h_{i_N}^{1-r}\delta_{N,j}\partial_h \hat{f}_h|_{h=h_{i_N}} + r_{N,j}, \end{aligned}$$

where $u_{N,j} \in L^2$ is the remainder with $\|u_{N,j}\|_2 \leq |\tau_{N,j}|\mu_{[h_{\min},h_{\max}]}(|\tau_{N,j}|)$ according to Lemma 7 and $r_{N,j} := u_{N,j} + \alpha_{N,j}\partial_h \hat{f}_{h_{i_N}}$, where $|\alpha_{N,j}| \leq C\delta_{N,j}^2$ for some constant

$C > 0$ and all N sufficiently large.

We have $|\tau_{N,j}| \leq h_{\max}^{1-r}|\delta_{N,j}|$ and $|\delta_{N,j}| \leq A\lambda_N^{-1/2}$ for all $j \in I_N(A)$. Since $h \mapsto \partial_h \hat{f}_h$ is continuous in $L^2(\mathbb{R})$, it follows that $h \mapsto \|\partial_h \hat{f}_h\|_2$ is continuous in $\mathbb{R}$. Thus, $M_0 := \max_{h\in[h_{\min},h_{\max}]} \|\partial_h \hat{f}_h\|_2$ is finite.

We have $\|r_{N,j}\|_2 \leq |\delta_{N,j}|a_N(A)$ for some sequence $a_N(A) \to 0$ for $N \to \infty$, because

$$\begin{aligned}
\|r_{N,j}\|_2 &\leq \|u_{N,j}\|_2 + C\delta_{N,j}^2 \|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2 \\
&\leq |\tau_{N,j}|\mu_{[h_{\min},h_{\max}]}(|\tau_{N,j}|) + C\delta_{N,j}^2 M_0 \\
&\leq |\delta_{N,j}| \left(h_{\max}^{1-r}\mu_{[h_{\min},h_{\max}]}(h_{\max}^{1-r}|\delta_{N,j}|) + CM_0|\delta_{N,j}|\right) \\
&\leq |\delta_{N,j}| \left(\underbrace{h_{\max}^{1-r}\mu_{[h_{\min},h_{\max}]}(h_{\max}^{1-r}A\lambda_N^{-1/2}) + CM_0 A\lambda_N^{-1/2}}_{=:a_N(A)} \right)
\end{aligned}$$

and $a_N(A) \to 0$ since $\lambda_N^{-1/2} \to 0$ by assumption and $\mu_{[h_{\min},h_{\max}]}(s) \to 0$ for $s \downarrow 0$ by Lemma 7.

Consequently,

$$D_{i_N j} = \frac{1}{2}\|\hat{f}_{h_j} - \hat{f}_{h_{i_N}}\|_2^2 = \frac{1}{2}h_{i_N}^{2(1-r)}(\delta_{N,j})^2 \|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2^2 + \rho_{N,j} = (\delta_{N,j})^2 \hat{Q}_r(h_{i_N}) + \rho_{N,j},$$

where $\rho_{N,j} = h_{i_N}^{1-r}\delta_{N,j}\langle \partial_h \hat{f}_h|_{h=h_{i_N}}, r_{N,j}\rangle + (1/2)\|r_{N,j}\|_2^2$ is yet another remainder term for which we have

$$\begin{aligned}
|\rho_{N,j}| &\leq |\delta_{N,j}|h_{i_N}^{1-r}|\langle \partial_h \hat{f}_h|_{h=h_{i_N}}, r_{N,j}\rangle| + (1/2)\|r_{N,j}\|_2^2 \\
&\leq |\delta_{N,j}|h_{\max}^{1-r} \underbrace{\|\partial_h \hat{f}_h|_{h=h_{i_N}}\|_2}_{\leq M_0} \underbrace{\|r_{N,j}\|_2}_{\leq |\delta_{N,j}|a_N(A)} + (1/2) \underbrace{\|r_{N,j}\|_2^2}_{\leq (\delta_{N,j})^2 a_N(A)^2} \\
&\leq (\delta_{N,j})^2 \underbrace{\left(h_{\max}^{1-r}M_0 a_N(A) + (1/2)a_N(A)^2\right)}_{=:b_N(A)}.
\end{aligned}$$

Since $a_N(A) \to 0$, we have shown that $\rho_{N,j} = \mathbf{o}(\delta_{N,j}^2)$. Thus,

$$S_{i_N j} = e^{-\lambda_N D_{i_N j}} = e^{-\lambda_N \delta_{N,j}^2 \hat{Q}_r(h_{i_N})} e^{-\lambda_N \rho_{N,j}}.$$

Define $\eta_{N,j} = e^{-\lambda_N \rho_{N,j}} - 1$. It follows $S_{i_N j} = e^{-\lambda_N (\delta_{N,j})^2 \hat{Q}_r(h_{i_N})}(1 + \eta_{N,j})$. Now write

$$\epsilon_N \sum_{j \in I_N(A)} S^{(N)}_{i_N j} = \epsilon_N \sum_{j \in I_N(A)} e^{-\lambda_N \delta^2_{N,j} \hat{Q}_r(h_{i_N})} + \epsilon_N \sum_{j \in I_N(A)} e^{-\lambda_N \delta^2_{N,j} \hat{Q}_r(h_{i_N})} \eta_{N,j}.$$

The local error can be bounded:

$$\left| \epsilon_N \sum_{j \in I_N(A)} \underbrace{e^{-\lambda_N \delta^2_{N,j} \hat{Q}_r(h_{i_N})}}_{\leq 1} \eta_{N,j} \right| \leq \underbrace{\epsilon_N \# I_N(A)}_{\leq 2A + \epsilon_N} \sup_{j \in I_N(A)} |\eta_{N,j}| \to 0, \quad N \to \infty.$$

We must show that $\sup_{j \in I_N(A)} |\eta_{N,j}| \to 0$ for $N \to \infty$. We have $\lambda_N |\rho_{N,j}| \leq \lambda_N (\delta_{N,j})^2 b_N(A) \leq A^2 b_N(A)$ for all $j \in I_N(A)$. Consequently, $|\eta_{N,j}| = |e^{-\lambda_N \rho_{N,j}} - 1| \leq \max\{e^{A^2 b_N(A)} - 1, 1 - e^{-A^2 b_N(A)}\} \to 0$ as $N \to \infty$.

Since $\hat{Q}_r$ is continuous in h_0, we have $\hat{Q}_r(h_{i_N}) \to \hat{Q}_r(h_0)$ for $N \to \infty$. Therefore

$$\left| \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_{i_N})} - \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_0)} \right| \leq (2A + \epsilon_N) A^2 |\hat{Q}_r(h_{i_N}) - \hat{Q}_r(h_0)| \to 0.$$

We continue working with the remaining non-vanishing term:

$$\begin{aligned} &\left| \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_{i_N})} - \int_{-A}^{A} e^{-t^2 \hat{Q}_r(h_0)} \, dt \right| \\ &\leq \underbrace{\left| \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_{i_N})} - \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_0)} \right|}_{\to 0} + \left| \epsilon_N \sum_{j \in I_N(A)} e^{-t^2_{N,j} \hat{Q}_r(h_0)} - \int_{-A}^{A} e^{-t^2 \hat{Q}_r(h_0)} \, dt \right|. \end{aligned}$$

In order to show that the second summand above also converges to 0 for $N \to \infty$ we first rewrite the sum over $I_N(A)$. Before that, set $q = \hat{Q}_r(h_0)$, $g(t) := e^{-qt^2}$, $m_N := \lfloor A \epsilon_N^{-1} \rfloor$ and $x_k := \epsilon_N(-m_N + k - 1)$. For sufficiently large N we can now

write

$$\begin{aligned}\epsilon_N \sum_{j\in I_N(A)} g(t_{N,j}) &= \epsilon_N \sum_{j=1,\dots,N:-A\epsilon_N^{-1}+i_N\le j\le A\epsilon_N^{-1}+i_N} g(t_{N,j}) \\ &= \epsilon_N \sum_{k=-\lfloor A\epsilon_N^{-1}\rfloor}^{\lfloor A\epsilon_N^{-1}\rfloor} g(k\epsilon_N) \\ &= \sum_{k=-m_N}^{m_N} (k\epsilon_N-(k-1)\epsilon_N)g(k\epsilon_N) \\ &= \sum_{k=1}^{2m_N+1} (x_k-x_{k-1})g(x_k)\end{aligned}$$

We have used $t_{N,1}=\sqrt{\lambda_N}\delta_{N,1}=\sqrt{\lambda_N}r^{-1}(h_{\min}^r-h_{i_N}^r)\to-\infty$ and $t_{N,N}\to+\infty$ for $N\to\infty$ because we assumed that $h_{i_N}\to h_0$ for some interior scale h_0 and because $\lambda_N\to\infty$.

Now $\{x_0,\dots,x_{2m_N+1}\}$ is a partition of the interval $[-\epsilon_N m_N-\epsilon_N,\epsilon_N m_N]$.

Thus,

$$\begin{aligned}\left|\sum_{k=1}^{2m_N+1}(x_k-x_{k-1})g(x_k)-\int_{-\epsilon_N m_N-\epsilon_N}^{\epsilon_N m_N} g(x)\,dx\right| &\le \sum_{k=1}^{2m_N+1}\int_{x_{k-1}}^{x_k}|g(x_k)-g(x)|\,dx \\ &\le 2A\hat{Q}_r(h_0)(2A\epsilon_N+\epsilon_N^2)\to 0,\quad N\to\infty,\end{aligned}$$

because $\epsilon_N\to 0$ since we have assumed $\lambda_N/N^2\to 0$ and

$$\left|\int_{-\epsilon_N m_N-\epsilon_N}^{\epsilon_N m_N} g(t)\,dt-\int_{-A}^{A} g(t)\,dt\right|\to 0,\quad N\to\infty,$$

because $-\epsilon_N(m_N+1)\to -A$ and $\epsilon_N m_N\to A$.

We have now shown that the aforementioned second summand also converges to zero.

Finally,

$$\begin{aligned} L_A &= \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} \le \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} + \liminf_{N\to\infty} \epsilon_N T_N(A) \\ &\le \liminf_{N\to\infty} \epsilon_N R^{(N)}_{i_N} \le \limsup_{N\to\infty} \epsilon_N R^{(N)}_{i_N} \le \lim_{N\to\infty} \epsilon_N \sum_{j\in I_N(A)} S^{(N)}_{i_N j} + \limsup_{N\to\infty} \epsilon_N T_N(A) \\ &\le L_A + \frac{e^{-cA^2}}{cA} \end{aligned}$$

Letting $A \to \infty$, we see that the limit of $\epsilon_N R^{(N)}_{i_N}$ as $N \to \infty$ exists and is given by

$$\lim_{N\to\infty} \epsilon_N R^{(N)}_{i_N} = \lim_{A\to\infty} L_A = \int_{-\infty}^{\infty} e^{-t^2 \hat{Q}_r(h_0)}\, dt = \sqrt{\frac{\pi}{\hat{Q}_r(h_0)}}.$$

We must make sure that $\hat{Q}_r(h_0) > 0$. This follows from previously established results inside this proof. For all sufficiently large N we have $0 < c \le \delta^{-2}_{N,i_N+1} D_{i_N,i_N+1} = \hat{Q}_r(h_{i_N}) + \delta^{-2}_{N,i_N+1}\rho_{N,i_N+1}$ and taking the limit gives $\hat{Q}_r(h_0) \ge c > 0$. □

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Erklärung

Ich erkläre, dass ich die Arbeit selbständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe.

Ulm, den .....

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